Build On or Branch Out?

Patent Grants and Peers' Innovation Policies*

Seiwoong Hong†

October 31, 2026

[Link to the Most Recent Version]

Abstract

When a firm is granted a patent, its technologically related peers can gain from the disclosed knowledge or lose from the patentee's stronger exclusion rights. Instrumenting grants with patent examiner leniency, I find that granting a patent on an important innovation raises R&D expenses among technologically close peers and shifts the patenting of close product-market rivals toward new technologies. Stock market reactions on the grant date reveal which force dominates for a given peer. When the stock prices of both the patentee and the peer rise, the peer raises R&D expenses by about 7% and its acquisition probability by 3 percentage points and builds on the granted patent's technology; when the peer's stock price falls, it moves its patenting away from that technology. Financial flexibility amplifies these responses, and patent enforcement limits expansion.

Keywords: Patent; Technological Complementarity; Product-Market Rivalry; Innovation; Stock Price Informativeness; Absorptive Capacity; Financial Flexibility; Knowledge Diffusion.

* I am grateful to Tom Meling, René Stulz (Chair), and Mike Wittry for their invaluable guidance and support. For helpful comments, suggestions, and discussions, I thank Julian Atanassov, Ezra Daniel, Yicheng Liu, Zack Liu, Jungu Myung, David Robinson, Sergey Sarkisyan, Kris Shen, Daisy Wang, and Yufeng Wu. I also thank session participants at the 2026 FMA Annual Meeting, as well as participants in the brown bag seminar at The Ohio State University for helpful comments. All remaining errors are my own.

† Fisher College of Business, The Ohio State University, 2100 Neil Avenue, Columbus, OH 43210. E-mail: hong.1017@osu.edu.

1. Introduction

Most patents granted to a firm have no noticeable effect on the stock prices of related firms. However, some patent grants coincide with contrasting market reactions. Around Geron's October 1, 2002 grant of U.S. Patent 6,458,589 for liver cells derived from pluripotent stem cells, its five-day cumulative abnormal return (CAR) was +3.75%, while StemCells' was −19.19%. Around Netflix's June 24, 2003 grant of U.S. Patent 6,584,450 for its subscription rental service, Netflix and Blockbuster had five-day CARs of +11.91% and +9.28%, respectively (Figure 4). This paper asks what technologically related peers do after patent grants—whether they build on the disclosed technology or move away from it—and whether stock prices around the grant date reveal which response will follow.[1]

Theory offers competing predictions. Knowledge spillovers create opportunities for technologically related firms, whereas exclusion rights and product-market rivalry create threats (Bloom et al., 2013). I organize these forces in a diffusion–constraint framework: a grant expands a peer's opportunities when technological complementarity dominates and constrains them when rivalry dominates. Because both forces operate at once, average effects can be close to zero even when individual grants matter a great deal. Which force prevails has implications beyond the firms involved: the diffusion of knowledge across firms is a central source of aggregate growth, and recent evidence points to a slowdown in that diffusion (Akcigit and Ates, 2023).

Patent grants provide a useful setting for examining how disclosure and exclusion shape peers' innovation. The patent document makes the invention's technical content publicly available, potentially helping peers build on that knowledge, while the grant gives the patentee the right to exclude others from using the protected invention. Although disclosure often precedes the grant, the grant establishes patent protection over the disclosed technology and thus shapes the conditions under which peers can use and build on it. Stock prices aggregate investors' assessment of both forces on the grant date. If these reactions merely reflected temporary mispricing (Haddad et al., 2022), they should not predict peers' subsequent real

[1] I refer to the patent recipient as the focal firm and to technologically related firms as peers.

decisions; if they contain information about technological opportunities and competitive threats, they should.

Two facts motivate the design. Most patent grants produce no measurable market reaction. Among grants that significantly raise the patentee's market capitalization, peers' stock price reactions fall on both sides of zero.[2] The sign of the peer's reaction can thus reveal whether investors view a given grant as an opportunity or a threat for that particular peer. I classify each grant–peer pair by the joint reaction over the five trading days around the grant date: a grant is AR^+ when both the focal firm's and the peer's standardized abnormal returns exceed two, and AR^- when the focal firm's exceeds two and the peer's falls below minus two. AR^+ and AR^- joint reactions are 3.6 and 4.0 times as frequent, respectively, on grant days as on days with equally large abnormal returns for the focal firm but no grant. Of the 924,125 patents granted to public firms in the sample, 6,341 meet the AR^+ or AR^- criterion, corresponding to 3,129 AR^+ and 1,815 AR^- peer-focal-year observations.

Using U.S. public firms from 2001 to 2021, I identify technological peers through shared Cooperative Patent Classification (CPC) subclasses and direct citation links. I then separate a causal claim from a predictive one. For the causal claim, I instrument the focal firm's grant with the leniency of the examiner who reviewed the application (Farre-Mensa et al., 2020), focusing on applications with high predicted importance, based on information available before the examiner's decision, and interacting the instrument with technological and product-market relationships measured before the grant. The first stage is strong: a ten-percentage-point higher examiner grant rate raises the probability of a grant by 9.3 percentage points. When such an application is granted rather than rejected, annual R&D expenditures rise by 0.5% of assets and the probability of making an acquisition rises by 1.3 percentage points (significant only at the 10% level) among technologically close peers, while the exploration ratio—the share of a peer's new patent applications that fall outside its three largest CPC classes in the five years before the grant—rises among

[2] Empirically, positive peer valuation shocks are substantially more common than negative ones. By construction, the focal firm's reaction is significantly positive in every selected event; conditional on this, peers' reactions are positive about two-thirds of the time.

close rivals. The corresponding estimates for other peers and for grants with low predicted importance are statistically insignificant. For the predictive claim, I ask whether grant-day peer reactions carry information about the scale and direction of the peer's subsequent innovation beyond the pre-grant relationship between the two firms. Because these reactions are realized at the grant, I interpret the AR^+ and AR^- estimates as conditional responses rather than causal effects.

The results are consistent with both the opportunity and the threat channels. When market reactions indicate complementarity (AR^+), peers increase both R&D investment and acquisition activity in the years following the focal firm's patent grant. The magnitude is economically meaningful: R&D investment rises by 0.6 percent of assets, roughly 7 percent of the sample mean, while the probability of engaging in an acquisition increases by about 3 percentage points. These patterns are consistent with diffusion-driven expansion, where knowledge flows between technologically connected firms lead to co-investment in innovation.

In contrast, when market reactions indicate rivalry (AR^-), peers exhibit a different kind of adaptive behavior. Instead of withdrawing from innovation, they redirect their inventive efforts toward new technological domains. Specifically, the number of patent filings does not change significantly after AR^- patent grants, but the exploration ratio rises by roughly its sample mean, indicating that peers diversify into unfamiliar technological spaces. Their new patents also become about 12.5% less similar in text to the focal firm's patent, whereas after AR^+ grants they become about 16% more similar. Thus, AR^- does not imply passivity but a reallocation of innovative effort away from direct rivalry.

These responses unfold over several years. R&D peaks at about 0.8 percentage points of assets one year after an AR^+ grant, and the acquisition response reaches about 4.5 percentage points in year two, although the yearly estimates are imprecise. Applying the estimated R&D response to the average affected peer implies about $21.6 million of additional R&D over four years, or about $5.4 million per year—roughly 7% of annual R&D for a peer with average R&D intensity. Responses also tend to be larger after larger peer reactions, although the gradient is statistically significant only for acquisitions.

Three pieces of evidence suggest that these responses reflect the information in the patent rather than

the market reaction itself. First, equally large joint price movements of the focal firm and its peer on days without a patent grant are followed by no comparable change in R&D, acquisitions, or the direction of innovation. Second, a weaker version of the same pattern appears when significant grants are classified only by technological and product-market proximity measured before the grant, and market reactions keep their explanatory power when both classifications are included. Third, the responses are tied to the patented technology itself. After AR^+ grants, the share of peers' new patents in the technology group of the focal firm's patent rises by 24% and their citations to the focal firm's patents in that group by 43%, whereas citations to the focal firm's other patents do not change; after AR^- grants, peers move out of that group. Net equity issuance is the exception: it rises after AR^+ grants but by a similar amount after equally large price movements without a grant, consistent with market timing (Baker et al., 2003) rather than patent-specific financing.

The R&D results are robust to alternative peer definitions, the exclusion of grants near earnings announcements, overlapping network exposure, and alternative expected-return models. The acquisition estimates keep their sign but lose statistical significance under alternative peer definitions and in the smaller samples that restrict overlapping exposure. Application publication dates have no statistically significant R&D or acquisition responses. Placebo grant dates have no significant acquisition response, although the placebo R&D estimates are too imprecise to rule out an effect of the baseline size. Section 5 reports the implementation and results of these tests.

Peers' ability and freedom to act shape these responses. Responses to AR^+ grants are concentrated among technologically close peers rather than product-market rivals. Peers with more financial flexibility, such as those with high cash holdings and low leverage, and peers with greater innovative capability, a proxy for proximity to the technological frontier (Akcigit and Ates, 2023), respond more strongly, consistent with absorptive capacity (Cohen and Levinthal, 1990). Aggressive post-grant filing by the focal firm, high litigation risk, and concentrated patent ownership are associated with weaker expansion after AR^+ grants, and stronger enforcement is associated with stronger redirection after AR^- grants.

This paper relates to three literatures. First, it contributes to work on how one firm's innovation affects

others. Bloom et al. (2013) separate technology spillovers from product-market rivalry using firm-level proximity, Galasso and Schankerman (2015) show that invalidating patents increases follow-on innovation by other firms, and Akcigit and Ates (2023) document a slowdown in knowledge diffusion from frontier firms. Studies emphasizing positive peer effects and those finding negative competitive effects (e.g., Megna and Klock, 1993) can be reconciled once one recognizes that the balance between spillovers and rivalry varies across individual grants. Grant-day market reactions provide information about this balance at the event level, and they explain peers' responses beyond pre-grant proximity.

Second, the paper relates to research that measures the value of innovation from stock prices. Kogan et al. (2017) value patents using the patentee's grant-day return, Kelly et al. (2021) measure patent importance from text, and Testoni (2022) studies value spillovers from technological acquisitions. Haddad et al. (2022) argue that market reactions to innovation may partly reflect mispricing. I extend the use of grant-day returns from the patentee to its technological peers and show that these reactions predict peers' subsequent real decisions, which mispricing alone would not produce.

Third, the paper relates to the real effects of financial markets (Bond et al., 2012). Baker et al. (2003) show that stock prices affect the investment of equity-dependent firms through the cost of external finance. Peer reactions to patent grants predict R&D, acquisitions, and the direction of patenting, whereas equally large price movements without a grant do not. This contrast separates an information channel, through which prices reveal investment opportunities created by others' innovation, from a financing channel.

The remainder of this paper is structured as follows: Section 2 develops the conceptual framework and the main hypotheses. Section 3 describes the data sources and outlines the empirical methodology. Section 4 presents empirical results. Section 5 provides robustness tests for empirical analyses. Section 6 concludes. The Internet Appendix contains supporting analyses.

2. Conceptual Framework and Hypotheses

Innovation occurs in an environment characterized by interdependence among firms. Firms rarely

innovate in isolation; their technological progress both influences and is influenced by the inventive activity of others. Patents formalize this interdependence. Each grant discloses new knowledge that may stimulate complementary research, but it also defines boundaries that constrain imitation and intensify competition. The dual nature of patents—as conduits for knowledge diffusion and as instruments of exclusion—underlies the framework developed in this paper. I conceptualize a focal firm's patent grant as a discrete innovation shock whose impact on peers depends on (i) the nature of their technological and product-market relationships, and (ii) the extent of their absorptive and financial capacity to respond. Market reactions around grant events, observed through abnormal returns, provide real-time indicators of whether these relationships are characterized by complementarity or rivalry.

Classical theories offer conflicting predictions about how firms respond to the innovation of others. Arrow's (1962) replacement effect implies that competition strengthens incentives to innovate, whereas Schumpeter (1942) emphasizes that market power allows innovators to appropriate the returns to innovation. Aghion et al. (2005) reconcile these views through an inverted-U relationship between competition and innovation. These frameworks concern how competition shapes a firm's own incentives to innovate. The framework below asks how a patent granted to one firm shifts these incentives for its peers, which depends on both technological complementarity and product-market rivalry.

The framework builds on three well-established ideas. First, the diffusion of knowledge depends on proximity in technology space. Cohen and Levinthal (1990) define absorptive capacity as a firm's ability to recognize the value of external information, assimilate it, and apply it to commercial ends. Peers with similar technological expertise are better positioned to understand and exploit the knowledge revealed in a focal firm's patent. Second, diffusion is constrained by rivalry in product markets. Bloom et al. (2013) decompose cross-firm externalities into positive spillovers and negative business-stealing effects using measures of technological proximity and product-market rivalry, respectively. The balance between these forces determines whether innovation spreads or stalls. Third, peers' actions are bounded by financial capacity and strategic flexibility. Even when opportunities exist, exploiting them requires resources and commitment. Peers with stronger balance sheets can respond more aggressively to positive opportunities,

while those facing leverage constraints or financing frictions may not.

To integrate these forces, I describe the peer response as a function of two latent dimensions: technological complementarity and product-market rivalry. The first dimension captures the extent to which two firms share or can leverage related technological capabilities; the second reflects the degree to which they compete in overlapping product markets. A focal firm's patent grant alters the peers' expected returns from innovation by affecting both dimensions. The net effect depends on which channel dominates. When technological complementarity prevails, peers perceive the grant as an opportunity for diffusion; when rivalry dominates, the grant constitutes a constraint that encourages redirection. The resulting strategic response—expansion or reallocation—can be inferred from market reactions that aggregate investors' assessments of these relationships.

Figure 1 illustrates the diffusion–constraint space. Technological complementarity, which reflects technological overlap and absorptive capacity, is on the vertical axis, and product-market rivalry is on the horizontal axis. What shapes a peer's response is not the level of either force but their relative importance. Above the diagonal, complementarity dominates: the focal firm's patent expands the peer's opportunity set, and the peer builds on the granted technology through R&D investment and acquisitions. Below the diagonal, rivalry dominates: the grant constrains the peer, which shifts its innovation toward other technology domains to reduce its exposure to direct competition. Two peers with the same complementarity can therefore respond in opposite ways if they differ in rivalry, as peers A and B in the figure illustrate. Appendix B formalizes this space by extending the framework of Bloom et al. (2013) to a peer that allocates R&D between the technology covered by the focal firm's patent and other fields.

2.1 Opportunity or Threat

When a patent granted to the focal firm generates positive abnormal returns for both the focal firm and the peer, these reactions indicate that markets forecast diffusion through technological complementarities. The disclosure of new knowledge reveals potential synergies in production or research processes, and the peer expands its innovative activity accordingly. Two mechanisms drive this response.

1. Learning and absorptive capacity: peers with high technological overlap are better able to interpret and integrate the knowledge revealed by a focal firm's patent. The proximity of technological expertise allows these peers to recombine existing capabilities with newly disclosed ideas, leading to incremental or follow-on innovation. The canonical formulation of absorptive capacity (Cohen and Levinthal, 1990) predicts precisely this pattern—firms invest more in innovation when they are close enough to external knowledge sources to understand and use them effectively.

2. Complementarity in strategic assets: the focal firm's patents may create new opportunities for collaboration, licensing, or acquisition. When technologies are complementary, joint or sequential development can enhance returns for peers. This mechanism aligns with theories of innovation networks and the accumulation of innovation capital (Bena and Li, 2014). Positive market reactions thus forecast that new knowledge will diffuse through cooperative or parallel efforts rather than through competitive displacement.

Empirically, opportunity responses should manifest as increases in R&D spending and acquisition activity following AR^+ patent grants, and these responses should be stronger when the peer is technologically closer to the focal firm. Acquisitions are a direct way to act on such an opportunity: buying a firm that already holds related technology lets a peer absorb new knowledge faster than developing it through internal R&D (Higgins and Rodriguez, 2006; Phillips and Zhdanov, 2013; Bena and Li, 2014). Opportunity should also affect where peers innovate. When a grant reveals that the new invention is compatible with the peer's capabilities, the peer has stronger incentives to build on the focal firm's technological trajectory, so its subsequent patents should move toward the technology covered by the focal firm's patent and cite similar prior art (Jaffe et al., 1993; Azoulay et al., 2010).

<u>Hypothesis 1 (Opportunity Response)</u>*: *Following patent grants associated with significantly positive abnormal returns for both the focal firm and the peer (AR^+), peers increase R&D and acquisitions and shift their patenting toward the technology covered by the focal firm's

patent. These responses are stronger for peers that are technologically closer to the focal firm.

When a focal firm's patent triggers a positive reaction for the innovator but a negative reaction for the peer, markets forecast competitive displacement. The patent is viewed as strengthening the focal firm's position in a shared market or technological domain, reducing the peer's expected profitability. In this context, the peer faces a choice between contesting the domain—by investing defensively in similar technologies—or redirecting its efforts toward less contested areas.

If redirection dominates, peers should shift patenting away from the technology covered by the patent granted to the focal firm and toward fields outside their three largest pre-existing CPC classes. Such a response is consistent with theories of strategic reorientation under threat (Christensen, 1997; Adner and Snow, 2010). Redirection reflects an attempt to restore differentiation by moving into less contested territories rather than directly imitating the protected innovation. This prediction concerns where peers innovate; it does not require an increase or decrease in total innovation.

<u>Hypothesis 2 (Redirection Response)</u>: Following patent grants associated with a significant positive abnormal return for the focal firm and a significant negative one for the peer (AR^-), peers shift their patenting away from the technology covered by the patent granted to the focal firm and toward new technological domains.

2.2 What Market Reactions Reveal

The market reactions to a focal firm's patent grant serve as an empirical map of the diffusion–constraint space. Positive abnormal returns for both the focal firm and the peer (AR^+) indicate that complementarity dominates rivalry, forecasting diffusion and co-investment in innovation. A negative abnormal return for the peer (AR^-), when the focal firm's reaction is positive, indicates that rivalry dominates, forecasting

displacement and redirection. The absence of a significant reaction suggests neutral or distant relationships where neither force is strong. Appendix B shows that the sign of the peer's reaction depends on whether technological proximity, weighted by the peer's exposure and absorptive capacity, exceeds product-market rivalry, and that both the reaction and the subsequent responses rise with the economic importance of the patent.

Alternative interpretations of the reaction deliver different predictions. If the peer's reaction merely reflected a change in its valuation, equally large joint price movements without patent news should produce the same responses, for example through cheaper equity financing (Baker et al., 2003). If the reaction merely reflected the average relationship between the two firms, classifying significant grants by technological and product-market proximity measured before the grant should explain the responses as well as market reactions do. If instead the reaction carries information specific to the patent, neither should be the case, and responses should rise with the size of the reaction. These predictions are tested in Sections 4.7 and 4.8 and Internet Appendix IA.4.

Hypothesis 3 (Information in Market Reactions)*: *Grant-day peer reactions predict the responses in Hypotheses 1 and 2 beyond the pre-grant technological and product-market relationship between the two firms, the responses rise with the size of the reaction, and equally large joint price movements unrelated to patent grants do not predict such responses.

2.3 Who Can Respond

Even when opportunities arise, exploiting them requires resources. Financial flexibility—captured by internal cash holdings and unused debt capacity—shapes how effectively peers can adjust their real activities following a focal firm's innovation. When a focal firm's patent is interpreted as favorable, financially flexible peers are better positioned to expand R&D, pursue follow-on innovation within the same technological domains, and acquire firms within their own industry to obtain complementary

capabilities. When a focal firm's patent is interpreted as unfavorable, financial flexibility similarly enables peers to absorb risk and redirect inventive effort toward new technological domains. Thus, financial flexibility governs both peers' ability to capitalize on diffusion opportunities and their capacity to adapt to competitive threats.

Not all peers are equally positioned to benefit from diffusion. Recent research documents a slowdown in knowledge diffusion and an increasing concentration of innovation among technological leaders (Akcigit and Ates, 2023). This asymmetry aligns with the broader theoretical insight—emphasized in Aghion et al. (2009) on frontier competition—that a firm's proximity to the technological frontier fundamentally shapes its innovation incentives. In Aghion et al. (2005), competition raises innovation among firms that are close to one another at the frontier but discourages laggards, whose expected rents from catching up are small. Aghion et al. (2009) show directly that firms close to the technological frontier respond to entry threats by innovating more, whereas firms far from the frontier innovate less. Within the diffusion–constraint framework, these patterns emerge naturally: frontier peers, endowed with recent knowledge stocks and more adaptive R&D structures, can absorb and leverage the focal firm's innovations more effectively, while laggard peers face higher costs of assimilation and weaker expected returns.

Consequently, diffusion following AR^{+} grants should be strongest among frontier peers and weakest among laggard peers. This heterogeneity also implies an asymmetry in the propagation of innovation: knowledge disclosure may reinforce existing technological hierarchies rather than leveling them.

<u>Hypothesis 4 (Capacity to Respond)</u>*: *Peers with greater financial flexibility—higher cash holdings or lower leverage—and greater innovative capability respond more strongly to both AR^{+} and AR^{-} patent grants.

Legal constraints shape how peers can respond. When a focal firm's patent strengthens exclusionary rights over a technological area, peers facing greater exposure to infringement risk or narrower freedom to

operate have fewer viable options for competing along their existing trajectory. Expanding investment in the protected domain becomes more costly, and redirecting inventive effort toward alternative technologies becomes a more attractive—and sometimes necessary—strategic response. Concentrated patent ownership in the peer's industry further raises the likelihood that follow-on work runs into a small number of large rights holders.

Hypothesis 5 (External Constraints)*: *Stronger patent enforcement and more concentrated patent ownership dampen the expansion that follows AR^+ patent grants, and stronger enforcement intensifies the redirection that follows AR^- patent grants.

Taken together, these hypotheses articulate a diffusion–constraint framework of peer responses to innovation. A focal firm's patent grant constitutes a technological shock whose implications vary across firm relationships and capacities. I use market reactions around the grant to classify the state of the world—whether complementarity or rivalry dominates—and examine how peers adjust the scale and direction of their innovation. Peers with high technological overlap and absorptive capacity expand innovation following positive shocks, while peers facing rivalry or constraints primarily redirect their efforts. Financial flexibility governs the ability to act on either margin, amplifying expansion or enabling redirection. Appendix B derives Hypotheses 1–5 from the same model.

3. Data and Empirical Design

This section describes the data sources, peer definitions, and key variables used in the empirical analysis, and outlines the empirical design. I first describe the sample construction and the definition of technological peer relationships. I then present the construction of the main variables, including abnormal returns around patent grant dates.

For the empirical design, I begin with the baseline stacked difference-in-differences (DiD) specification,

which incorporates the market-based classification of significant patent grants, followed by an instrumental variable approach to address endogeneity concerns, and conclude with a triple-difference design used to examine heterogeneity across firms and environments.

3.1 Sample Construction using the Patent Dataset

I construct the patent dataset by merging information from three complementary sources: (1) the Duke Innovation & Scientific Enterprises Research Network (DISCERN) database (Arora et al., 2021), (2) the PatentsView database, and (3) the Patent Examination Research Dataset (PatEx) (Graham et al., 2018).

The DISCERN Version 2.0 dataset includes patents granted by the U.S. Patent and Trademark Office (USPTO) between 1980 and 2021. A key advantage of this dataset is that it provides firm-level linkages to Compustat, enabling accurate assignment of patents to publicly traded firms. The database accounts for both firm name changes and patent ownership transfers, which are central challenges in identifying the correct owner of a patent over time. From the PatentsView database, I obtain detailed patent citation links, which allow me to trace the technological relationship between patents and to construct measures of technological proximity and innovative capability. The PatEx dataset provides detailed information on the patent application and examination process, drawing from the Public Patent Application Information Retrieval (Public PAIR) system. PatEx includes variables covering the characteristics of inventions, applicants, attorneys, and examiners, along with a full history of procedural events and status codes throughout the examination process. This dataset is crucial for constructing measures of patent examiner leniency, which will later serve in the construction of an instrumental variable. I merge these three datasets using patent identifiers and restrict the sample to patents granted after the year 2000. Before the American Inventors Protection Act (AIPA), applications remained confidential until grant. Applications filed on or after November 29, 2000, are published 18 months after filing, unless the applicant forgoes foreign filing and opts out. Because many grants in the early 2000s stem from applications filed before this date, analyses that require application-level data, such as the instrumental variable analysis, are restricted to applications filed on or after November 29, 2000.

I obtain firm-level financial and organizational characteristics from Compustat and merge with the patent data using firm identifiers provided in the DISCERN database. These variables include standard controls such as firm size (log of total assets), firm age, Tobin's Q, leverage, and cash holdings. Detailed variable definitions are provided in Appendix A. Incorporating Compustat data allows me to link patenting activity to firms' balance sheet characteristics and to control for heterogeneity in firms' capacity and incentives to respond to the focal firm's innovation news throughout the analysis.

The final sample consists of all U.S.-based publicly traded firms that were granted at least one patent between 2001 and 2021.[3] Consistent with prior literature, I exclude firms in highly regulated or structurally distinct industries, including utilities (Standard Industrial Classification [SIC] codes 4900–4999), financial institutions (SIC 6000–6999), and organizations classified under public services, international affairs, or non-operating establishments (SIC 9000 and above). These exclusions help ensure comparability across firms in terms of innovation incentives and competitive dynamics. All asset-based and financial variables are winsorized at the 1st and 99th percentiles to mitigate the influence of extreme outliers. After applying these filters and merging across datasets, the final panel includes 445,451 peer-focal-year observations, representing 3,569 unique firms.

3.2 Defining Peer Relationships

To identify peer relationships between firms, I rely on the Cooperative Patent Classification (CPC) system, which categorizes patents based on technological content using a hierarchical structure. The CPC system was jointly developed by the United States Patent and Trademark Office (USPTO) and the European Patent Office (EPO), introduced in 2013, and provides a fine-grained taxonomy of technological domains. I use the CPC codes that PatentsView assigns retroactively to patents granted before 2013.

The CPC hierarchy consists of multiple levels. At the top level are sections, which are broad technology domains. Each section is divided into classes, and each class is further divided into one or more subclasses,

[3] I require at least one patent grant during the sample period to identify peer firms.

representing more granular areas of technological specialization. In my sample, there are 105 unique classes and 517 unique subclasses. Figure 2 illustrates the structure of the CPC hierarchy and provides a representative snapshot of the classification breakdown.

Using this classification framework, I define a technological peer relationship between two firms based on two criteria, both evaluated over a five-year rolling window prior to the focal firm's patent grant. First, both firms must have been granted at least one patent in the same CPC subclass during the previous five years, indicating sustained activity within a similar technological space rather than a transitory or incidental overlap. Second, within this same five-year window, at least one patent from one firm must cite a patent granted to the other firm, indicating a realized and economically meaningful technological linkage through prior innovation activity. This dual condition helps ensure that identified peer firms are not only active in related domains but are also technologically connected through innovation activity.

This definition follows from the paper's focus on innovation. The question is how a patent grant affects the innovation of firms that work on related technologies, so the relevant peers are firms that are active in the same technological domains and draw on each other's knowledge. First, requiring overlap in CPC subclasses ensures that peer relationships are formed within narrowly defined technological domains. Because CPC subclasses are designed by patent examiners to capture fine-grained technological content, this criterion maps firms' innovative focus closely. Industry and product-market classifications capture other relationships between firms, such as competition for the same customers, and are suited to other questions.

Second, incorporating patent citation links strengthens the peer definition by capturing realized technological connections rather than potential similarity alone. Patent citations reflect instances in which prior inventions are directly relevant to subsequent innovation, providing evidence of knowledge flows, technological dependence, or cumulative innovation processes. Section 5.4 examines how the results change when peers are instead defined by product-market similarity or by membership in the same Fama–French 48 industry.

3.3 Variable Construction

This subsection describes the construction of the core variables used in the analysis. I begin by introducing abnormal stock returns around the focal firm's patent grant dates, which serve as a market-based measure of the economic significance and direction of innovation shocks. I then describe the instrumental variable—patent examiner leniency—used to address endogeneity in the focal firm's patent grants. Finally, I outline the peers' patent outcome variables following these shocks. In subsequent subsections, I introduce additional measures that capture heterogeneity in technological complementarity, product-market rivalry, and legal enforcement risk, as well as acquisition activity and firm characteristics, which are used to study mechanisms and cross-sectional variation in peers' responses.

3.3.1 Patent Stock Price Reactions

This paper measures the economic significance of patent grants using stock market reactions around the grant date. A growing literature shows that abnormal stock returns provide a forward-looking measure of the value of innovation by capturing how investors update expectations about future cash flows and competitive positioning. In particular, Kogan et al. (2017) demonstrate that firm-value changes around innovation events contain information about the economic importance of patents.

An important advantage of this approach is that it is observable in real time, precisely when the patent grant occurs. In contrast, commonly used measures of patent importance—such as citation-based metrics or textual novelty measures (e.g., Kelly et al., 2021)—are realized only ex post and are not available at the time of the grant.

I use cumulative abnormal stock returns (CAR) around the focal firm's patent grant as a market-based measure of significance. Following Testoni (2022), who measures rivals' reactions to technological acquisitions, CAR is measured over a window starting two trading days before the patent grant and ending two trading days after. I estimate the market model over a 250-trading-day window that ends 42 trading

days prior to the patent grant, using the CRSP value-weighted market return as the market return.[4] Abnormal returns are the residuals from this predicted return, and cumulative abnormal returns are calculated by summing the daily abnormal returns over a five-trading-day window surrounding the patent grant date. For each firm, I standardize the CAR as $z = (CAR - \mu)/\sigma$, where μ and σ are the mean and standard deviation of five-day CARs over the estimation window. A firm's reaction is significantly positive if $z > 2$ and significantly negative if $z < -2$. Section 5.8 reconstructs the classifications using the Fama–French five-factor and q-factor models; Internet Appendix IA.1 also examines a shorter event window and industry-adjusted returns.

For each grant day of the focal firm, I compute cumulative abnormal returns (CARs) for both the patenting firm and its technologically related peer firms. The focus on the patent grant date—rather than the disclosure date—is deliberate. A grant formally certifies novelty and enforceability, converting a previously uncertain application into a legally protected technological asset. As a result, grant dates are the moments at which both legal rights and competitive implications become most salient to investors, making them natural candidates for identifying economically meaningful innovation shocks.[5,6] Section 5.1 tests for peer responses around application publication, and Section 5.6 examines whether responses vary with application-to-grant delays.

For the majority of patent grants, CARs for both the patenting firm and its peers are economically small and statistically indistinguishable from zero. This pattern reflects the fact that most patents are incremental, narrowly scoped, or strategically marginal, and therefore do not materially alter market expectations about firm value or competitive dynamics.

However, a subset of patent grants generates statistically significant positive or negative abnormal

---

[4] I require a minimum of 100 non-missing return observations during the estimation window and complete return data for all five days in the event window.

[5] One potential concern is that peers may instead respond when patent applications are disclosed rather than when patents are granted. I examine this possibility in Section 5.1 and find no evidence of comparable responses around disclosure dates.

[6] Because abnormal returns may also reflect other contemporaneous corporate news, Section 5.5 re-estimates the analysis after excluding patent grants occurring near earnings announcements.

returns, both for the patenting firm and for its technologically related peers. They correspond to patents that markets interpret as meaningfully shifting technological trajectories, competitive positions, or expected future cash flows. Importantly, these reactions are heterogeneous: large positive reactions for the patenting firm are sometimes accompanied by positive reactions for peers, but in other cases coincide with sharply negative peer responses.

This heterogeneity is central to the paper's approach. Treating all patent grants as homogeneous shocks would average over fundamentally different events and obscure the mechanisms through which a focal firm's innovation affects peers' behavior. Instead, firm-level abnormal returns reveal which patent grants are salient and how their implications differ across firms.

Abnormal returns around the focal firm's patent grants can be interpreted as the market's assessment of two competing forces inherent in innovation: knowledge diffusion and competitive displacement. Positive abnormal returns for the focal firm indicate that investors expect the patent to enhance the innovator's future profitability, either through superior technology, stronger exclusion rights, or improved product positioning. The reaction of the peer's stock price then reveals how that same innovation is expected to affect related firms.

When the peer's abnormal return is also positive, markets appear to anticipate that a focal firm's innovation creates opportunities for diffusion, complementarities, or shared growth. Such anticipation is expected when firms operate in adjacent technological spaces, rely on similar scientific foundations, or benefit from common downstream demand. In contrast, when the peer's abnormal return is negative, markets interpret the focal firm's patent as a competitive threat—one that is expected to erode the peer's rents, constrain its technological options, or intensify rivalry in overlapping product markets. These interpretations are inherently forward-looking. Unlike citation counts or realized product outcomes, abnormal returns reflect expectations about future strategic interactions, legal constraints, and the reallocation of innovative effort.

To operationalize these insights, I use the joint distribution of abnormal returns for the focal firm and its peer to classify the focal firm's patent grants into economically meaningful categories. Specifically, I

define two types of patent grant events based on statistically significant abnormal returns:

1. **AR^+ patent grants**

   A focal firm's patent grant is classified as AR^+ if both the focal firm and its peer experience significantly positive reactions ($z > 2$) over the $[-2, +2]$ window. These events indicate that markets expect the innovation to create value for both firms, consistent with technological complementarity, knowledge spillovers, or mutually reinforcing innovation paths.

2. **AR^- patent grants**

   A focal firm's patent grant is classified as AR^- if the focal firm experiences a significantly positive abnormal return while its peer experiences a significantly negative abnormal return—two standard deviations below the mean—over the same event window. These events indicate that markets view the innovation as strengthening the focal firm at the expense of its peer, consistent with business-stealing effects, exclusion, or intensified product-market competition.

Events are defined at the patent-peer level; patents granted to the same focal firm on the same day share the same classification. The classification relies on statistically significant abnormal returns at the firm level, assessed relative to each firm's own pre-event return variation rather than the cross-sectional return distribution. I focus on such returns because they are more likely to be noticeable to management than ordinary stock-price fluctuations, which may just be noise.[7] The two-standard-deviation threshold is a convenient cutoff rather than a discontinuity: Internet Appendix IA.4 shows that peer responses rise continuously with the size of the peer's reaction.

Abnormal returns play a central role in the empirical design of the paper. First, they provide a market-

[7] On grant days on which the focal firm's standardized CAR exceeds 2, joint positive and negative reactions of peers are 3.6 and 4.0 times as frequent as on days with equally large focal firm returns but no grant (Section 4.7), so most AR^+ and AR^- joint reactions are unlikely to arise by chance.

based measure of economic significance, allowing the analysis to distinguish between routine patent grants and those that meaningfully affect expectations of the firm's future cash flows and their risk. Second, they offer a directional classification that separates the focal firm's innovations perceived as opportunities from those perceived as threats.

This approach avoids imposing assumptions about whether a focal firm's innovation is beneficial or harmful ex ante. Instead, it conditions on how markets interpret each event in real time. Because the classification uses returns realized at the grant, I use it to describe how responses differ across grants and rely on pre-grant relationships for causal estimates (Section 3.4.2). Subsequent analyses examine how peers adjust R&D investment, acquisition activity, financing decisions, and the direction of innovation following AR^+ versus AR^- patent grants. By doing so, the paper links market-based assessments of innovation to real economic adjustments, providing a unified framework for understanding when and how the focal firm's innovations diffuse or constrain peers' behavior.

3.3.2 Instrument: Examiner Leniency

To address endogeneity concerns, I instrument for the focal firm's patent grants using the leniency of the patent examiners assigned to review the focal firm's patent applications. The idea is that some examiners are systematically more likely than others to approve patents, even within the same art unit and year.

Following Farre-Mensa et al. (2020), I use this quasi-random variation in examiner approval tendencies to construct an instrument for whether the focal firm receives a patent grant. Specifically, I measure examiner leniency at the examiner–art unit–year level as the fraction of applications reviewed by examiner k in art unit a and year t that are granted, excluding the focal firm's own applications (a leave-one-out measure, as in Farre-Mensa et al., 2020):

$$EL_{kat}^{-j} = \frac{n_{kat}^{granted} - n_{kat,j}^{granted}}{n_{kat}^{reviewed} - n_{kat,j}^{reviewed}}$$

where EL_{kat}^{-j} denotes the leniency of examiner k in art unit a during year t. $n_{kat}^{granted}$ is the number of patent applications granted by examiner k in art unit a in year t. $n_{kat}^{reviewed}$ is the total number of applications

reviewed by the examiner in the same unit and year. For each application in the instrumental variable sample (Section 3.4.2), I use the leniency of the examiner who reviewed that application, computed excluding the focal firm's own applications.

The first stage relates whether the focal application is granted to the leniency of its examiner. Intuitively, applications assigned to more lenient examiners should be more likely to be granted, holding fixed other determinants of innovation outcomes. The identifying variation therefore comes from differences in the grant propensity of the examiners reviewing the focal firm's applications, rather than from the peers' innovation choices or financing policies.

This instrument is appealing because examiner assignment generates variation in the focal firm's patent grant outcomes that is plausibly exogenous to the peers' activities. The second stage then uses the predicted component of the grant decision from examiner leniency to estimate how peers adjust their R&D and acquisition activity in response to the focal firm's innovation shocks.

3.3.3 Peers' Patent Outcome Variables

I next construct variables that capture peers' patent outcomes. These variables are designed to reflect both the intensity and the direction of innovation output. The first variable, *PatentFiling*, measures the number of patent applications filed by a peer. The second variable, *CitShareOverlap*, captures the directional alignment of innovation by measuring the citation share overlap between the focal firm's patents and the peer's patents. For each patent granted to the peer, I calculate the proportion of backward citations that overlap with those made by the focal firm's patents. I then average this measure over the six years from t to t+5. That is,

$$CitShareOverlap_{\ ij[t,t+5]} = \frac{1}{N}\sum_{p=1}^{N}\left[\frac{\left(\left|Citations_{i,p} \cap Citations_{j}\right|\right)}{\left|Citations_{i,p}\right|}\right]$$

where $CitShareOverlap_{\ ij[t,t+5]}$ is the citation share overlap between peer *i* and focal firm *j* over [t, t+5]. *N* is the number of patents granted to peer *i* over these six years. $Citations_{i,p}$ refers to the set of backward

citations made by peer i's patent p. $Citations_j$ refers to the set of backward citations made by focal firm j's patents.

The citation share overlap variable reflects whether peers shift their R&D focus toward similar technological domains. A higher degree of citation share overlap indicates stronger technological convergence and suggests that peers perceive the focal firm's innovation as strategically relevant or complementary to their own innovation paths.

To measure whether peers move into or out of the specific technology covered by the focal firm's patent, I construct three additional outcomes from the peer's patent applications filed in each event year. *FocalGroupShare* is the share of the peer's applications classified in the CPC main group of the focal firm's significant patent. *CitesFocalField* is the share of the peer's applications that cite at least one patent of the focal firm in that CPC main group. As a placebo, *CitesFocalOther* is the share that cites at least one patent of the focal firm outside that main group. Because the outcomes are defined by the technology covered by the focal firm's patent rather than by the patent itself, they are observed both before and after the grant, and the stacked specification compares their change with that of control pairs. All three are defined only for peer-years with at least one patent application.

3.3.4 Technological Complementarity and Product-Market Rivalry Measures

To capture technological complementarity between the focal firm and its peers, I use the citation share overlap measure defined in the previous subsection, computed over the five years before the grant, [$\tau-5$, $\tau-1$], where τ denotes the year in which the focal firm receives the patent grant. A higher lagged citation share overlap indicates that two firms draw on closely related technological knowledge, enhancing the peer's ability to interpret, absorb, and potentially build upon the focal firm's newly disclosed innovation.

To measure product-market rivalry, I use the Text-based Network Industry Classification (TNIC-2) of Hoberg and Phillips (2010, 2016). TNIC-2 identifies pairs of firms with sufficiently similar product descriptions in their 10-K filings, providing a dynamic, text-based measure of product-market relatedness.

For each peer-focal pair, I construct an indicator equal to one if the two firms are classified as TNIC-2 peers in year τ−1, and zero otherwise. Using the lagged classification ensures that product-market rivalry is measured prior to the focal firm's patent grant. A value of one therefore indicates that the focal firm and peer operate in sufficiently similar product markets to be classified as direct product-market peers.

In Table 7, I also use two continuous ex ante measures: *ProductSimilarity*, the pairwise TNIC product similarity score in year τ−1 (Hoberg and Phillips, 2016), and *TextSimilarity*, the TF-IDF cosine similarity between the two firms' patents granted over [τ−3, τ−1] (constructed as in Section IA.3). This pre-grant pair-level measure differs from the outcome variable TextSimilarity in Section 4.10, which compares the focal firm's patent with the peer's subsequent filings.

3.3.5 Legal Enforcement Risk and Market Structure Variables

To capture the legal constraints surrounding follow-on innovation, I construct two measures: *Thicketing* and *LitigationRisk*. The first measure, *Thicketing*, captures the extent to which a focal firm becomes more active in filing patents following a patent grant—potentially to protect or consolidate its technological position. The measure is calculated as the change in the focal firm's share of all filings in its dominant CPC subclass, comparing an 18-month window after the grant [t,t+18] to an 18-month window before [t−18,t]:

$$Thicketing_{jt} = \frac{Firm\ j's\ Patent\ filings\ in\ CPC_{[t,t+18]}}{Total\ Patent\ filings\ in\ CPC_{[t,t+18]}} - \frac{Firm\ j's\ Patent\ filings\ in\ CPC_{[t-18,t]}}{Total\ Patent\ filings\ in\ CPC_{[t-18,t]}}$$

A higher value of this measure indicates more aggressive patent filing activity in the dominant technological space, suggesting greater IP congestion or defensive behavior by the focal firm. Because *Thicketing* is measured after the grant, I interpret it as a characteristic of the focal firm's post-grant enforcement posture rather than as a pre-determined moderator.

The second measure, *LitigationRisk*, is constructed at the industry level using data from the Patent Litigation Docket Reports. It is defined as the number of patent litigations filed among firms in the same industry in a given year, per 100 patent-holding firms in the industry:

$$LitigationRisk_{st} = 100 * \frac{Total\ patent\ litigations\ in\ industry\ s\ at\ time\ t}{Total\ number\ of\ patent\ holding\ firms\ in\ industry\ s\ at\ time\ t}$$

This measure captures the general legal environment and intensity of patent enforcement in a given industry. Higher litigation risk reflects industries where patent-related disputes are more frequent, potentially deterring innovation due to elevated enforcement uncertainty and costs.

To capture structural barriers to diffusion, I also measure the concentration of patenting within the peer's industry. To test whether concentrated patent ownership constrains peers' responses (Hypothesis 5, External Constraints), I define *PatentConc_Ind* as the Herfindahl–Hirschman index of patent-grant shares across firms in the peer's Fama–French 48 industry, computed over the three years before the grant year:

$PatentConc_Ind_{\ s,\tau} = \sum_k (n_{k,s,[\tau-3,\tau-1]} / N_{s,[\tau-3,\tau-1]})^2$,

where $n_{k,s}$ is the number of patents granted to firm k in industry s and N_s is the total number of patents granted to firms in industry s. Higher values indicate that patenting in the industry is concentrated among a few leader firms. Measuring the index before the grant year ensures that it is not affected by the focal firm's grant itself.

3.3.6 Acquisitions

Data on acquisition activity are obtained from the SDC Platinum Mergers and Acquisitions database. Following Moeller et al. (2004), I include completed acquisitions announced since 2001 in which the acquirer is a publicly traded U.S. firm that controls less than 50% of the target before the announcement and owns 100% of it after the deal, the deal value is at least $1 million and at least 1% of the acquirer's market value, and the deal is completed within 1,000 days of the announcement. These criteria exclude purchases of minority stakes and clean-up acquisitions of the remaining interest in firms that the acquirer already controls. Because the focus is on how peers expand their capabilities in response to related innovation, I further restrict attention to within-industry acquisitions, in which the target and the acquirer share the same two-digit SIC code. Within-industry targets are more likely to hold technologies and assets that are complementary to the acquirer's existing operations (Bena and Li, 2014). Applying these filters, I

identify a total of 3,524 completed acquisitions carried out by public firms. Acquisition is an indicator equal to one if the peer completes at least one such acquisition in a given year.

3.3.7 Ex Ante Technological Proximity and Product-Market Rivalry

The AR^+ and AR^- classifications rely on the peer's realized return. To examine whether the same patterns arise from relationships that are fixed before the grant, I also classify peer-focal pairs using only pre-grant information. $TechProximity_{ij,\tau}$ is the average of the standardized *CitShareOverlap* over [τ−5, τ−1] and the standardized *TextSimilarity* between the two firms' patents granted over [τ−3, τ−1], the two measures of technological complementarity in Table 7. *HighTech* equals one if *TechProximity* is in the top tercile of its distribution in cohort τ. *HighRival* equals one if the two firms are TNIC-2 peers in year τ−1 and their pairwise product similarity is above the median among TNIC-2 pairs in that year. By construction, 33.3% of pair-years are *HighTech* and 9.0% are *HighRival*. In this analysis, I keep the focal firm's own reaction to select grants with statistically significant abnormal returns for the focal firm: $FocalSigGrant_{j,\tau}$ equals one in the post-grant years of a cohort in which the focal firm receives at least one patent grant with a standardized CAR above 2. These grants cover 18,912 peer-focal-years, including the 3,129 AR^+ and 1,815 AR^- peer-focal-years; 34.2% of them are *HighTech* and 9.1% are *HighRival*.

3.4 Empirical Methods

To examine how peers respond to the focal firm's patent grants, I estimate peer-focal-year panel regressions that exploit the timing of the focal firm's patent grants associated with statistically significant abnormal returns. The main outcome variables include R&D investment, acquisition activity, innovation output, and financing decisions.

Grant timing differs across firm pairs, and a peer can be exposed to several focal firms. With staggered treatment timing and heterogeneous effects, two-way fixed effects regressions implicitly use already-treated units as controls and can be biased (de Chaisemartin and D'Haultfœuille, 2020; Goodman-Bacon, 2021;

Baker et al., 2022). I therefore use stacked cohort panels, which compare treated pairs only with clean controls within a common event window (Gormley and Matsa, 2011; Cengiz et al., 2019; Wing et al., 2024); Section 5.7 separately examines overlapping exposure and repeated peer-year observations.

3.4.1 Baseline Specification: Stacked DiD with Market-Based Classification

For each peer-focal pair *(i, j)*, let τ denote the year in which the focal firm *j* receives a patent grant. I construct cohort-specific event-time panels centered around τ, and compare treated pairs with clean control pairs within the same cohort (Cengiz et al., 2019; Baker et al., 2022). The baseline specification is:

$$Y_{ijt,\tau} = \Sigma_{k \neq -1} \beta_{k,\tau} \left[1\{t = \tau + k\} \times 1\{Grant_{j,\tau} = 1\} \right] + \Gamma' X_{i,t-1} + \delta_{i,\tau} + \mu_{st,\tau} + \varepsilon_{ijt,\tau}, \quad (1)$$

where $Y_{ijt,\tau}$ denotes the outcome of peer *i* in year *t* linked to the focal firm *j*, and *k* indexes event time relative to the focal firm's patent grant. The omitted category is $k = -1$. The coefficients for k = −3 and k = −2 test for differential pre-trends between treated and control pairs (Sun and Abraham, 2021; Roth et al., 2023). $X_{i,t-1}$ is a vector of lagged firm-level controls, $\delta_{i,\tau}$ denotes firm fixed effects, and $\mu_{st,\tau}$ denotes industry × time fixed effects. τ represents a cohort defined by the focal firm's patent grant year. Separate event-time panels are constructed for each cohort and then stacked together to estimate the overall treatment effect. Within each cohort, treated pairs are compared with clean control pairs whose focal firm receives no patent grant in year τ and that experience no AR^+ or AR^- grant within the [τ−3, τ+3] window, including both pairs that are treated only in later years and pairs that are never treated (Callaway and Sant'Anna, 2021). Because a control pair can enter several cohorts, the same peer-focal-year can appear in more than one cohort. Section 5.7 addresses overlapping network exposure by using clean event windows, one event per peer-year, and inverse duplication weights that give repeated peer-year observations a total weight of one, in the spirit of the weighted stacked regressions in Wing et al. (2024).

A central feature of the empirical design is that not all the focal firm's patent grants have the same

economic implications. To capture this heterogeneity, I classify patent grant events using abnormal stock returns around the grant date. Specifically, I distinguish between the two types of the focal firm's patent grants with statistically significant abnormal returns.

First, AR^+ events correspond to cases in which both the focal firm and its peer experience strongly positive abnormal returns around the grant date. These events are interpreted as opportunity shocks, consistent with technological complementarity, knowledge diffusion, or shared growth.

Second, AR^- events correspond to cases in which the focal firm experiences a strongly positive abnormal return while its peer experiences a strongly negative abnormal return. These events are interpreted as threat shocks, consistent with competitive displacement or exclusion.

I incorporate this classification directly into the baseline specification by interacting the treatment indicator with AR^+ and AR^- indicators:

$$Y_{ijt,\tau} = \beta_{1,\tau}[Grant_{j,\tau} \times AR^{+}{}_{ij\tau}] + \beta_{2,\tau}[Grant_{j,\tau} \times AR^{-}{}_{ij\tau}] + \beta_{3,\tau} Grant_{j,\tau} + \Gamma' X_{i,t-1} + \delta_{i,\tau} + \mu_{st,\tau} + \varepsilon_{ijt,\tau}, \quad (2)$$

The indicator $AR^{+}{}_{ij\tau}$ equals one if both the focal firm j and its peer i experience abnormal stock returns at least two standard deviations above the mean over the five-day window [−2, +2] around the grant date, and zero otherwise. Analogously, $AR^{-}{}_{ij\tau}$ equals one if the focal firm j experiences an abnormal stock return at least two standard deviations above the mean and its peer i experiences an abnormal stock return at least two standard deviations below the mean over the same five-day window, and zero otherwise. $Grant_{j,\tau}$ equals one for treated pairs in the post-grant years of cohort τ and zero otherwise. Treated pairs in cohort τ are pairs whose focal firm receives at least one patent grant in year τ; AR^+ and AR^- further classify these grants by the market reaction. The coefficient β_3 therefore captures the response to grants without significant market reactions, and β_1 and β_2 capture the additional responses to AR^+ and AR^- patent grants. Section 5.3 reports the placebo-date test and its limits. Standard errors are double clustered by peer firm and year (Petersen, 2009; Cameron et al., 2011).

3.4.2 Instrumental Variable Estimation

The timing of a focal firm's grant may coincide with unobserved shocks that also affect its peers. To address this concern, I instrument the grant decision for the selected application with the leniency of the examiner who reviewed the application. Within an art unit and year, applications are assigned to examiners quasi-randomly, and examiners differ systematically in their propensity to allow applications (Farre-Mensa et al., 2020).

Two features of the setting shape the design. First, most grants are economically unimportant, so the average grant need not affect peers. Second, the importance of a grant as measured by the grant-day return is not defined for applications that are abandoned. I therefore measure importance before the examiner's decision. Using the model in Table IA6-1, I predict for each application the probability that it would result in a significant grant, using only information available before the decision. For each focal firm-year, I take the application decided in that year with the highest predicted significance, keep it if it lies in the top quintile of predicted significance, and pair it with all of the focal firm's technological peers defined in Section 3.2. Because abandoned applications are observable only if they were published, the sample is restricted to applications filed after November 29, 2000 that did not request non-publication (Table IA6-1, Panel B).

The first stage consists of three regressions, one for each endogenous variable, each of which includes all three excluded instruments:

$$D^k_{ij\tau} = \pi^k_0 EL_{j\tau} + \pi^k_1 \left(EL_{j\tau} \times HighTech_{ij\tau}\right) + \pi^k_2 \left(EL_{j\tau} \times HighRival_{ij\tau}\right) + \theta^k_1 HighTech_{ij\tau} + \theta^k_2 HighRival_{ij\tau} + \Gamma^{k\prime} X_{i,\tau-1} + \delta^k_i + \mu^k_{s\tau} + \lambda^k_{u\tau} + u^k_{ij\tau} \quad (3)$$

where D^1 = *FocalAppGrant,* D^2 = *FocalAppGrant* × *HighTech*, and D^3 = *FocalAppGrant* × *HighRival*. *FocalAppGrant* equals one if the application of focal firm j decided in year τ is granted and zero if it is abandoned. *EL* is the leave-one-out grant rate of the application's examiner within the same art unit and year (Section 3.3.2). *HighTech* and *HighRival* are measured before the decision (Section 3.3.7). δ denotes peer firm fixed effects, μ industry-by-year fixed effects, and λ art unit-by-decision-year fixed effects, so

that leniency is compared only among applications assigned within the same art unit and year. The second stage is:

$$Y_{ij,[\tau+1,\tau+3]} = \beta_1 \widehat{D}^1_{ij\tau} + \beta_2 \widehat{D}^2_{ij\tau} + \beta_3 \widehat{D}^3_{ij\tau} + \theta_1 HighTech_{ij\tau} + \theta_2 HighRival_{ij\tau} \quad (4)$$
$$+\Gamma' X_{i,\tau-1} + \delta_i + \mu_{s\tau} + \lambda_{u\tau} + \varepsilon_{ij\tau}$$

where Y is the peer's R&D, acquisition, or exploration ratio averaged over the three years after the decision. β_1 is the effect of granting, rather than abandoning, an important application on peers that are neither technologically close nor close product-market rivals; $\beta_1 + \beta_2$ is the effect on technologically close non-rivals, and $\beta_1 + \beta_3$ is the effect on close rivals that are not technologically close. For peers in both groups, the specification implies $\beta_1 + \beta_2 + \beta_3$.

Identification requires examiner leniency to affect peers only through the focal application's grant outcome, conditional on the controls, art unit, and decision year. Using leniency interactions additionally requires this exclusion restriction to hold across the pre-grant *HighTech* and *HighRival* groups. Measuring these groups before the decision avoids conditioning on grant-day returns, but does not by itself establish instrument validity. I therefore use the IV design for causal estimates conditional on these assumptions and interpret the AR^+ and AR^- estimates as conditional responses. Standard errors are double clustered by focal firm and year. I report Sanderson–Windmeijer and Kleibergen–Paap statistics (Sanderson and Windmeijer, 2016; Kleibergen and Paap, 2006) and Anderson–Rubin confidence intervals (Anderson and Rubin, 1949). Grants of applications in the bottom tercile of predicted significance provide a falsification test.

3.4.3 Triple Difference for Heterogeneity Analyses

Finally, I explore heterogeneity in peers' responses to the focal firm's patent grants using a triple-difference (DDD) framework (Olden and Møen, 2022). This approach allows me to test how the impact of the focal firm's innovation depends on firm-specific capabilities as well as external constraints. The general

form of the regression is:

$$Y_{ijt,\tau} = \alpha + \beta_1 \mathbf{1}_{j\tau} + \beta_2 \mathbf{1}_{j\tau} \times AR^{+}_{ij\tau} + \beta_3 \mathbf{1}_{j\tau} \times AR^{-}_{ij\tau} + \beta_4 \mathbf{1}_{j\tau} \times Z_{ij\tau} + \beta^{+} \mathbf{1}_{j\tau} \times AR^{+}_{ij\tau} \times Z_{ij\tau} + \beta^{-} \mathbf{1}_{j\tau} \times AR^{-}_{ij\tau} \times Z_{ij\tau} + \Gamma' X_{i,t-1} + \delta_{i,\tau} + \mu_{st,\tau} + \varepsilon_{ijt,\tau}, \quad (5)$$

where $Z_{ij\tau}$ (or in some cases just $Z_{i\tau}$) is a variable that moderates the effect of the focal firm's patent grant. To make magnitudes comparable across moderators, I standardize each continuous moderator to have zero mean and unit standard deviation; *PMR* is an indicator and is mean-centered. All moderators except *Thicketing* are measured before the grant. The coefficients β_2 and β_3 measure responses to AR^{+} and AR^{-} patent grants at the sample mean of *Z*, and β^{+} (β^{-}) measures how these responses change with a one-standard-deviation increase in *Z*. I examine four different types of variables:

1. Firm relationship – including technological complementarity (measured using prior citation-based relatedness with the peer) and product-market rivalry (proxied by an indicator for sharing a TNIC-2 industry from Hoberg and Phillips (2016)), which tests the second part of Hypothesis 1.
2. Internal capabilities – including financial flexibility (proxied by cash holdings and leverage) and innovative capabilities (measured as the log of one plus the number of forward citations the firm's patents receive from its technological peers over the past five years), which tests Hypothesis 4.
3. Legal enforcement risk – proxied by (i) industry-level litigation intensity and (ii) the focal firm's patent thicketing behavior, as described in Section 3.3.5, which test the external-constraints prediction (Hypothesis 5).
4. Market structure – proxied by the concentration of patenting among leader firms in the peer's industry (*PatentConc_Ind*), which also tests the external-constraints prediction (Hypothesis 5).

4. Empirical Results

After describing the sample, I present the baseline estimates and the redirection response (Sections 4.3–

4.4), followed by the IV and dynamic estimates (Sections 4.5–4.6). Sections 4.7–4.8 examine the information in market reactions. Sections 4.9–4.10 examine heterogeneity and innovation output. Supporting analyses are reported in the Internet Appendix.

4.1 Data Description

To better understand the landscape of peer relationships and innovation dynamics in my sample, I begin by characterizing the structure of technological peer networks. On average, each firm has approximately 7.6 peer firms per year, but this masks significant variation: about 7% of firm-years involve only one peer, while about 37% involve more than seven (Figure 3). The resulting variation in network exposure motivates the overlap and weighting checks in Section 5.7.

Figure 4 plots cumulative abnormal returns around two patent grants. Panel A shows Geron's U.S. Patent 6,458,589, granted on October 1, 2002, covering hepatocyte lineage cells derived from pluripotent stem cells. Over [−2,+2], Geron's CAR was +3.75%, while StemCells, which also pursued liver cell therapies, had a CAR of −19.19%. Panel B shows Netflix's U.S. Patent 6,584,450, granted on June 24, 2003, covering its subscription rental service. Netflix's CAR was +11.91%, and its competitor Blockbuster's was +9.28%. The paths accumulate daily market-model abnormal returns from day −2, with zero at the preceding close. These approximate estimates use the S&P 500 as the market proxy and illustrate contrasting return patterns without identifying the causal effect of the grants.

The examples in Figure 4 illustrate the signs of peer CARs. The empirical AR^{+} and AR^{-} classifications require a standardized five-day CAR above two for the patenting firm and above two or below minus two for its technological peer, using the CRSP value-weighted market benchmark and the peer criteria in Section 3. Out of the 924,125 patents granted to public firms in my dataset, 6,341 patents meet this dual-market reaction criterion. Because a focal firm often receives several such patents in a year, these patents correspond to 3,129 AR^{+} and 1,815 AR^{-} peer-focal-year observations in Table 1. By zeroing in on these high-salience events, I aim to isolate innovation news that is immediately perceived by the market as

strategically consequential. These events provide a setting in which to test for firm-level strategic responses in innovation behavior.

4.2 Summary Statistics

Table 1 provides descriptive statistics for the main variables used throughout the empirical analysis. The sample includes 3,569 unique firms between 2001 and 2021.

Starting with the dependent variables, the average R&D investment is 8.4% of total assets, with a median of 0%, reflecting the highly skewed distribution of R&D activity. The median of zero reflects how R&D is reported rather than an absence of research. Following common practice, I set R&D to zero when Compustat reports it as missing; Koh and Reeb (2015) show that many firms with missing R&D nonetheless patent. The acquisition indicator has a mean of 0.035, so a peer completes at least one acquisition in about 3.5% of firm-years. The variable *PatentFiling* has a mean of 2.31 patents filed annually, but a highly right-skewed distribution with a 90th percentile of 12 filings. *CitShareOverlap*, which measures the average share of overlapping citations between the focal firm's and its peer's patents over the six years from t to t+5, has a relatively low mean of 0.038 and a median of 0.021, but a long right tail—highlighting the heterogeneity in firms' technological convergence with their peers. The *ExplorationRatio*—the share of a peer's patent applications in a given year that fall outside its three largest CPC classes in the five years before the grant—has a median of 0.008, with a heavy right tail. This pattern suggests that most firms continue to innovate within their established technological domains, while a small subset engages in meaningful exploration. The *NetEquityIssuance* variable, defined as stock sales minus repurchases scaled by lagged total assets, is heavily centered around zero with a long right tail, indicating that large external-equity financing events are infrequent but occasionally substantial.

TextSimilarity, the average TF-IDF cosine similarity between the focal firm's patent and the patents the peer files in a given event year, has a mean of 0.120, a median of 0.108, and a 90th percentile of 0.201. These values are comparable to the pairwise patent similarities reported by Kelly et al. (2021) and indicate that a typical peer patent shares only a modest fraction of its weighted vocabulary with the focal firm's

patent. The technology-specific outcomes are sparse: on average, 4.6% of a peer's new applications fall in the CPC main group of the focal firm's patent and 2.1% cite the focal firm's patents in that group, and both have a median of zero.

Two variables capture the relationship between the focal firm and its peers. The first, Technological Complementarity, is measured using the lagged *CitShareOverlap*, which reflects the degree of technological similarity and potential knowledge spillovers between the two firms based on shared patent citations. The second, Product-Market Rivalry (*PMR*), is a dummy variable equal to one if the focal firm and its peer belong to the same industry according to the TNIC-2 industry classification from Hoberg and Phillips (2016). The mean of *PMR* is 0.18, indicating that 18% of peer-focal pairs in my sample operate in the same product market, as defined by the TNIC-2 industry classification.

The table also reports the two event-classification indicators. The AR^{+} variable (equal to one when both the peer and the focal firm experience significant positive abnormal stock returns) has a mean of 0.0070, reflecting the rarity of significant market responses. The AR^{-} variable (equal to one when the focal firm experiences a significant positive abnormal return while the peer's abnormal return is significantly negative) has a mean of 0.0041, indicating that opposite-sign reactions are less frequent but economically meaningful when they occur. The limited variation in both AR^{+} and AR^{-} highlights that statistically significant market responses to the focal firm's patent grants are infrequent events, consistent with the notion that only a subset of innovations move stock prices significantly. The ex ante proximity indicators classify 33.3% of pair-years as technologically close (*HighTech*) and 9.0% as close product-market rivals (*HighRival*).

Panel B of Table 1 compares the pairs behind significant grants, those with a standardized CAR above 2 for the focal firm, by the sign of the peer's reaction. AR^{+} and AR^{-} pairs differ most in product-market rivalry. AR^{-} pairs share a TNIC-2 market in 31.2% of cases, compared with 17.6% of AR^{+} pairs and 16.9% of pairs whose peer does not react, and their product similarity is about twice as high. Technological overlap shows the opposite ordering: citation overlap is 0.061 for AR^{+} pairs, 0.047 for AR^{-} pairs, and 0.041 for pairs without a reaction. AR^{-} grants also come from larger focal firms and carry higher patent value (Kogan et al., 2017). A positive peer reaction thus tends to mark a technologically related firm that does not compete

directly with the patentee, and a negative reaction a related firm that does. Compared with peers that do not react, AR^+ peers are smaller, younger, more R&D-intensive, and hold more cash, consistent with patent news mattering more for firms whose value depends on technology. Section 4.8 examines these relationships in a regression framework.

Three variables proxy for external constraints. The *Thicketing* measure has a mean near zero but a notable spread, suggesting time variation in the focal firm's patenting intensity. *LitigationRisk*, constructed at the industry level, averages 1.41 litigations per 100 patent-holding firms per year and is right-skewed (median 0.87, 90th percentile 3.39), indicating that some industries are particularly litigious. *PatentConc_Ind* averages 0.178 with a 90th percentile of 0.367, so patenting is highly concentrated among a few leaders in a meaningful subset of industries.

For internal firm capabilities, *Cash* as a share of total assets averages 12.7%, with a wide distribution that reflects variation in financial flexibility. The *InnovCap* variable (the log of one plus forward citations received from technological peers) averages 0.615 with a median of 0.405, and its 90th percentile exceeds 2.4, highlighting stark differences in firms' technological prominence within their networks.

Lastly, the control variables show familiar patterns. The average firm in the sample has a Tobin's Q of 3.51 and a mean firm age of about 16 years. Profitability and leverage are within expected ranges. Profitability is left-skewed, with a median of 0.089 but a 10th percentile of −0.181, indicating that a substantial minority of firms report negative operating income before depreciation in a given year.

4.3 Baseline Estimates: Focal Firm's Patent Grants and Peer Innovative Efforts

Table 2 presents the estimation results for the baseline specification (Equation (2)), which examines how the focal firm's patent grant affects a peer's innovative efforts, measured by R&D investment and the probability of engaging in acquisitions.

To capture heterogeneity in responses, I interact the focal firm's patent grant indicator with indicators for AR^+ and AR^- grants, where AR^+ corresponds to cases in which both the focal firm and the peer

experience significantly positive abnormal returns, and AR^- corresponds to cases in which the focal firm experiences a significantly positive and the peer a significantly negative abnormal return.

The results reveal a strong asymmetric pattern. The coefficient on *FocalPatentGrant* × AR^+ is positive and statistically significant across all specifications, although only at the 10% level for acquisitions in Column (2). In Columns (1) and (2), AR^+ grants are associated with increases in R&D (0.007, t = 2.32) and acquisition activity (0.030, t = 1.71). These effects remain robust when firm-level controls are included in Columns (3) and (4), with coefficients of 0.006 (t = 2.21) for R&D and 0.029 (t = 2.17) for acquisitions. These estimates associate favorable peer reactions with subsequent R&D and acquisition expansion. Internet Appendix IA.4 reports the response-size analysis and scales the R&D response to about $21.6 million per affected peer over four years. Section 5 presents the robustness checks.

In contrast, the coefficients on *FocalPatentGrant* × AR^- are small and statistically insignificant across all specifications. This provides no evidence that peers increase innovative investment along either internal or external margins when a focal firm's innovation is perceived as unfavorable. The absence of a strong response in these cases points to alternative adjustment channels, such as changes in the direction of innovation, which I explore in later sections.

The main effect of *FocalPatentGrant*, which captures responses to grants without significant market reactions, is positive but statistically insignificant across all columns. Turning to the control variables in Columns (3) and (4), the results are broadly consistent with prior literature. Cash holdings are positively associated with both R&D and acquisitions, but the association is significant only for R&D (at the 10% level), suggesting that financially flexible firms are more likely to make innovation investments (Brown and Petersen, 2011). Tobin's Q is positively related to R&D (at the 10% level) but is not significantly related to acquisitions. Other controls, including profitability, leverage, and firm age, do not exhibit consistent or statistically significant effects in this baseline specification.

4.4 Peer Adjustments Following Focal Firm's Adverse Patent News

Because peers experiencing AR^- patent grants show no statistically significant change in the level of innovative effort, it is natural to ask whether they instead respond along a different margin—namely, by changing the direction of their innovation. A decline in market valuation following the focal firm's patent grant may imply that the peers' existing technological path has become less attractive, raising the possibility that peers respond by exploring new areas rather than intensifying activity in their current domains.

To examine this, I compute the exploration ratio—the share of the peer's patent applications filed in each event year that fall outside its three largest CPC classes over $[\tau-5, \tau-1]$. Because these classes are defined in the same way for treated and control pairs, the mechanically lower ratio in pre-grant years, which fall inside this window, is differenced out in the stacked comparison. This measure captures the extent to which a peer shifts its post-grant patenting toward technological domains that were not central to its portfolio during the pre-grant five-year period. An increase in the exploration ratio would indicate movement into unfamiliar or previously underutilized technology classes.

Table 3 reports the regression estimates for both AR^+ and AR^- patent grants. The results show a clear asymmetry. For AR^+ patent grants, the interaction term $FocalPatentGrant \times AR^+$ is small and statistically indistinguishable from zero across specifications. This indicates that the focal firm's positively interpreted innovations do not prompt peers to branch out into new technological areas. This is consistent with the evidence in Section 4.10 that peers move toward the technology covered by the focal firm's patent after AR^+ patent grants.

In sharp contrast, the coefficient on $FocalPatentGrant \times AR^-$ is large (0.012–0.016) and statistically significant. Given that the mean exploration ratio is approximately 0.012, these estimates represent an increase of roughly 100 to 130 percent relative to the baseline level. This substantial rise indicates that peers experiencing AR^- patent grants meaningfully redirect their limited post-grant patenting activity toward new technological fields. Although peers exposed to AR^- patent grants show no significant change in the overall volume of innovation, they exhibit a pronounced shift in technological focus in response to patent grant news that has negative implications.

Taken together, these findings reveal a sharp directional asymmetry in peers' responses to the focal firm's patent news. Peers experiencing AR^+ patent grants do not expand into new technological domains and instead continue along their established innovation paths, consistent with the evidence that the focal firm's positively interpreted innovations reinforce existing trajectories rather than prompting exploratory shifts. In contrast, peers experiencing AR^- patent grants, despite showing no increase in the level of innovative effort, substantially reallocate their limited post-grant patenting toward new technology classes, indicating a meaningful redirection of innovative activity. With these directional responses established, Section 4.9 examines how external environments, internal capabilities, and peer-focal relationships further condition the responsiveness to the focal firm's innovation shocks.

4.5 Instrumental Variable Estimation

Table 4 reports the first stage of the instrumental variable strategy described in Section 3.4.2. The sample consists of 8,126 focal applications with high predicted significance, each paired with all of the focal firm's technological peers, for 61,840 peer-focal observations. Because each focal application is linked to several peers, the same instrument appears in multiple observations; I account for this by double clustering standard errors by focal firm and year.

Panel A of Table 4 shows that examiner leniency strongly predicts whether an important application is granted. A ten-percentage-point higher examiner grant rate raises the probability of grant by 9.3 percentage points ($t = 9.42$). The interactions predict the corresponding endogenous interactions, with coefficients of 0.95 ($t = 8.07$) for technologically close pairs and 0.88 ($t = 5.31$) for close rivals. Sanderson–Windmeijer F-statistics range from 28.2 to 88.7, and the Kleibergen–Paap F-statistic is 26.4. Because conventional critical values (Stock and Yogo, 2005) are not designed for clustered errors with multiple endogenous regressors, I base inference on Anderson–Rubin confidence intervals, which are robust to weak instruments. Panel B shows that leniency is unrelated to the application's predicted significance, its initial claims, the

novelty of its text, and peers' R&D, acquisitions, and exploration before the decision, which supports the assumption that examiners are assigned quasi-randomly within an art unit and year.

Table 5 reports the second stage. Granting an important application, rather than abandoning it, has no statistically significant estimated effect on peers that are neither technologically close nor close rivals. It raises R&D by 0.5% of assets ($t = 2.79$) and the acquisition probability by 1.3 percentage points ($t = 1.68$) among technologically close non-rivals, and raises the exploration ratio of close rivals by 0.008 ($t = 2.65$). Anderson–Rubin confidence intervals, which remain valid if the instruments are weak, exclude zero for the R&D and exploration-ratio coefficients but not for the acquisition coefficient ($p = 0.15$ for the Anderson–Rubin test).

Panel B of Table 5 applies the same procedure to applications in the bottom tercile of predicted significance. Despite an equally strong first stage, grants of these applications have no effect on technologically close peers or close rivals. This placebo is consistent with the view that the average grant has little effect on peers because most grants are economically unimportant. It is also consistent with the exclusion restriction: a direct effect of examiner leniency on peers that did not depend on the application's importance would also appear in this sample.

The causal estimates are smaller than the AR^{+} and AR^{-} estimates, particularly for acquisitions, where the IV estimate is less than half of the AR^{+} estimate. This is expected: the instrumental variable estimates average over all important applications for a given pre-grant relationship, whereas grant-day reactions also distinguish grants that affect the same pair of firms differently. Because the instrument compares grants with rejections, it captures the effect of the right to exclude and of the certified disclosure even when the grant itself was anticipated by investors.

4.6 Dynamic Peer Response to Focal Firm's Significant Patents

To explore the timing and persistence of peer responses more explicitly, I examine the dynamic effects of the focal firm's patent grants on peers' internal and external innovation activities. While earlier analyses establish that peers react to the focal firm's innovation, they do not clarify when these responses occur or

how long they persist. Because R&D and acquisitions may operate on different timelines, the dynamic analysis distinguishes immediate adjustments from delayed ones.

To estimate this, I use the baseline stacked DiD framework and run the following regression:

$$Y_{ijt+k,\tau} = \sum_{k=-3,k\neq-1}^{k=3} \beta_{1,t+k,\tau} \left[Post_{t+k} \times Grant_{j,\tau} \times AR^{+}_{ij\tau}\right] + \sum_{k=-3,k\neq-1}^{k=3} \beta_{2,t+k,\tau} \left[Post_{t+k} \times Grant_{j,\tau} \times AR^{-}_{ij\tau}\right] + \Gamma' X_{i,t-1} + \delta_{i,\tau} + \mu_{st,\tau} + \varepsilon_{ijt,\tau},$$

where $Post_{t+k}$ are dummy variables for k periods before/after the event. The coefficients β_{t+k} trace out how the outcome evolves in the years before and after the focal firm's patent is granted, relative to a baseline year (year −1).

The left plot in Figure 5 shows the dynamic response of R&D investment. The pre-grant coefficients are small and statistically insignificant, supporting the parallel-trends assumption. Following AR^{+} patent grants, R&D rises in year 0 and becomes statistically significant in years +1 and +2, peaking at about 0.8 percentage points of assets in year +1, before tapering off in year +3. The average post-grant coefficient (0.006) matches the static estimate in Table 2. This pattern suggests that peers increase internal R&D within a year or two of the grant, consistent with a response to the opportunities implied by the focal firm's innovation. By contrast, the coefficients for AR^{-} patent grants remain close to zero throughout, and their confidence intervals include zero in every year. The increase in R&D is therefore concentrated among patents that the market interprets as creating opportunities for peer firms rather than those perceived as competitive threats.

The right plot in Figure 5 presents the dynamic response of acquisition activity. Unlike the relatively fast R&D response, acquisitions respond more slowly. The increase is modest in years 0 and +1, but by year +2 the acquisition likelihood rises by about 4.5 percentage points and remains at about 4 percentage points in year +3. The average post-grant coefficient (0.029) matches the static estimate in Table 2, although the 95% confidence interval includes zero in each individual year. This delayed response aligns with the idea that acquisitions require longer lead times due to the complexity of deal-making and due diligence. It

suggests that after absorbing and evaluating the strategic implications of the patent, peers may acquire firms within their own industry to obtain complementary technologies or shore up their competitive position. These distinct temporal patterns underscore that the focal firm's patent grants shape peer behavior through multiple innovation channels on different timelines. As with R&D, the estimates for AR^- grants remain close to zero throughout the post-grant period, well below those for AR^+ grants. Thus, the delayed increase in external innovation activity appears specific to positively interpreted patent grants to the focal firm.

4.7 Patent News or Price News?

The previous sections establish that peers respond to AR^+ and AR^- patent grants and that the IV estimates identify responses among peers classified using pre-grant relationships (Section 4.5). A natural concern is that the AR^+ classification conditions on a positive return for the peer. Any large positive price shock may raise investment, acquisitions, and equity issuance through a valuation or financing channel (Baker et al., 2003), regardless of whether the shock originates from a patent. If so, the AR^+ results would reflect the peer's price increase rather than the information contained in the focal firm's patent grant.

To separate responses to patent-specific information from responses to stock-price changes alone, I construct price-shock (PS) pseudo-events: days on which the focal firm and its peer experience the same joint return pattern as in AR^+ and AR^- grants, but the focal firm receives no patent grant. Specifically, I take all trading days on which the focal firm's standardized five-day CAR exceeds 2 and the focal firm receives no patent grant within five trading days. PS^+, the price-shock counterpart of AR^+, equals one if the peer's standardized CAR over the same window exceeds 2; PS^-, the counterpart of AR^-, equals one if it falls below -2. $PriceShock_{j,\tau}$, the counterpart of $Grant_{j,\tau}$, equals one in the post-event years of a cohort in which the focal firm has at least one such day. I aggregate these events to the peer-focal-year level with the same rule used for AR^+ and AR^-, stack them with the same clean-control rule, and estimate both sets of events in one regression:

$$Y_{ijt,\tau} = \beta_1 G_{j\tau} AR^{+}_{ij\tau} + \beta_2 G_{j\tau} AR^{-}_{ij\tau} + \beta_3 G_{j\tau} + \gamma_1 S_{j\tau} PS^{+}_{ij\tau} + \gamma_2 S_{j\tau} PS^{-}_{ij\tau} + \gamma_3 S_{j\tau} + \Gamma' X_{i,t-1} + \delta_{i,\tau} + \mu_{st,\tau} + \varepsilon_{ijt,\tau} \quad (6)$$

where $G_{j\tau}$ is $Grant_{j,\tau}$ and $S_{j\tau}$ is $PriceShock_{j,\tau}$. If the baseline results reflect price shocks alone, γ_1 should equal β_1.

Panel A of Table 6 shows that the opportunity response is specific to patent-linked price changes. *FocalPatentGrant* × AR^{+} remains 0.006 for R&D and 0.029 for acquisitions, both significant at the 5% level, whereas the corresponding pseudo-event coefficients are close to zero and insignificant (0.000 and 0.003), even though pseudo-events are almost twice as frequent (5,846 versus 3,129 peer-focal-years). Equality of the two coefficients is rejected at the 10% level for R&D (p = 0.055) but not for acquisitions (p = 0.11). Likewise, the shift toward new technological fields follows AR^{-} patent grants (0.012, significant at the 5% level) but not negative joint price movements without a grant (0.001; p = 0.087 for equality). Net equity issuance behaves differently: it rises by a similar amount after AR^{+} patent grants (0.004) and after PS^{+} pseudo-events (0.003), both significant at the 5% level (p = 0.65 for equality). The equity response is therefore better interpreted as a response to higher valuations, consistent with market timing, than as patent-specific financing of follow-on innovation. Internet Appendix IA.2 examines the dynamic responses of cash holdings, leverage, equity issuance, and capital expenditures (Figure IA2-1).

Panel B asks how many AR^{+} and AR^{-} joint reactions would arise by chance. On grant days on which the focal firm's standardized CAR exceeds 2, 16.4% of peers also have a standardized CAR above 2 and 9.5% have one below −2. On days with an equally large focal return but no grant, the corresponding shares are 4.6% and 2.4%. Patent grants therefore raise the frequency of joint positive and negative peer reactions by factors of 3.6 and 4.0. If joint reactions occurred on grant days only as often as on no-grant days, they would account for roughly a quarter of AR^{+} and AR^{-} joint reactions, so most of these joint reactions are unlikely to arise by chance.

4.8 What Do Peer Market Reactions Capture?

The results so far show that peers respond very differently depending on whether a focal firm's patent grant is accompanied by a positive (AR^+) or negative (AR^-) peer market reaction. Throughout the paper, I interpret these market reactions as forward-looking indicators of technological opportunity and competitive threat. This interpretation is central to the empirical framework because it motivates the distinction between complementary and rivalrous innovation shocks.

An important question, however, is whether these market-based classifications indeed capture underlying technological and competitive relationships rather than reflecting unrelated information incorporated into stock prices. To address this question, I examine whether AR^+ and AR^- systematically align with observable firm characteristics measured prior to the patent grant. The model in Appendix B shows why the peer's reaction summarizes both dimensions: its sign depends on whether technological proximity, weighted by the peer's exposure and absorptive capacity, exceeds product-market rivalry.

Specifically, I relate AR^+ and AR^- classifications to three ex ante measures of firm relationships. The first two—citation overlap and patent-text similarity—capture technological complementarity between the focal firm and the peer. The third measure, TNIC product similarity, captures product-market rivalry. If AR^+ represents technological opportunity, it should be concentrated among firms with stronger technological complementarity. Conversely, if AR^- reflects competitive threat, it should primarily arise among firms that compete more directly in product markets.

Panel B of Table 1 shows these patterns in raw comparisons; Table 7 examines them in a regression framework. Two patterns emerge. First, AR^+ grants are significantly associated with both citation overlap and patent-text similarity, indicating that positive peer market reactions occur primarily among technologically complementary firms. Second, TNIC product similarity has little association with AR^+ but is strongly related to AR^- grants, suggesting that negative peer market reactions arise when firms compete more directly in product markets. Because these measures are all constructed using information available before the patent grant, the observed relationships are not driven by ex post outcomes.

An additional test asks whether these pre-grant relationships alone produce the same real responses,

without using the peer's return at all. I therefore focus on grants with a significant focal-firm reaction, those with a standardized CAR above 2 for the focal firm, and replace the AR indicators, which depend on the peer's return, with the ex ante indicators defined in Section 3.3.7:

$$Y_{ijt,\tau} = \beta_1 SG_{j\tau} HighTech_{ij\tau} + \beta_2 SG_{j\tau} HighRival_{ij\tau} + \beta_3 SG_{j\tau} + \Gamma' X_{i,t-1} + \delta_{i,\tau} + \mu_{st,\tau} + \varepsilon_{ijt,\tau} \quad (7)$$

where $SG_{j\tau}$ is $FocalSigGrant_{j,\tau}$. Conditioning on the focal firm's reaction selects the same economically important grants that underlie AR^+ and AR^-, but the classification no longer uses the peer's return.

Columns (1)–(3) of Table 8 show the same pattern, although less precisely. Significant patent grants to the focal firm raise R&D by 0.003 among technologically close peers and the exploration ratio by 0.005 among close product-market rivals, both significant only at the 10% level; the acquisition estimate for technologically close peers (1.2 percentage points) is positive but insignificant. These effects are about half of the baseline AR^+ and AR^- estimates (0.006, 0.029, and 0.012), consistent with ex ante proximity being a noisy, average measure of how a given grant affects the pair. Neither ex ante dimension affects the other margin.

Columns (4)–(6) add the AR^+ and AR^- interactions. Their coefficients remain close to the baseline and significant (0.005 and 0.026 for AR^+; 0.011 for AR^- in the exploration ratio), while the ex ante coefficients shrink further and become insignificant. Market reactions therefore contain substantial event-specific information beyond the average relationship between the two firms. For this reason, the analysis uses market reactions as the main classification and ex ante proximity as validation.

Overall, the evidence supports interpreting positive peer market reactions as indicating technological opportunity and negative peer market reactions as indicating competitive threat. Weaker versions of the same real responses arise when grants with a significant focal-firm reaction are classified only by the pre-grant relationship between the two firms, and market reactions sharpen this classification at the level of individual grants. This interpretation provides the economic foundation for the subsequent analyses.

4.9 Heterogeneous Responses: Triple Difference Analyses

To understand why peers react so differently to the focal firm's patent grants depending on whether the market views the focal firm's innovation favorably (AR^+) or unfavorably (AR^-), I estimate a triple-difference specification that interacts the focal firm's patent grant indicator with both the valuation shock and a set of moderating variables capturing peer-focal relationships, external constraints, and internal firm capabilities. This framework allows for a structured comparison of heterogeneous responses across economically relevant firm characteristics.

The first source of heterogeneity arises from the relationship between the focal firm and its peer, measured along two economically distinct dimensions: technological complementarity and product-market rivalry. Technological complementarity is captured by *CitShareOverlap* measured over the five years before the grant, $[\tau-5, \tau-1]$, which measures the extent to which the focal firm's patents cite similar prior art as the peer's patents. Product-market rivalry is proxied by *PMR*, an indicator equal to one when the two firms operate in the same TNIC-2 product market.

Table 9 shows that technological complementarity, rather than product-market rivalry, is the primary dimension of peer-focal relatedness that amplifies responses to AR^+ patent grants. At the sample mean of the moderators, the coefficients on $FocalPatentGrant \times AR^+$ are 0.006 for R&D and 0.028 for acquisitions, nearly identical to the baseline estimates in Table 2. The triple interaction $FocalPatentGrant \times CitShareOverlap \times AR^+$ is positive and significant at the 5% level for R&D (0.004) and positive but insignificant for acquisitions (0.016). A one-standard-deviation increase in *CitShareOverlap* thus raises the R&D response to AR^+ patent grants by about two-thirds of its average size and, as a point estimate, the acquisition response by 1.6 percentage points, about 46% of the sample mean acquisition rate. By contrast, the corresponding coefficient in the *ExplorationRatio* specification is small and insignificant (0.001), suggesting that opportunity-type responses show up in the intensity of innovation investment and acquisitions rather than in shifts toward exploratory innovation.

In contrast, none of the triple interactions involving AR^- is statistically significant, and the product-market rivalry interactions are small and insignificant for both AR^+ and AR^- patent grants (all $|t| < 1$). The redirection response itself remains: at the mean of the moderators, *FocalPatentGrant* × AR^- in the *ExplorationRatio* specification is 0.012 and significant at the 5% level, the same as in Table 3. Peers exposed to AR^- patent grants thus shift their innovation toward new technological fields regardless of how technologically or competitively close they are to the focal firm, which reinforces the earlier conclusion that AR^- patent grants change the direction rather than the level of innovation. Together with Table 7, these results suggest that product-market rivalry determines which grants are perceived as threats but does not further moderate the responses once the market reaction is taken into account.

The second dimension of heterogeneity concerns external constraints. Panel A of Table 10 shows that stronger enforcement environments and more concentrated patent ownership dampen the responses to AR^+ patent grants. A one-standard-deviation increase in *Thicketing* is associated with a smaller R&D response by 0.003, half of the average effect and significant at the 5% level, and a smaller acquisition response by 1.2 percentage points, which is not statistically significant. LitigationRisk has a similar effect on R&D (0.003, significant only at the 10% level), and its acquisition estimate (1.1 percentage points) is insignificant. Industry patent concentration works in the same direction: a one-standard-deviation increase in *PatentConc_Ind* reduces the R&D response by 0.003, significant at the 5% level, and the acquisition response by an insignificant 1.2 percentage points, consistent with the external-constraints prediction (Hypothesis 5). None of these constraints significantly affects R&D or acquisition responses to AR^- patent grants.

This pattern is consistent with the interpretation that AR^+ patent grants represent technological opportunities that peers seek to build upon. When a focal firm's innovation is viewed as complementary or value-enhancing, peers often respond by increasing R&D investment and acquiring firms within their industry that operate in technologically related areas. However, these activities bring firms closer to the focal firm's intellectual-property boundary. In environments with aggressive post-grant filing by the focal firm or elevated litigation risk, the expected costs of operating in these technologically adjacent spaces

increase substantially. As a result, peers become less willing to intensify innovation investment or acquisition activity, leading to the observed dampening of AR^+ responses.

External constraints also shape where peers innovate after AR^- patent grants. Although neither patent thicketing nor litigation risk alters R&D or acquisition responses in these cases, both raise the exploration response: a one-standard-deviation increase in *Thicketing* raises it by 0.005, and a one-standard-deviation increase in *LitigationRisk* raises it by 0.006, both significant at the 5% level and roughly 40–50% of the average AR^- response. When the focal firm's negatively interpreted patent grants create competitive threats, stronger enforcement is thus associated with a further shift toward more distant and less contested technological areas. Enforcement thus affects not only whether knowledge diffuses to technologically close peers but also where threatened peers direct their innovation. Patent concentration shows a weaker pattern in the opposite direction (−0.002, significant at the 10% level), consistent with fewer uncontested areas being available when patent ownership is concentrated.

The third source of heterogeneity relates to internal firm capabilities. Panel B of Table 10 shows that peers with greater cash holdings respond more strongly in R&D to both AR^+ and AR^- patent grants: a one-standard-deviation increase in cash raises the R&D response by 0.004 and 0.003, respectively, both significant at the 1% level. For peers exposed to AR^- patent grants, cash also raises the exploration response by 0.003, significant at the 1% level and about a quarter of the average AR^- response; the corresponding acquisition interaction is positive (1.4 percentage points) but insignificant. This suggests that financial resources facilitate strategic adaptation under competitive pressure. The acquisition interaction for AR^+ patent grants is positive but insignificant (0.007).

Leverage has the opposite effect. A one-standard-deviation increase in leverage reduces the R&D response by 0.003 for AR^+ and 0.002 for AR^- patent grants, both significant at the 5% level. The acquisition interactions are also negative (−1.3 and −1.1 percentage points) but insignificant. For AR^- patent grants, more levered peers also show a smaller exploration response (−0.003, significant at the 5% level). Access to financial resources thus appears to shape whether peers can turn related firms' patent grants into strategic responses.

Innovative capability similarly amplifies responsiveness, consistent with the capacity-to-respond prediction (Hypothesis 4). A one-standard-deviation increase in *InnovCap* raises the R&D response to AR^+ patent grants by 0.004 and the acquisition response by 1.7 percentage points, significant at the 5% and 10% levels, respectively. For AR^- patent grants, the effects on R&D and acquisitions are weaker (0.002, significant only at the 10% level, and an insignificant 0.005), but a one-standard-deviation increase in *InnovCap* raises the exploration response by 0.007, significant at the 1% level and more than half of the average AR^- response. Technologically capable peers thus appear better able to redirect their innovation toward alternative domains when they face competitive threats.

Taken together, these findings show a clear asymmetry in how peers adjust after the focal firm's patent grants. AR^+ patent grants coincide with increases in R&D and acquisition activity, and these increases, especially in R&D, are concentrated among peers that are technologically close to the focal firm, face weaker enforcement frictions and less concentrated patent ownership, and have the liquidity and absorptive capacity to pursue related opportunities. AR^- patent grants coincide with little change in innovation intensity across all moderators, except that peers with stronger balance sheets show somewhat larger adjustments. Instead, the heterogeneity in AR^- responses appears in their direction: peers facing stronger enforcement, and peers with more cash and greater innovative capability, redirect more of their innovation toward new technological fields. Overall, increases in innovation effort arise chiefly when the focal firm's innovations expand relevant opportunities and peers can act on them, whereas negatively valued grants mainly change where peers innovate.

4.10 Innovation Output: Patenting, Alignment, and the Technology Covered by the Focal Firm's Patent

In this section, I investigate whether peers' responses to the focal firm's patent grants extend beyond increased innovative efforts (R&D and acquisitions) and translate into measurable innovation output. Specifically, I examine three outcomes: (1) whether peers file more patents in response to the focal firm's

patent grants associated with statistically significant abnormal returns (*PatentFiling*), (2) whether these patents become more technologically aligned with the focal firm's innovation, as measured by overlapping backward citations (*CitShareOverlap*) and by textual similarity to the focal firm's patent (*TextSimilarity*), and (3) whether peers move into or out of the specific technology covered by the focal firm's patent (*FocalGroupShare* and *CitesFocalField*).

Table 11 presents the results, where the key variables of interest are *FocalPatentGrant* × AR^{+} and *FocalPatentGrant* × AR^{-}. These interactions capture peer responses to the focal firm's patent grants that are interpreted by the market as positive or negative shocks. The results reveal a clear asymmetry in both the level and direction of innovation. For AR^{+} patent grants, Column (1) shows that peers significantly increase patenting activity, with a coefficient of 0.208, significant at the 5% level. Relative to the mean of *PatentFiling* (2.311), this implies a 9% increase. Column (2) further shows a positive and statistically significant increase in *CitShareOverlap* (0.003, significant at the 5% level), corresponding to an 8% rise relative to its mean (0.038). These results indicate that peers not only innovate more but also direct their innovation toward the focal firm's technological domain, consistent with knowledge spillovers and technological complementarity. Columns (3) and (4) confirm the alignment result; with controls, the patenting result is weaker (0.142, significant only at the 10% level).

In contrast, for AR^{-} patent grants, the response differs sharply. The effect on patenting activity remains small and statistically insignificant in both specifications (0.048 and 0.028), indicating no meaningful increase in overall innovation output. However, the effect on directional alignment is negative and statistically significant. In Column (2), the coefficient on *CitShareOverlap* is −0.004, and in Column (4) it is −0.005, both significant at the 5% level. Relative to the mean of 0.038, these estimates imply a decline of approximately 10–13%, indicating that peers shift their innovation away from the focal firm's technological space. An alternative text-based measure gives the same directional pattern: TextSimilarity rises by about 16% after AR^{+} grants and falls by about 12.5% after AR^{-} grants. Internet Appendix IA.3 reports the construction and estimates (Table IA3-1).

Table 12 asks whether peers move into or out of the specific technology covered by the focal firm's patent. After AR^+ patent grants, the share of the peer's new applications in the CPC main group of the focal firm's patent rises by 1.1 percentage points (24% of the sample mean), and the share that cites the focal firm's patents in that group rises by 0.9 percentage points (43% of the mean), significant at the 5% and 1% levels, respectively. After AR^- patent grants, the share in the CPC main group of the focal firm's patent falls by 1.0 percentage point, significant at the 5% level, and citations to the focal firm's patents in that group decline (−0.004, significant at the 10% level).

The coefficients for citations to the focal firm's other technologies are 0.001 and 0.000 and statistically insignificant. Together with Table 3, these results are consistent with peers moving toward the technology covered by the focal firm's patent after AR^+ grants and away from it after AR^- grants.

Taken together, these findings show that a focal firm's innovation affects not only how much peers innovate, but also where they innovate. Opportunity-type shocks (AR^+) are followed by more patenting and by convergence toward the technology covered by the focal firm's patent, whereas threat-type shocks (AR^-) are followed by peers leaving that technology without increasing overall innovation intensity. This asymmetry highlights that peers respond to competitive threats not by scaling up innovation broadly, but by strategically repositioning their innovation efforts away from areas dominated by the focal firm.

5. Robustness Tests

The supporting tests address event timing, peer definitions, concurrent news, and measurement. Publication-date and placebo-date estimates are statistically insignificant; the placebo R&D estimates are too imprecise to rule out the baseline effect. Responses also remain positive under TNIC and industry peer definitions, although magnitudes and significance are weaker (Tables 13–16).

The AR^+ R&D pattern persists after excluding nearby earnings announcements, restricting overlapping exposure, applying inverse duplication weights, and reconstructing returns using the Fama–French five-factor and q-factor models. The acquisition estimates keep a similar magnitude but are statistically

insignificant in the smaller samples that restrict overlapping exposure. Above-median application-to-grant delays do not produce a statistically significant difference in responses, but delay is an imperfect uncertainty proxy (Tables 17–20).

The following subsections report each specification, its estimates, and the limits of the test. These checks support the stability of the main findings without replacing the identifying assumptions of the baseline and IV designs.

5.1 Innovation Response around the Patent Disclosure Date

Patent applications are published 18 months after the earliest priority date unless the applicant opts out, making previously private R&D projects publicly observable for the first time. In principle, this early visibility into the focal firm's technological direction could inform peers about upcoming threats, reveal areas of potential competition, or signal future product positioning. For firms operating in fast-moving or IP-intensive sectors, such early information might help shape strategic expectations and motivate preemptive innovation or acquisitions (Gans et al., 2008; Hegde and Luo, 2018). From this perspective, patent filing disclosures could serve as early, forward-looking signals of competitive intent.

However, several reasons suggest that the patent grant date is more likely to drive meaningful peer responses than the disclosure date. First, disclosures are inherently noisier and more ambiguous. Many disclosed applications are never granted (Farre-Mensa et al., 2020), and the scope or enforceability of the innovation remains uncertain until examination is complete. Peers may therefore discount disclosures as weak or tentative signals. In contrast, a grant certifies that the patent has satisfied novelty and non-obviousness requirements, providing a clear legal right to exclude and a credible assertion of technological capability. This formal recognition may be what prompts peers to reassess their strategic positioning.

Second, market reactions are more pronounced around the grant date. Kogan et al. (2017) show that stock-price reactions contain valuable information about the economic significance of innovations. Because a patent grant resolves uncertainty regarding patentability and legal protection, investors are better able to assess the economic value of the underlying innovation at the grant stage than at disclosure. Moreover,

investors extract value-relevant information from patent documents (Zheng, 2025). If the market responds strongly to a grant (e.g., through abnormal returns), it reveals that investors perceive the patent to have economic value (Kogan et al., 2017). The market reaction thus measures how the grant changes the prospects of the focal firm's innovation and the peer's relative position. Disclosures, by contrast, receive little market attention and are unlikely to generate sharp updates to focal firm valuations.

Finally, a patent grant unlocks legal enforceability. From the perspective of strategic investment decisions—particularly in R&D or M&A—firms must consider not just technological competition but legal risk (Galasso and Schankerman, 2015; Shapiro, 2000). The threat of litigation or exclusion from a key technology space only becomes credible upon grant. Thus, the grant event represents a sharper strategic inflection point, both in terms of competitive threat and option value of follow-on innovation. This motivates the focus on the grant date as the primary event of interest and sets the stage for testing whether the earlier disclosure event elicits any meaningful response. Table 13 reports no statistically significant R&D or acquisition responses around application publication. Neither disclosure itself nor disclosures that trigger positive or negative abnormal returns are associated with significant changes in peers' R&D investment or acquisition activity. These effects are small in magnitude and statistically indistinguishable from zero, even when accounting for firm-level controls.

This finding suggests that the information relevant for peers' decisions becomes available at the grant rather than at disclosure. Although the technical content is already public at publication, the grant determines whether and how broadly that content is protected, and grant-day returns measure its economic value. Section 5.6 shows that the responses do not depend on a proxy for patentability uncertainty, which suggests that the grant matters through the value and scope of the protection it establishes rather than through the resolution of uncertainty alone.

5.2 Focal Firm's Patent Grant Intensity and Innovative Efforts

In this section, I explore whether peers respond more strongly when they experience multiple significant patent events by the focal firm within the same year. To do so, I construct a peer-year level measure—

#[SignificantFocalPatent]—which counts the number of the focal firm's AR^+ and AR^- patent grants that a peer experiences. The peer-year regressions use the controls and fixed effects reported in Table 14.

Using this measure, I estimate peer-year level regressions of R&D investment and acquisition activity, controlling for firm characteristics and fixed effects as in the baseline model. As shown in Table 14, I find that an increase in the number of the focal firm's significant patent grants is positively associated with innovation inputs (R&D), whereas its association with external innovation efforts (acquisitions) is positive but statistically insignificant. With firm-level controls, each additional significant patent grant is associated with 0.003 higher R&D (significant at the 10% level), about half of the average response to an AR^+ patent grant in Table 2, and a 1.5 percentage point higher probability of an acquisition (not statistically significant). These findings suggest that peers do not respond to patenting activity in a binary manner; rather, they intensify their response as they experience more significant patent grant events that the market views as impactful.

This intensity analysis helps buttress the core mechanism of my paper—that peers interpret the focal firm's patent grants as strategically relevant news and respond with innovation. The fact that the response scales with the frequency of such events is consistent with this interpretation.

However, this analysis is not suitable as the main empirical specification for identification reasons. The regressions are conducted at the peer-year level using a count of significant grants as the main independent variable, whereas the instrumental variable strategy in Section 3.4.2 is defined for individual applications: it compares the grant and the abandonment of a given application using the leniency of its examiner. Aggregating grants to counts at the peer-year level removes this application-level variation, so the instrument cannot be applied. Therefore, while this analysis provides useful supporting evidence, it lacks a causal identification strategy and should be viewed as complementary rather than standalone.

5.3 Placebo Grant Dates

As an additional robustness test, I construct a placebo event by shifting each significant focal firm's patent grant date forward by 90 calendar days. I then re-estimate abnormal returns around these placebo

dates using the same event-study procedure and reconstruct the AR^+ and AR^- classifications based on the placebo CARs. Finally, I re-estimate the baseline specifications using these placebo treatment indicators. Because the outcomes are annual, the shift moves the event year only for grants in October through December, so the test mainly checks whether the effects depend on the returns realized on the actual grant dates. The rationale is that if peers respond specifically to the informational and legal implications of the focal firm's patent grant, shifting the event date away from the true grant should eliminate the observed effects.

Table 15 reports the results. Consistent with the identification strategy, neither the placebo AR^+ nor the placebo AR^- interactions generate statistically significant effects on peers' R&D investment, acquisition activity, innovation output, or innovation direction. The placebo coefficients in Table 15 are statistically indistinguishable from zero. The test has power to detect an effect of the baseline size for acquisitions, but the R&D placebo estimates are too imprecise to rule out an effect of that size. This stands in sharp contrast to the main results, where AR^+ patent grants are followed by significant increases in R&D, acquisitions, patent filings, citation overlap, and patent text similarity, while AR^- patent grants are followed by greater technological exploration and directional divergence.

The absence of a detectable acquisition response around placebo dates suggests that the acquisition results depend on the returns realized on the actual grant dates. Because the R&D placebo estimates are imprecise, the test is less informative for R&D.

5.4 Alternative Peer Definitions

Throughout the paper, peers are identified using technological proximity, requiring both overlap in CPC subclasses and direct patent citation links, because the paper studies how a patent grant affects the innovation of technologically related firms. Other peer definitions capture different relationships between firms. This section examines how sensitive the results are to the definition of peers.

I re-estimate the baseline specifications using two alternative peer definitions that do not rely on patent-based technological relationships. First, I identify peers using the TNIC product-market network developed

by Hoberg and Phillips (2016), which groups firms according to similarities in product descriptions disclosed in their 10-K filings. Second, I define peers more broadly as firms operating within the same Fama–French 48 industry (Fama and French, 1997). These alternative definitions capture product-market and industry relatedness rather than technological relatedness.

Table 16 reports the results. Across both alternative peer definitions, the central pattern is qualitatively similar. The focal firm's patent grants associated with AR^{+} continue to be followed by higher R&D investment; the acquisition coefficients keep their positive sign but are not statistically significant. The AR^{+} coefficients are 0.004 and 0.020 for TNIC peers and 0.003 and 0.015 for industry peers, about 50–70% of the baseline estimates; the R&D coefficients are significant at the 10% level, whereas the acquisition coefficients are not statistically significant. The effects retain the same direction, although their magnitudes and statistical significance are weaker. This attenuation is economically intuitive. Firms identified through product-market similarity or industry classifications are, on average, less connected through direct knowledge flows than technologically related peers and therefore have fewer opportunities to benefit from knowledge diffusion.

In contrast, the coefficients on $FocalPatentGrant \times AR^{-}$ remain economically small and statistically insignificant across all specifications. As in the baseline results, the focal firm's negatively interpreted patent grants do not induce meaningful changes in the overall level of innovative effort. This suggests that the asymmetry between AR^{+} and AR^{-} responses is not specific to a particular peer-network construction but instead reflects a broader difference in how firms respond to perceived technological opportunities versus perceived competitive threats.

Overall, the R&D results are robust to alternative definitions of peer firms, whereas the acquisition results keep their sign but not their statistical significance. While the strongest responses occur among technologically connected firms, similar qualitative patterns emerge when peers are defined using broader product-market and industry relationships. These findings provide additional confidence that the documented opportunity and threat effects are not artifacts of the baseline technological-peer definition.

5.5 Excluding Nearby Earnings Announcements

Another potential concern with the market-based classification of the focal firm's patent grants is that abnormal stock returns around grant dates may reflect information unrelated to the patent itself. For example, earnings announcements, acquisition announcements, seasoned equity offerings, or other major corporate disclosures occurring near the patent grant date could influence stock prices and contaminate the AR^+ and AR^- classifications.

To address this concern, I re-estimate the baseline specifications after excluding all the focal firm's patent grants that occur within a five-trading-day window of a firm's earnings announcement. Earnings announcements are particularly important because they are among the most information-rich corporate events and can generate substantial stock-price movements independent of patent-related news.

Table 17 retains positive, statistically significant AR^+ R&D and acquisition responses, while AR^- estimates remain small and insignificant. Although the sample size declines due to the exclusion of overlapping events, the magnitude and significance of the estimated effects are broadly similar to those reported in the baseline analysis.

These findings suggest that the market reactions used to construct the AR^+ and AR^- classifications are not driven by earnings-related information releases. Instead, the results are consistent with the interpretation that stock-price reactions around the focal firm's patent grants contain information about the technological and competitive implications of the patent itself. More broadly, the persistence of the results after removing grants near earnings announcements strengthens the validity of the market-based classification strategy employed throughout the paper.

5.6 Grant Delay and Patentability Uncertainty

The interpretation of patent-grant effects depends on the type of information revealed at the grant date. By the time a patent is granted, much of the technological information may already be publicly available through the disclosure process. If the grant event matters mainly because it resolves uncertainty regarding

patentability and legal protection, one would expect peer responses to be concentrated among patents whose approval outcomes were more uncertain ex ante. In contrast, if peer responses reflect the market's assessment of the innovation's economic significance, the effects should not depend strongly on the degree of patentability uncertainty surrounding the grant decision.

To examine this issue, I proxy for patentability uncertainty using the application-to-grant delay, defined as the number of days between the patent filing date and the patent grant date. Patents with longer delays are more likely to have faced greater uncertainty during the examination process. I classify patents with above-median application-to-grant delays as high-uncertainty patents and interact this indicator with the baseline AR^+ and AR^- treatment variables.

Table 18 reports the results. The triple interactions are small relative to the main effects and statistically insignificant: the AR^+ effect on R&D differs by only 0.001 between patents with above- and below-median grant delays. The effects of AR^+ patent grants remain positive and economically meaningful regardless of the degree of patentability uncertainty, while the corresponding AR^- effects remain limited. These findings suggest that the responses documented throughout the paper are not driven primarily by the resolution of patentability uncertainty at the grant date.

Instead, the results are more consistent with the interpretation that peers respond to the economic implications of the focal firm's innovations. Although patent grants undoubtedly resolve uncertainty regarding legal protection, the absence of systematically stronger responses among high-uncertainty patents suggests that uncertainty resolution alone cannot explain the observed patterns. Rather, market reactions appear to capture information about future technological opportunities and competitive threats that is relevant for peers' subsequent innovation decisions.

5.7 Robustness to Overlapping Network Exposure

The empirical design for the baseline analysis is implemented at the peer-focal-year level. Consequently, a given peer-year observation may appear multiple times in the estimation sample. This can occur for two

reasons. First, the same focal firm may receive multiple patent grants within a year. Second, a peer firm may be technologically connected to multiple focal firms, causing the same peer-year outcome to be linked to several distinct patent grants by the focal firm. As a result, the same peer-year outcome may contribute to multiple observations in the estimation sample. Such repetition does not invalidate the empirical design, but it may mechanically increase the influence of highly connected peer firms or peer-years that are associated with multiple patent grants by the focal firm. If the estimated effects are primarily driven by these repeated observations rather than by the underlying economic relationship, the baseline estimates could overstate the importance of particular peer-year outcomes.

Table 19 compares the baseline with clean event windows that exclude other connected grants, a restriction retaining the event with the largest absolute peer reaction for each peer-year, and inverse duplication weights that give repeated peer-year observations a total weight of one.

AR^+ R&D coefficients range from 0.005 to 0.006 and acquisition coefficients from 0.025 to 0.029. The R&D coefficients remain statistically significant, although only at the 10% level in some restricted samples; the acquisition coefficients are significant in the baseline and inverse-weight samples but not in the smaller CleanWindow and one-event samples. AR^- coefficients remain insignificant. The checks support the stability of the main pattern under these treatments of network exposure.

5.8 Alternative Asset Pricing Models

A potential concern is that the classification of the focal firm's patent grants into AR^+ and AR^- grants may depend on the choice of asset-pricing model used to estimate abnormal returns. Since the market-based classification plays a central role throughout the paper, it is important to verify that the results are not driven by a particular specification of expected returns.

At the same time, there are reasons to expect limited sensitivity to the choice of asset-pricing model in this setting. The abnormal returns are measured over a short five-trading-day event window centered on the patent grant date. Over such a narrow horizon, differences in expected returns implied by alternative asset-

pricing models are generally small relative to the firm-specific valuation effects generated by major information events. Consequently, event-study estimates based on short-window abnormal returns are typically less sensitive to the precise specification of expected returns than analyses conducted over longer horizons.

Nevertheless, I formally examine this issue by reconstructing abnormal returns using two alternative asset-pricing models. First, I employ the Fama–French five-factor model (Fama and French, 2015), which controls for market, size, value, profitability, and investment factors. Second, I use the q-factor model (Hou et al., 2015), which controls for market, size, investment, and profitability factors. I then reclassify the focal firm's patent grants into AR^+ and AR^- grants using the abnormal returns estimated from each model and re-estimate the baseline specifications.

Table 20 reports the results. Consistent with the intuition above, the distribution of AR^+ and AR^- patent grants remains remarkably stable across specifications. More importantly, the main findings are virtually unchanged. The focal firm's patent grants classified as AR^+ continue to generate statistically significant increases in R&D investment and acquisition activity, whereas AR^- patent grants remain economically small and statistically insignificant. The magnitudes of the estimated coefficients are highly similar to those obtained under the baseline market-model specification.

These findings suggest that the paper's conclusions are not driven by a particular choice of expected-return model. Instead, the market reactions surrounding the focal firm's patent grants appear sufficiently large that alternative asset-pricing models generate nearly identical classifications and inferences. This robustness provides additional confidence that the AR^+ and AR^- measures capture economically meaningful differences in how markets interpret the focal firm's innovations rather than artifacts of a specific abnormal-return methodology.

6. Conclusion

This paper examines how peers' real investment responds when a technologically related firm obtains

a patent grant, and whether stock prices on the grant date anticipate these responses. Using a comprehensive dataset that links patent grants, firm financials, acquisitions, market valuations, and patent citation patterns for U.S. public firms from 2001 to 2021, I study whether the focal firm's patent grants are followed by systematic changes in peers' innovative effort, innovation output, and firm-level financial choices. I provide novel evidence that the focal firm's significant patent grants—those perceived to be strategically important—are followed by meaningful changes in peers' innovation-related behavior.

The evidence shows that the average focal firm's patent grant has little effect on peers' R&D or acquisition activity, and instrumented grants of applications with low predicted importance have no detectable effect. Instrumenting grants of applications with high predicted importance, however, shows that grants raise R&D (and, less precisely, acquisitions) among technologically close peers and redirect the patenting of close rivals. This suggests that many of the focal firm's patent grants do not lead to detectable adjustments in peers' innovation behavior, masking substantial heterogeneity in responses. In contrast, a subset of grants—those associated with large, positive abnormal stock returns for both the peer and the focal firm (AR^+)—coincide with sizable increases in R&D investment and the likelihood of acquiring firms within the same industry. Dynamic event-time estimates indicate that these adjustments unfold along distinct timelines: R&D rises within a year of the grant, peaks in year +1, and fades by year +3, whereas acquisition activity rises more gradually, with effects peaking a few years after the grant. By comparison, grants associated with negative abnormal returns for the peer firm (AR^-) show no meaningful change in the level of innovative effort. Instead, peers exposed to AR^- patent grants exhibit pronounced shifts in the direction of their limited subsequent patenting, moving into new technological fields rather than scaling up activity along existing trajectories. These responses do not follow equally large stock price movements without a patent grant, are tied to the specific technology covered by the focal firm's patent, and tend to be larger after larger peer reactions.

The analysis of broader firm-level outcomes shows that, among financial outcomes, only net equity issuance responds to AR^+ patent grants. Around AR^+ patent grants, net equity issuance increases markedly, while cash holdings, leverage, and capital expenditures display no consistent movements. Because a similar

increase follows equally large price movements without a grant, this response is better interpreted as market timing than as patent-specific financing of follow-on innovation. In contrast, AR^- patent grants are not followed by systematic changes in liquidity, capital structure, or CAPEX, reinforcing the view that the focal firm's adverse patent news is associated primarily with directional shifts in patenting rather than broad-based changes in financial or real investment policies.

The triple-difference analyses reveal that these effects are highly heterogeneous across peer-focal relationships, legal environments, and internal capabilities. First, technological complementarity—measured via citation-based overlap—is the key dimension along which AR^+ patent grants coincide with stronger increases in R&D (the corresponding acquisition estimate is positive but imprecise), while product-market rivalry plays little systematic role. Second, stronger enforcement environments and more concentrated patent ownership attenuate the effects of AR^+ patent grants. In AR^- settings, where changes in innovation intensity are small, stronger enforcement instead pushes peers further away from the focal firm's technological space, whereas concentrated patent ownership slightly weakens this redirection, consistent with fewer uncontested areas to move into. Third, internal resources matter: peers with greater cash holdings and stronger innovative capability exhibit much larger AR^+ effects, and even in AR^- cases, financially flexible peers show somewhat stronger adjustments. In sum, these patterns underscore that substantial post-grant adjustments arise mainly when the focal firm's innovations expand technologically relevant opportunities, enforcement risk is not prohibitively high, and peers possess the financial and technological capacity to exploit those opportunities.

This study makes four contributions. It shows that stock prices anticipate how a firm's innovation propagates to related firms, extending the use of grant-day returns from the patentee (Kogan et al., 2017) to its peers and separating an information channel from a financing channel in the real effects of financial markets. It provides causal evidence that grants of applications with high predicted importance raise investment by technologically close peers and redirect the innovation of close rivals. It also reconciles studies that find positive and negative peer effects of innovation by showing that the balance between spillovers and rivalry varies across individual grants. Finally, it shows that financial flexibility and the legal

environment determine whether peers can respond to new patent grants to related firms.

From a policy perspective, these findings highlight the importance of understanding how intellectual property protection interacts with market competition. If certain firms are systematically discouraged from innovating due to high litigation risk or aggressive patenting by dominant players, this may reduce the overall dynamism of innovation ecosystems. Policymakers may therefore need to consider how the legal environment and patenting norms shape the incentives of firms to innovate. More broadly, when knowledge diffusion is constrained by rivalry or concentrated among a few frontier firms, the aggregate pace of innovation may slow, a pattern related to the decline in business dynamism documented by Akcigit and Ates (2023).

Future research can build on this work in several directions. First, while I focus on R&D and acquisitions as primary innovative responses, future studies could explore how the related firm's patenting affects other dimensions of the peer's strategic behavior—such as talent acquisition, product development, or market entry. Second, it would be valuable to examine whether the responses documented here ultimately translate into superior long-term innovation outcomes or competitive positioning.

In sum, this paper shows that patent grants shape both the scale and the direction of peers' innovation, and that stock prices on the grant date reveal which response will follow. Understanding these dynamics is essential for researchers studying how financial markets shape real investment and for policymakers seeking to foster competitive innovation.

References

Acharya, V., and Xu, Z. 2017. Financial dependence and innovation: The case of public versus private firms. Journal of Financial Economics, 124(2), 223-243.

Adner, R., and Snow, D. 2010. Old technology responses to new technology threats: Demand heterogeneity and technology retreats. Industrial and Corporate Change, 19(5), 1655-1675.

Aghion, P., Bloom, N., Blundell, R., Griffith, R., and Howitt, P. 2005. Competition and innovation: An inverted-U relationship. Quarterly Journal of Economics, 120(2), 701-728.

Aghion, P., Blundell, R., Griffith, R., Howitt, P., and Prantl, S. 2009. The effects of entry on incumbent innovation and productivity. The Review of Economics and Statistics, 91(1), 20-32.

Akcigit, U., and Ates, S. T. 2023. What happened to US business dynamism? Journal of Political Economy, 131(8), 2059-2124.

Anderson, T. W., and Rubin, H. 1949. Estimation of the parameters of a single equation in a complete system of stochastic equations. Annals of Mathematical Statistics, 20(1), 46-63.

Arora, A., Belenzon, S., and Sheer, L. 2021. Matching patents to Compustat firms, 1980–2015: Dynamic reassignment, name changes, and ownership structures. Research Policy, 50(5), 104217.

Arrow, K. J. 1962. Economic welfare and the allocation of resources for invention. In The rate and direction of inventive activity: Economic and social factors (pp. 609–626). Princeton University Press.

Azoulay, P., Graff Zivin, J. S., and Wang, J. 2010. Superstar extinction. Quarterly Journal of Economics, 125(2), 549-589.

Baker, A. C., Larcker, D. F., and Wang, C. C. 2022. How much should we trust staggered difference-in-differences estimates? Journal of Financial Economics, 144(2), 370-395.

Baker, M., Stein, J. C., and Wurgler, J. 2003. When does the market matter? Stock prices and the investment of equity-dependent firms. Quarterly Journal of Economics, 118(3), 969–1005.

Bena, J., and Li, K. 2014. Corporate innovations and mergers and acquisitions. Journal of Finance, 69(5), 1923-1960.

Bloom, N., Schankerman, M., and Van Reenen, J. 2013. Identifying technology spillovers and product market rivalry. Econometrica, 81(4), 1347-1393.

Bond, P., Edmans, A., and Goldstein, I. 2012. The real effects of financial markets. Annual Review of Financial Economics, 4, 339-360.

Brown, J. R., Fazzari, S. M., and Petersen, B. C. 2009. Financing innovation and growth: Cash flow, external equity, and the 1990s R&D boom. Journal of Finance, 64(1), 151-185.

Brown, J. R., and Petersen, B. C. 2011. Cash holdings and R&D smoothing. Journal of Corporate Finance, 17(3), 694-709.

Callaway, B., and Sant'Anna, P. H. C. 2021. Difference-in-differences with multiple time periods. Journal of Econometrics, 225(2), 200-230.

Cameron, A. C., Gelbach, J. B., and Miller, D. L. 2011. Robust inference with multiway clustering. Journal of Business & Economic Statistics, 29(2), 238-249.

Cengiz, D., Dube, A., Lindner, A., and Zipperer, B. 2019. The effect of minimum wages on low-wage jobs. Quarterly Journal of Economics, 134(3), 1405-1454.

Christensen, C. M. 1997. The innovator's dilemma: When new technologies cause great firms to fail. Boston, MA: Harvard Business School Press.

Cohen, W. M., and Levinthal, D. A. 1990. Absorptive capacity: A new perspective on learning and innovation. Administrative Science Quarterly, 35(1), 128-152.

de Chaisemartin, C., and D'Haultfœuille, X. 2020. Two-way fixed effects estimators with heterogeneous treatment effects. American Economic Review, 110(9), 2964-2996.

Fama, E. F., and French, K. R. 1997. Industry costs of equity. Journal of Financial Economics, 43(2), 153-193.

Fama, E. F., and French, K. R. 2015. A five-factor asset pricing model. Journal of Financial Economics, 116(1), 1-22.

Farre-Mensa, J., Hegde, D., and Ljungqvist, A. 2020. What is a patent worth? Evidence from the US patent "lottery". Journal of Finance, 75(2), 639-682.

Galasso, A., and Schankerman, M. 2015. Patents and cumulative innovation: Causal evidence from the courts. Quarterly Journal of Economics, 130(1), 317-369.

Gans, J. S., Hsu, D. H., and Stern, S. 2008. The impact of uncertain intellectual property rights on the market for ideas: Evidence from patent grant delays. Management Science, 54(5), 982-997.

Goodman-Bacon, A. 2021. Difference-in-differences with variation in treatment timing. Journal of Econometrics, 225(2), 254-277.

Gormley, T. A., and Matsa, D. A. 2011. Growing out of trouble? Corporate responses to liability risk. Review of Financial Studies, 24(8), 2781-2821.

Graham, S. J., Marco, A. C., and Miller, R. 2018. The USPTO patent examination research dataset: A window on patent processing. Journal of Economics & Management Strategy, 27(3), 554-578.

Haddad, V., Ho, P., and Loualiche, E. 2022. Bubbles and the value of innovation. Journal of Financial Economics, 145(1), 69-84.

Hall, B. H. 2002. The financing of research and development. Oxford Review of Economic Policy, 18(1), 35-51.

Hall, B. H., Jaffe, A., and Trajtenberg, M. 2005. Market value and patent citations. RAND Journal of Economics, 36(1), 16-38.

Harhoff, D., Scherer, F. M., and Vopel, K. 2003. Citations, family size, opposition and the value of patent rights. Research Policy, 32(8), 1343-1363.

Hegde, D., and Luo, H. 2018. Patent publication and the market for ideas. Management Science, 64(2), 652-672.

Higgins, M. J., and Rodriguez, D. 2006. The outsourcing of R&D through acquisitions in the pharmaceutical industry. Journal of Financial Economics, 80(2), 351-383.

Hoberg, G., and Phillips, G. M. 2010. Product market synergies and competition in mergers and acquisitions: A text-based analysis. Review of Financial Studies, 23(10), 3773-3811.

Hoberg, G., and Phillips, G. M. 2016. Text-based network industries and endogenous product differentiation. Journal of Political Economy, 124(5), 1423-1465.

Hoberg, G., and Phillips, G. M. 2025. Scope, scale, and concentration: The 21st-century firm. Journal of Finance, 80(1), 415-466.

Hou, K., Xue, C., and Zhang, L. 2015. Digesting anomalies: An investment approach. Review of Financial Studies, 28(3), 650-705.

Jaffe, A. B., Trajtenberg, M., and Henderson, R. 1993. Geographic localization of knowledge spillovers as evidenced by patent citations. Quarterly Journal of Economics, 108(3), 577-598.

Kelly, B., Papanikolaou, D., Seru, A., and Taddy, M. 2021. Measuring technological innovation over the long run. American Economic Review: Insights, 3(3), 303-320.

Kleibergen, F., and Paap, R. 2006. Generalized reduced rank tests using the singular value decomposition. Journal of Econometrics, 133(1), 97-126.

Kogan, L., Papanikolaou, D., Seru, A., and Stoffman, N. 2017. Technological innovation, resource allocation, and growth. Quarterly Journal of Economics, 132(2), 665-712.

Koh, P. S., and Reeb, D. M. 2015. Missing R&D. Journal of Accounting and Economics, 60(1), 73-94.

Lanjouw, J. O., and Schankerman, M. 2001. Characteristics of patent litigation: A window on competition. RAND Journal of Economics, 32(1), 129-151.

Lanjouw, J. O., and Schankerman, M. 2004. Patent quality and research productivity: Measuring innovation with multiple indicators. Economic Journal, 114(495), 441-465.

Marco, A. C., Sarnoff, J. D., and deGrazia, C. A. W. 2019. Patent claims and patent scope. Research Policy, 48(9), 103790.

Megna, P., and Klock, M. 1993. The impact of intangible capital on Tobin's q in the semiconductor industry. American Economic Review, 83(2), 265-269.

Moeller, S. B., Schlingemann, F. P., and Stulz, R. M. 2004. Firm size and the gains from acquisitions. Journal of Financial Economics, 73(2), 201-228.

Olden, A., and Møen, J. 2022. The triple difference estimator. Econometrics Journal, 25(3), 531-553.

Petersen, M. A. 2009. Estimating standard errors in finance panel data sets: Comparing approaches. Review of Financial Studies, 22(1), 435-480.

Phillips, G. M., and Zhdanov, A. 2013. R&D and the incentives from merger and acquisition activity. Review of Financial Studies, 26(1), 34-78.

Putnam, J. 1996. The value of international patent rights. Ph.D. dissertation, Yale University.

Roth, J., Sant'Anna, P. H. C., Bilinski, A., and Poe, J. 2023. What's trending in difference-in-differences? A synthesis of the recent econometrics literature. Journal of Econometrics, 235(2), 2218-2244.

Sanderson, E., and Windmeijer, F. 2016. A weak instrument F-test in linear IV models with multiple endogenous variables. Journal of Econometrics, 190(2), 212-221.

Schumpeter, J. 1942. Capitalism, Socialism and Democracy. Harper and Brothers.

Shapiro, C. 2000. Navigating the patent thicket: Cross licenses, patent pools, and standard setting. Innovation Policy and the Economy, 1, 119-150.

Stock, J. H., and Yogo, M. 2005. Testing for weak instruments in linear IV regression. In D. W. K. Andrews and J. H. Stock (Eds.), Identification and Inference for Econometric Models: Essays in Honor of Thomas Rothenberg (pp. 80-108). Cambridge University Press.

Sun, L., and Abraham, S. 2021. Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. Journal of Econometrics, 225(2), 175-199.

Testoni, M. 2022. The market value spillovers of technological acquisitions: Evidence from patent-text analysis. Strategic Management Journal, 43(5), 964-985.

Wing, C., Freedman, S. M., and Hollingsworth, A. 2024. Stacked difference-in-differences. NBER Working Paper No. 32054.

Zheng, Y. 2025. Can investors learn from patent documents? Evidence from textual analysis. Contemporary Accounting Research, 42(2), 1331-1358.

Appendix A: Variable Definitions

All continuous variables are winsorized at the 1st and 99th percentiles.

Variable	Definition
Dependent variables	
R&D	R&D expenditures scaled by total assets; set to 0 when missing; (XRD/AT).
Acquisition	Dummy variable equal to one if the firm completes at least one acquisition in year t in which the target shares the same two-digit SIC code as the acquirer. Deals are from SDC Platinum and include completed deals in which the acquirer controls less than 50% of the target before the deal and owns 100% after it, with a deal value of at least $1 million and at least 1% of the acquirer's market value (Moeller et al., 2004).
PatentFiling	Number of patent applications filed by the peer in year t.
CitShareOverlap	Average share of the backward citations of the peer's patents granted over [t, t+5] that overlap with the backward citations of the focal firm's patents (Section 3.3.3). As a moderator (Table 9), the same measure computed over [$\tau-5$, $\tau-1$].
TextSimilarity	Average TF-IDF cosine similarity between the focal firm's significant patent and the patents filed by the peer in each event year $t \in [\tau-3, \tau+3]$. Text consists of patent abstracts and claims; IDF weights are computed from patents granted before the focal firm's patent grant date (Kelly et al., 2021). Missing for peer-years without patent filings.
FocalGroupShare	Share of the peer's patent applications filed in year t that are classified in the CPC main group of the focal firm's significant patent.
CitesFocalField	Share of the peer's patent applications filed in year t that cite at least one patent of the focal firm in the CPC main group of the significant patent.
CitesFocalOther	Share of the peer's patent applications filed in year t that cite at least one patent of the focal firm outside that CPC main group.
ExplorationRatio	Share of the peer's patent applications filed in year t that fall outside its three largest CPC classes, where the largest classes are defined by the peer's filings over [$\tau-5$, $\tau-1$].
ExplorationRatio ($\tau-3$ to $\tau-1$)	Pre-grant counterpart of ExplorationRatio: share of the peer's new patent filings over [$\tau-3$, $\tau-1$] that fall outside its three largest CPC classes, where the largest classes are defined by filings over the five years before that window (Table 1, Panel B).
NetEquityIssuance	Sale of common and preferred stock minus purchases of common and preferred stock, scaled by lagged total assets; (SSTK − PRSTKC)/lagged AT.
ΔCash, ΔLeverage, CAPEX	Change in CHE/AT, change in (DLC+DLTT)/AT, and CAPX/lagged AT, respectively (Figure IA2-1).
Independent variables	
FocalPatentGrant	Dummy variable equal to one if the focal firm is granted at least one patent in year t.
FocalSigGrant	Dummy variable equal to one if the focal firm is granted at least one patent with a standardized five-day CAR above 2 in year t.
FocalPatentDisc	Dummy variable equal to one if at least one of the focal firm's patent applications is published by the USPTO in year t (pre-grant publication, typically 18 months after the earliest priority date).
Examiner Leniency (EL)	Leave-one-out grant rate of the examiner who reviewed the focal application, computed within the same art unit and year and excluding the focal firm's own applications (Section 3.3.2).

Variable	Definition
FocalAppGrant	Dummy variable equal to one if the focal application in the instrumental variable sample is granted and zero if it is abandoned (Section 3.4.2).
Predicted significance	Predicted probability that an application would be a significant grant (focal firm's standardized five-day CAR around the grant date above 2), from the logit model in Table IA6-1 estimated with information available before the examiner's decision; coefficients are estimated on other filing-year cohorts (leave-cohort-out).
#[SignificantFocalPatent]	Number of the focal firm's AR^+ and AR^- patent grants that the peer experiences in year t.
Event classification	
CAR	Sum of daily market-model abnormal returns over trading days [−2, +2] around the grant date. The market model is estimated over 250 trading days ending 42 trading days before the grant, using the CRSP value-weighted index. $z = (CAR - \mu)/\sigma$, where μ and σ are the mean and standard deviation of the firm's five-day CARs over the estimation window.
AR^+ (dummy)	Equal to one if, in year τ, at least one patent granted to the focal firm produces standardized CARs above 2 for both the focal firm and the peer.
AR^- (dummy)	Equal to one if, in year τ, at least one patent granted to the focal firm produces a standardized CAR above 2 for the focal firm and below −2 for the peer.
PS^+ / PS^- (dummy)	Price-shock counterparts of AR^+ and AR^-: equal to one if, in year τ, on a trading day on which the focal firm's standardized CAR exceeds 2 and the focal firm receives no patent grant within five trading days, the peer's standardized CAR exceeds 2 (falls below −2).
PriceShock	Counterpart of FocalPatentGrant for PS^+ and PS^-: equal to one in the post-event years of a cohort in which the focal firm has at least one such trading day.
Peer CAR	The peer's standardized five-day CAR around the focal firm's grant date.
No reaction	Peer-focal pairs around a patent grant with a standardized CAR above 2 for the focal firm in which the absolute value of the peer's standardized CAR is at most 2 (Table 1, Panel B).
RelativeSize	The focal firm's Log(Asset) minus the peer's Log(Asset), measured before the grant (Table 1, Panel B).
PatentValue (KPSS)	Market value of the focal firm's patents granted on the event date, in millions of dollars, from Kogan et al. (2017) (Table 1, Panel B).
Ex ante proximity	
ProductSimilarity	Pairwise TNIC product similarity score between the focal firm and the peer in year $\tau-1$ (Hoberg and Phillips, 2016).
SameIndustry (FF48)	Dummy variable equal to one if the focal firm and the peer belong to the same Fama–French 48 industry in year $\tau-1$ (Table 1, Panel B).
TechProximity	Average of the standardized CitShareOverlap over $[\tau-5, \tau-1]$ and the standardized TextSimilarity between the focal firm's and the peer's patents granted over $[\tau-3, \tau-1]$.
HighTech	Equal to one if TechProximity is in the top tercile of its distribution within cohort τ.
HighRival	Equal to one if the focal firm and the peer are TNIC-2 peers in year $\tau-1$ and their product similarity is above the median among TNIC-2 pairs in that year (Hoberg and Phillips, 2010, 2016).

Variable	Definition
Moderating variables	
Thicketing	Change in the focal firm's share of all patent filings in its dominant CPC subclass (the subclass with the most focal-firm filings over the five years before the grant) from the 18 months before to the 18 months after the grant month.
LitigationRisk	Number of patent litigation cases filed in the peer's industry in year t per 100 patent-holding firms in that industry (Patent Litigation Docket Reports).
PatentConc_Ind	Herfindahl–Hirschman index of patent-grant shares across firms in the peer's Fama–French 48 industry over [τ−3, τ−1].
InnovCap	Log of one plus the number of forward citations the peer's patents receive from its technological peers over the past five years.
PMR	Dummy variable equal to one if the focal firm and its peer belong to the same industry according to the TNIC-2 industry classification (Hoberg and Phillips, 2010, 2016).
HighUncertainty	Dummy variable equal to one if the application-to-grant delay of the focal firm's patent is above the sample median.
Control variables	
Log(Asset)	The logarithm of total assets; log(AT).
Tobin's Q	Market value of assets over total assets; (AT − (CEQ + TXDITC) + CSHO × PRCC_F)/AT.
Cash	The ratio of cash to total assets; CHE/AT.
Leverage	The ratio of total book debt to total assets; (DLC + DLTT)/AT.
Profitability	The ratio of operating income before depreciation to total assets; OIBDP/AT.
Age	The number of years since first appearing on the CRSP tapes.
Alternative peer definitions	
TNICPeer	Peers defined as TNIC-2 peers of the focal firm in year τ−1 (Hoberg and Phillips, 2010, 2016; Table 16).
IndustryPeer	Peers defined as firms in the same Fama–French 48 industry as the focal firm (Table 16).

Appendix B: A Model of Peer Responses

I extend the framework of Bloom et al. (2013), in which a firm's knowledge production benefits from the knowledge of technologically close firms and its product-market profits are reduced by the knowledge of its product-market rivals. I add three features: the peer allocates R&D between the technology covered by the focal firm's patent and other domains, the patent gives the focal firm enforceable exclusion rights in its technology, and rivalry in one domain raises the return to differentiating into others (Aghion et al., 2005).

A grant to the focal firm affects the peer through two channels. The first is diffusion: the patent discloses the focal firm's knowledge, which raises the productivity of the peer's research in the technology domain of the patent, more so when the peer is technologically closer and has greater absorptive capacity. The second is constraint: the grant strengthens the focal firm's position in that domain, which lowers the value of the peer's new knowledge there in proportion to product-market rivalry, more so when the focal firm can enforce its rights, while raising the value of research in other domains, where the peer can escape competition with the focal firm. The knowledge production function below captures the first channel, and the peer's profit function adds the second.

A focal firm j receives a patent in technology domain F whose economic importance is $q > 0$. A peer i has technological proximity $\kappa \geq 0$ and product-market rivalry $\lambda \geq 0$ with the focal firm, and absorptive capacity $\varphi \geq 0$. The peer chooses R&D in domain F, s, and R&D in other domains N, n. Following the knowledge production function of Bloom et al. (2013), the disclosure in the focal firm's patent raises the productivity of the peer's research in F:

$$k_F = (1 + \varphi\kappa q)\, s, \qquad k_N = n \tag{B1}$$

The peer's profits are

$$\pi_i = \bar{\pi}_i + m(\kappa - \lambda)q + [a - (1 + e)\lambda q]k_F + (b + \psi\lambda q)k_N \tag{B2}$$

The second term is the effect on the peer's existing products, with exposure $m > 0$: knowledge spillovers raise their profits and business stealing lowers them, as in Bloom et al. (2013). In domain F, the focal firm's stronger position lowers the value of the peer's new knowledge in proportion to rivalry, and more so when the focal firm can enforce its rights, as captured by $e \geq 0$ (Galasso and Schankerman, 2015). In other domains, rivalry raises the value of new knowledge by the factor $\psi \geq 0$ because differentiation lets the peer escape competition with the focal firm (Aghion et al., 2005). The peer solves

$$\max_{s,n} \pi_i - \frac{d}{2}(s^2 + n^2) - \frac{c}{2}(s + n)^2 \qquad \text{(B3)}$$

where $d > 0$ captures diminishing returns within each domain and $c > 0$ is the cost of financing total R&D. Up to terms of order q^2, the marginal returns to R&D in the two domains are

$$R_F(q) = a + [a\varphi\kappa - (1 + e)\lambda]q, \qquad R_N(q) = b + \psi\lambda q \qquad \text{(B4)}$$

and the first-order conditions give

$$s^* = \frac{(d + c)R_F - cR_N}{d(d + 2c)}, \qquad n^* = \frac{(d + c)R_N - cR_F}{d(d + 2c)} \qquad \text{(B5)}$$

$$X^* = s^* + n^* = \frac{R_F + R_N}{d + 2c}, \qquad n^* - s^* = \frac{R_N - R_F}{d} \qquad \text{(B6)}$$

I assume parameters such that s^* and n^* are positive. By the envelope theorem, the change in the peer's value at the grant date is, to first order,

$$\Delta V_i \approx q[\kappa\,(m + a\varphi s_0) - \lambda\,(m + (1 + e)s_0 - \psi n_0)] \qquad \text{(B7)}$$

where s_0 and n_0 are investments before the grant. The responses of total R&D and of its direction to the grant are

$$\frac{\partial X^*}{\partial q} = \frac{a\varphi\kappa - (1 + e - \psi)\lambda}{d + 2c}, \qquad \frac{\partial(n^* - s^*)}{\partial q} = \frac{(1 + e + \psi)\lambda - a\varphi\kappa}{d} \qquad \text{(B8)}$$

and their dependence on capacity and constraints is

$$\frac{\partial^2 X^*}{\partial q\ \partial \varphi} = \frac{a\kappa}{d+2c}, \qquad \frac{\partial^2 X^*}{\partial q\ \partial e} = -\frac{\lambda}{d+2c}, \qquad \frac{\partial^2 (n^* - s^*)}{\partial q\ \partial e} = \frac{\lambda}{d}, \qquad \frac{\partial^2 X^*}{\partial q\ \partial c} = -\frac{2}{d+2c}\frac{\partial X^*}{\partial q} \tag{B9}$$

Finally, the peer can acquire a target with knowledge g in domain F at a fixed cost f, financed at the same cost c. It acquires when the value of the target's knowledge covers the cost:

$$A = 1 \Leftrightarrow g\, R_F(q) \geq (1+c)f \tag{B10}$$

Six predictions follow.

(i) *Classification (Hypothesis 3).* By (B7), the peer's reaction is positive when technological proximity, weighted by the peer's exposure and absorptive capacity, exceeds rivalry. AR^+ therefore identifies grants for which complementarity dominates and AR^- grants for which rivalry dominates. Without the second domain and enforcement ($\psi = e = 0$, $n \equiv 0$), (B7) reproduces the result of Bloom et al. (2013) that technological proximity raises firm value and product-market proximity lowers it.

(ii) *Opportunity (Hypothesis 1).* When κ is large relative to λ, (B8) and (B5) imply that total R&D rises and investment shifts toward the technology covered by the focal firm's patent. By (B10), the threshold for acquiring a target in F falls, so acquisitions rise. Both responses are stronger when κ is higher and, as long as $\psi < 1 + e$, weaker when λ is higher.

(iii) *Redirection (Hypothesis 2).* When λ is large relative to κ, investment moves from F to other domains because the focal firm's exclusion rights lower the return in F while escaping competition raises the return elsewhere. The two forces have opposite effects on total R&D; when technological complementarity is negligible, they approximately offset when ψ is close to $1 + e$. Total R&D can therefore stay flat while the exploration ratio rises. This offsetting case is a possibility rather than a general prediction: total R&D may increase, decrease, or remain unchanged. Hypothesis 2 concerns the direction of innovation.

(iv) *Absorptive capacity (Hypothesis 4).* By (B9), the opportunity response is larger for peers with greater absorptive capacity φ, such as those closer to the technological frontier.

(v) *Legal constraints (Hypothesis 5).* By (B9), stronger enforcement e, which is more likely when litigation risk is high or the focal firm files aggressively after the grant, dampens the expansion of total R&D and strengthens the shift away from the technology covered by the focal firm's patent. Concentrated patent ownership in the peer's industry instead lowers ψ, because other domains are also more likely to be held by a few large rights holders; by (B8), this dampens both the expansion of total R&D and the redirection.

(vi) *Financing and dose-response (Hypotheses 3 and 4).* By (B9) and (B10), a higher financing cost c dampens the expansion of R&D and acquisitions after AR^+ grants. Because both the peer's reaction in (B7) and its responses in (B8) are proportional to q, responses also rise with the absolute size of the peer's market reaction.

A simple extension also links financing to redirection. If starting research in a domain outside the peer's current core requires a fixed cost f_N, financed at cost c, the peer redirects only when the gain from investing in N exceeds $(1 + c)f_N$. Since that gain rises with λq, a higher financing cost raises the threshold at which redirection begins, so financially flexible peers redirect more after AR^- grants.

Figure 1: Diffusion–Constraint Space (Conceptual)

This figure shows the conceptual diffusion–constraint space. The diagonal marks where technological complementarity and product-market rivalry are equally important. Peers A and B have the same complementarity but differ in rivalry. Table IA1-2 reports its empirical counterpart.

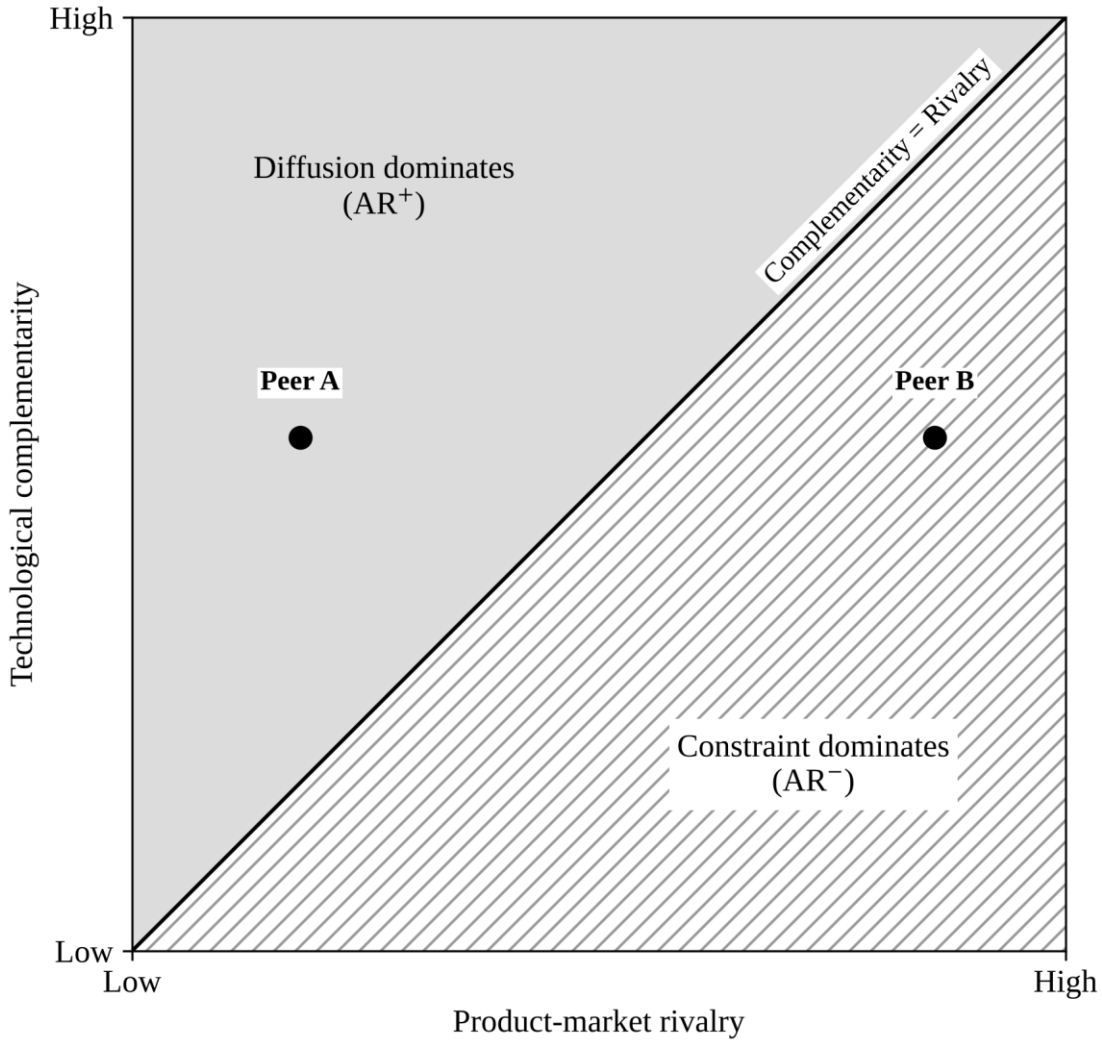


Figure 2: Hierarchical Arrays of CPC

This figure illustrates the hierarchy of the Cooperative Patent Classification (CPC), using Section C (Chemistry; Metallurgy) as an example. Each section is divided into classes (e.g., C03) and each class into subclasses (e.g., C03B). Technological peers are identified by co-location in CPC subclasses.

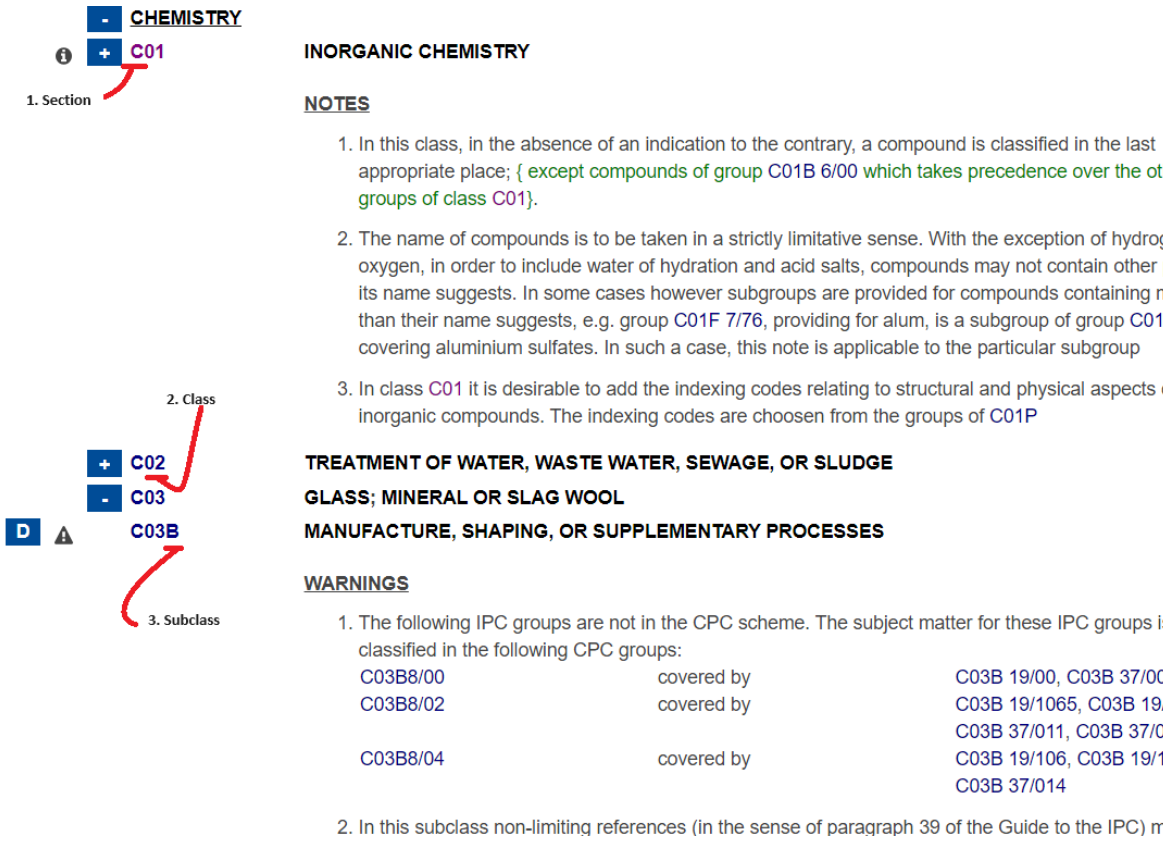


Figure 3: Distribution of Firms by Number of Technological Peers

This figure shows the distribution of firm-years by the number of technological peers. The mean is 7.6 peers per firm-year.

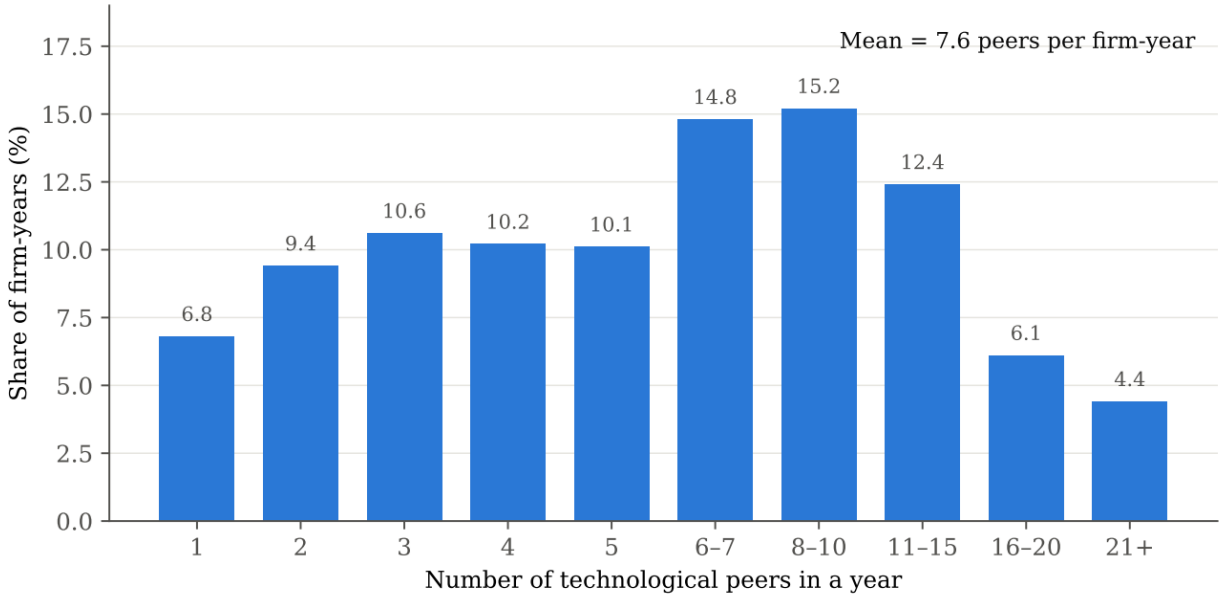
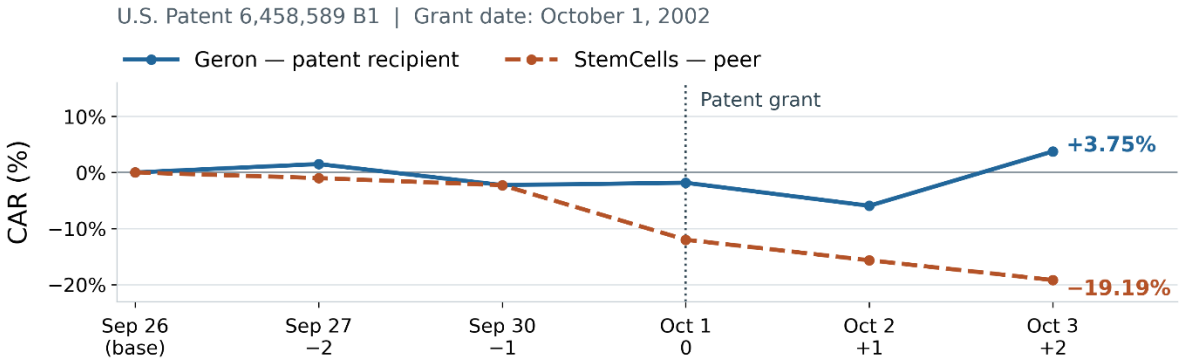


Figure 4: Five-Day Cumulative Abnormal Returns around Patent Grants

Panel A shows Geron's October 1, 2002 grant of U.S. Patent 6,458,589 B1; Panel B shows Netflix's June 24, 2003 grant of U.S. Patent 6,584,450 B1. CAR sums daily abnormal returns from day −2 through each plotted day; the last point is the five-day [−2,+2] CAR. Daily abnormal returns equal stock returns minus predictions from an OLS market model with an intercept and S&P 500 returns. Estimation uses 250 trading-day returns ending 42 trading days before the grant; Netflix uses its 231 available returns because of its shorter listing history. Returns are computed from Yahoo Finance adjusted closes. These S&P-based approximations illustrate return patterns and do not establish causality or satisfy the paper's standardized AR^{+}/AR^{-} selection thresholds.

Panel A: Geron and StemCells



Panel B: Netflix and Blockbuster

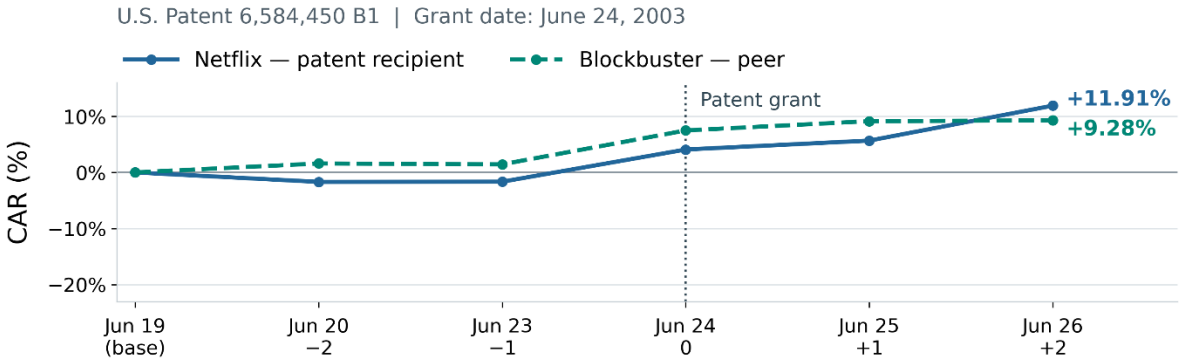


Figure 5: Dynamic Response of Peers' Innovative Efforts after Focal Firm's Patent Grant

This figure plots the coefficients from the event-time specification in Section 4.6 for peers' R&D (Panel a) and acquisition likelihood (Panel b), separately for AR^+ and AR^- patent grants. Year −1 is the omitted reference year. Shaded areas are 95% confidence intervals based on standard errors double clustered at the peer firm and year level.

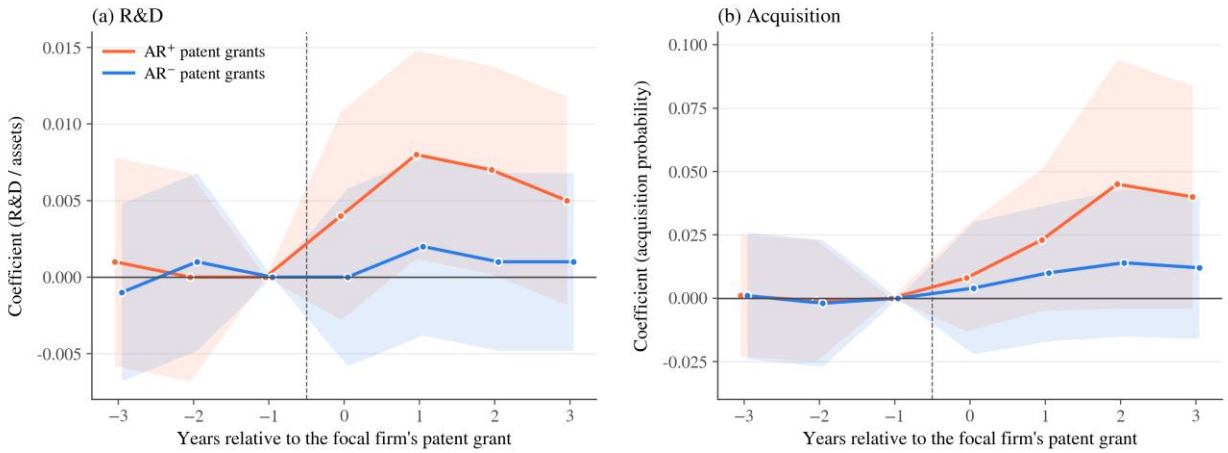


Table 1: Summary Statistics

Panel A presents summary statistics for the peer-focal-year panel. Variable definitions are provided in Appendix A. All continuous variables are winsorized at the 1st and 99th percentiles. *TextSimilarity, FocalGroupShare, CitesFocalField*, and *CitesFocalOther* are available only for peer-years in which the peer files at least one patent. *ProductSimilarity* is available for pairs covered by the TNIC data. Panel B compares peer-focal pairs around patent grants with a standardized CAR above 2 for the focal firm (18,912 peer-focal-years). AR^+ pairs are those in which the peer's standardized CAR exceeds 2, AR^- pairs those in which it falls below −2, and No reaction pairs those in which its absolute value is at most 2. The first block reports the relationship between the two firms, the second the peer's characteristics, and the third the focal firm's and the granted patent's characteristics, all measured before the grant. *RelativeSize* is the focal firm's Log(Asset) minus the peer's. Columns (4) and (5) report differences in means from regressions of each characteristic on group indicators with grant-year fixed effects; t-statistics based on standard errors clustered by focal firm are in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A: Peer-Focal-Year Panel

Variable	N	Mean	SD	P10	Median	P90
Dependent variables						
R&D	445,451	0.084	0.232	0.000	0.000	0.217
Acquisition	445,451	0.035	0.184	0.000	0.000	0.000
PatentFiling	421,636	2.311	5.637	0.000	1.000	12.000
CitShareOverlap	413,491	0.038	0.064	0.004	0.021	0.142
TextSimilarity	268,904	0.120	0.058	0.052	0.108	0.201
FocalGroupShare	268,904	0.046	0.118	0.000	0.000	0.167
CitesFocalField	268,904	0.021	0.074	0.000	0.000	0.063
CitesFocalOther	268,904	0.034	0.097	0.000	0.000	0.111
ExplorationRatio	445,451	0.012	0.078	0.000	0.008	0.036
NetEquityIssuance	445,451	0.008	0.023	−0.021	0.000	0.062
Independent variables						
FocalPatentGrant	445,451	0.430	0.495	0.000	0.000	1.000
FocalSigGrant	445,451	0.042	0.202	0.000	0.000	0.000
Event classification						
AR^+ (dummy)	445,451	0.0070	0.083	0.000	0.000	0.000
AR^- (dummy)	445,451	0.0041	0.064	0.000	0.000	0.000
PS^+ (dummy)	445,451	0.0131	0.114	0.000	0.000	0.000
PS^- (dummy)	445,451	0.0076	0.087	0.000	0.000	0.000
Ex ante proximity						
ProductSimilarity	398,214	0.021	0.038	0.000	0.006	0.061
HighTech	445,451	0.333	0.471	0.000	0.000	1.000
HighRival	445,451	0.090	0.286	0.000	0.000	0.000
Moderating variables						
Thicketing	423,219	0.013	0.051	−0.045	0.021	0.095
LitigationRisk	441,027	1.410	1.732	0.214	0.873	3.386
PatentConc_Ind	445,451	0.178	0.083	0.078	0.153	0.367

Variable	N	Mean	SD	P10	Median	P90
InnovCap	406,387	0.615	0.802	0.047	0.405	2.413
PMR	445,451	0.180	0.384	0.000	0.000	1.000
Control variables						
Log(Asset)	445,451	5.136	2.166	2.317	4.359	8.352
Tobin's Q	445,451	3.514	2.352	0.632	2.314	5.783
Cash	445,451	0.127	0.075	0.031	0.062	0.259
Leverage	445,451	0.244	0.139	0.000	0.196	0.352
Profitability	445,451	0.042	0.190	−0.181	0.089	0.198
Age	445,451	16.164	8.425	3.000	14.000	37.000

Panel B: Characteristics of AR^+, AR^-, and No-Reaction Pairs

Variable	AR^+ (1)	AR^- (2)	No reaction (3)	Diff. (1) − (2)	Diff. (1) − (3)
Relationship between the two firms					
CitShareOverlap	0.061	0.047	0.041	0.014*** (3.12)	0.020*** (5.85)
TextSimilarity	0.142	0.131	0.118	0.011** (2.41)	0.024*** (6.37)
ProductSimilarity	0.022	0.041	0.020	−0.019*** (−6.83)	0.002 (0.71)
PMR	0.176	0.312	0.169	−0.136*** (−7.94)	0.007 (0.62)
SameIndustry (FF48)	0.384	0.452	0.341	−0.068*** (−3.05)	0.043*** (2.88)
Peer characteristics					
Log(Asset)	4.82	4.97	5.31	−0.15 (−1.41)	−0.49*** (−5.12)
Tobin's Q	3.92	3.71	3.38	0.21 (1.58)	0.54*** (4.07)
Cash	0.146	0.131	0.124	0.015** (2.03)	0.022*** (3.66)
Leverage	0.228	0.249	0.243	−0.021** (−2.11)	−0.015* (−1.87)
Profitability	0.031	0.019	0.046	0.012 (1.12)	−0.015** (−2.04)
Age	14.9	15.6	16.4	−0.7 (−1.27)	−1.5*** (−3.21)
R&D	0.102	0.097	0.081	0.005 (0.61)	0.021*** (3.48)
InnovCap	0.71	0.64	0.60	0.07* (1.94)	0.11*** (3.02)

Variable	AR^{+} (1)	AR^{-} (2)	No reaction (3)	Diff. (1) − (2)	Diff. (1) − (3)
ExplorationRatio (τ−3 to τ−1)	0.011	0.012	0.012	−0.001 (−0.38)	−0.001 (−0.52)
Focal firm and granted patent					
Focal CAR (standardized)	2.71	2.84	2.63	−0.13* (−1.76)	0.08 (1.49)
Focal Log(Asset)	6.94	7.31	7.02	−0.37** (−2.48)	−0.08 (−0.84)
RelativeSize	2.12	2.34	1.71	−0.22* (−1.69)	0.41*** (3.37)
PatentValue (KPSS, $M)	18.4	23.7	17.9	−5.3** (−2.21)	0.5 (0.31)
Observations	3,129	1,815	13,968		

Table 2: Baseline Estimates: Focal Firm's Patent Grants and Peer Innovative Efforts

This table reports the estimation results for Equation (2). The dependent variables are *R&D* and *Acquisition* (innovative effort variables). The firm-level controls (i.e., *Log(Asset)*, *Tobin's Q*, *Cash, Leverage*, *Profitability,* and *Age*) are included in columns (3)-(4). Firm-cohort fixed effects as well as industry-year-cohort fixed effects are controlled for in the regression. Within each cohort, controls are pairs whose focal firm has no grant in year τ and that have no AR^+ or AR^- grant within $[\tau-3, \tau+3]$, including never-treated pairs. Because a control pair can enter several cohorts, the same peer-focal-year can appear more than once in the stacked sample. All standard errors are double clustered at peer firm and year level. The reported coefficients are cohort-size-weighted averages of the cohort-specific treatment effects. The t-statistics in parentheses are calculated based on the clustered standard errors. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Variable	*R&D*	*Acquisition*	*R&D*	*Acquisition*
	(1)	(2)	(3)	(4)
FocalPatentGrant	0.003	0.004	0.002	0.003
	(0.48)	(0.32)	(0.62)	(0.20)
*FocalPatentGrant * AR⁺*	0.007**	0.030*	0.006**	0.029**
	(2.32)	(1.71)	(2.21)	(2.17)
*FocalPatentGrant * AR⁻*	0.002	0.008	0.001	0.010
	(0.15)	(0.52)	(0.32)	(0.46)
Log(Asset)			0.013	0.001
			(1.56)	(0.69)
Tobin's Q			0.041*	−0.007
			(1.75)	(−0.47)
Cash			0.029*	0.021
			(1.92)	(1.42)
Leverage			−0.021	0.001
			(−0.29)	(0.36)
Profitability			0.013	0.004
			(0.85)	(0.20)
Age			0.001	0.001
			(0.41)	(1.02)
Controls	No	No	Yes	Yes
Firm x Cohort FE	Yes	Yes	Yes	Yes
Ind x Year x Cohort FE	Yes	Yes	Yes	Yes
Adj. R-squared	0.505	0.487	0.517	0.442
Observations	240,138	240,138	238,416	238,416

Table 3: Effect of Focal Firm's Adverse Patent News on Peers' Innovation Direction

This table reports the estimation results for changes in peers' innovation direction following patent grants associated with adverse peer reactions. The dependent variable is *ExplorationRatio, the share of the peer's patent applications filed in year t that fall outside its three largest CPC classes over [τ−5, τ−1]; these classes are defined identically for treated and control pairs.* The main explanatory variables are $FocalPatentGrant * AR^{+}(and\ AR^{-})$. All the firm-level controls (i.e., *Log(Asset), Tobin's Q, Cash, Leverage, Profitability,* and *Age*) are included in column (2). Firm-cohort fixed effects as well as industry-year-cohort fixed effects are controlled for in the regression. All standard errors are double clustered at peer firm and year level. The reported coefficients are cohort-size-weighted averages of the cohort-specific treatment effects. The t-statistics in parentheses are calculated based on the clustered standard errors. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Variable	*ExplorationRatio*	*ExplorationRatio*
	(1)	(2)
*FocalPatentGrant * AR⁺*	−0.001	0.004
	(−0.35)	(0.18)
*FocalPatentGrant * AR⁻*	0.016**	0.012**
	(2.29)	(2.04)
Controls	No	Yes
Firm x Cohort FE	Yes	Yes
Ind x Year x Cohort FE	Yes	Yes
Adj. R-squared	0.639	0.647
Observations	240,138	238,416

Table 4: First-Stage Results for the Instruments

This table reports the first stage of the instrumental variable strategy in Section 3.4.2. For each focal firm-year, the sample takes the application decided in that year with the highest predicted significance (Table IA6-1), if it is in the top quintile, and matches it to all technologically linked peers. *FocalAppGrant* equals one if the application is granted and zero if it is abandoned. *EL* is the leave-one-out grant rate of the application's examiner in the same art unit and year. *HighTech* and *HighRival* are defined in Section 3.3.7. Panel A reports the three first-stage regressions, each including all three excluded instruments; SW F is the Sanderson–Windmeijer F-statistic and KP rk Wald F the Kleibergen–Paap statistic. Panel B regresses pre-decision characteristics on *EL*. All specifications include the baseline controls and firm, industry-by-year, and art unit-by-decision-year fixed effects. Standard errors are double clustered at the focal firm and year level; t-statistics are in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Panel A: First Stage

Variable	*FocalAppGrant*	*FocalAppGrant* × *HighTech*	*FocalAppGrant* × *HighRival*
	(1)	(2)	(3)
EL	0.93***	0.004	0.002
	(9.42)	(0.37)	(0.21)
*EL * HighTech*	0.03	0.95***	−0.002
	(0.45)	(8.07)	(−0.19)
*EL * HighRival*	−0.04	0.003	0.88***
	(−0.39)	(0.24)	(5.31)
Controls	Yes	Yes	Yes
Firm, Ind x Year, Art unit x Decision-year FE	Yes	Yes	Yes
Mean of endogenous variable	0.72	0.24	0.065
Observations	61,840	61,840	61,840
Focal applications	8,126	8,126	8,126
SW F-statistic	88.7	65.1	28.2
KP rk Wald F		26.4	

Panel B: Balance of the Instrument

Dependent variable (measured before the decision)	*Coefficient on EL*	*t-statistic*
Predicted significance (Table IA6-1)	0.002	0.41
Log(1 + independent claims in published application)	−0.011	−0.52
Novelty of published application text	0.006	0.33
Peer ΔR&D (τ−3 to τ−1)	0.000	0.18
Peer acquisition rate (τ−3 to τ−1)	0.002	0.27
Peer ExplorationRatio (τ−3 to τ−1)	0.000	0.09

Table 5: Two-Stage Regressions for Peer Responses to Grants of Important Applications

This table reports second-stage estimates of the instrumental variable strategy in Section 3.4.2. *FocalAppGrant*, *FocalAppGrant * HighTech*, and *FocalAppGrant * HighRival* are instrumented with *EL*, *EL * HighTech*, and *EL * HighRival* (Table 4). Panel A uses applications in the top quintile of predicted significance; Panel B applies the same procedure to the bottom tercile. Dependent variables are averaged over the three years after the decision. Rows labeled "Effect for" report the sum of the main and interaction coefficients. AR denotes Anderson–Rubin 95% confidence intervals, which are robust to weak instruments. All columns include the baseline controls and firm, industry-by-year, and art unit-by-decision-year fixed effects. Standard errors are double clustered at the focal firm and year level; t-statistics are in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Panel A: Applications with High Predicted Significance

Variable	*R&D*	*Acquisition*	*ExplorationRatio*
	(1)	(2)	(3)
FocalAppGrant (fitted)	0.000	−0.001	0.000
	(0.12)	(−0.13)	(0.07)
*FocalAppGrant * HighTech (fitted)*	0.005***	0.014*	−0.001
	(2.94)	(1.91)	(−0.28)
*FocalAppGrant * HighRival (fitted)*	−0.002	−0.003	0.008***
	(−0.66)	(−0.29)	(2.83)
Effect for HighTech, non-rival pairs	0.005***	0.013*	−0.001
	(2.79)	(1.68)	(−0.24)
Effect for HighRival, non-HighTech pairs	−0.002	−0.004	0.008***
	(−0.61)	(−0.37)	(2.65)
AR 95% CI, * HighTech	[0.002, 0.009]	[−0.001, 0.030]	[−0.008, 0.006]
AR 95% CI, * HighRival	[−0.009, 0.005]	[−0.025, 0.019]	[0.002, 0.014]
AR joint test, p-value	<0.01	0.15	<0.01
Controls	Yes	Yes	Yes
Firm, Ind x Year, Art unit x Decision-year FE	Yes	Yes	Yes
Mean of dep. var.	0.091	0.037	0.012
Observations	61,840	61,840	61,840

Panel B: (Placebo) Applications with Low Predicted Significance

Variable	*R&D*	*Acquisition*	*ExplorationRatio*
	(1)	(2)	(3)
*FocalAppGrant * HighTech (fitted)*	0.000	0.001	0.000
	(0.11)	(0.13)	(0.06)
*FocalAppGrant * HighRival (fitted)*	0.000	−0.001	0.000
	(0.04)	(−0.05)	(0.08)
KP rk Wald F	31.9	31.9	31.9
Observations	97,215	97,215	97,215

Table 6: Patent-Linked versus Pure Price Shocks

Panel A estimates Equation (6). PS^{+} (PS^{-}), the price-shock counterpart of AR^{+} (AR^{-}), equals one if, on a day when the focal firm's standardized CAR exceeds 2 but it receives no patent grant within five trading days, the peer's standardized CAR exceeds 2 (falls below −2); *PriceShock* is the counterpart of *FocalPatentGrant*. The p-values test the equality of the grant-day and pseudo-event coefficients. Panel B compares joint-reaction frequencies on grant and non-grant days with a standardized CAR above 2 for the focal firm. Column (4) divides the non-grant share by the grant-day share; this ratio is not an identified misclassification rate. Return-measurement and event-attribution checks are reported in Table IA1-1. Controls and fixed effects are as in Table 2. Standard errors are double clustered at the peer firm and year level; t-statistics are in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Panel A: Grant-Day Events versus Non-Grant Joint Price Movements

Variable	*R&D*	*Acquisition*	*NetEquityIssuance*	*ExplorationRatio*
	(1)	(2)	(3)	(4)
*FocalPatentGrant * AR⁺*	0.006**	0.029**	0.004**	0.004
	(2.18)	(2.13)	(2.45)	(0.17)
*FocalPatentGrant * AR⁻*	0.001	0.009	0.000	0.012**
	(0.30)	(0.44)	(0.12)	(2.01)
*PriceShock * PS⁺*	0.000	0.003	0.003**	0.000
	(0.11)	(0.34)	(2.09)	(0.05)
*PriceShock * PS⁻*	−0.001	−0.003	−0.001	0.001
	(−0.47)	(−0.27)	(−0.66)	(0.42)
FocalPatentGrant	0.002	0.003	0.001	0.001
	(0.61)	(0.19)	(0.52)	(0.30)
PriceShock	0.000	0.002	0.000	0.000
	(0.13)	(0.16)	(0.09)	(0.11)
p-value: $AR^{+} = PS^{+}$	0.055	0.11	0.65	0.86
p-value: $AR^{-} = PS^{-}$	0.61	0.61	0.69	0.087
Controls	Yes	Yes	Yes	Yes
Firm x Cohort FE	Yes	Yes	Yes	Yes
Ind x Year x Cohort FE	Yes	Yes	Yes	Yes
Adj. R-squared	0.518	0.443	0.284	0.648
Observations	271,903	271,903	271,903	271,903

Panel B: Joint Reactions on Grant Days and Non-Grant Days

	Grant days	*Non-grant days*	*Ratio (1)/(2)*	*Non-grant / grant share*
	(1)	(2)	(3)	(4)
Observations (focal z > 2)	21,684	152,480		
Share with peer z > 2	16.4%	4.6%	3.6	28%
Share with peer z < −2	9.5%	2.4%	4.0	25%

Table 7: Interpretation of Positive and Negative Peer Market Reactions

This table examines whether positive and negative peer market reactions around the focal firm's patent grants are systematically associated with ex ante technological and competitive relationships between the focal firm and its peers. The sample consists of peer-year observations in which the focal firm's patent grant occurs. The dependent variables are indicator variables equal to one if the focal firm's patent grant is classified as an AR^+ grant (Column (1)) or an AR^- grant (Column (2)), and zero otherwise. Citation overlap and patent-text similarity measure ex ante technological complementarity between the focal firm and the peer, whereas TNIC product similarity measures ex ante product-market rivalry. All explanatory variables are measured prior to the patent grant. Firm-level controls are included in all specifications. All standard errors are double clustered at peer firm and year level. The t-statistics in parentheses are calculated based on the clustered standard errors. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Variable	AR^+	AR^-
	(1)	(2)
CitShareOverlap	0.006***	−0.001
	(3.21)	(−0.74)
TextSimilarity	0.004**	−0.002
	(2.49)	(−1.18)
ProductSimilarity	0.001	0.005***
	(0.63)	(2.89)
Controls	Yes	Yes
Firm FE	Yes	Yes
Ind x Year FE	Yes	Yes
Adj. R-squared	0.403	0.326
Observations	191,544	191,544

Table 8: Ex Ante Proximity and Peer Responses to Significant Patent Grants

This table classifies patent grants to the focal firm with statistically significant positive abnormal returns using only pre-grant information on the two firms. *FocalSigGrant* equals one in the post-grant years of a cohort in which the focal firm receives at least one patent grant with a standardized CAR above 2 (18,912 peer-focal-years). *HighTech* equals one if *TechProximity*, the average of the standardized *CitShareOverlap* over [τ−5, τ−1] and the standardized *TextSimilarity* between the two firms' patents granted over [τ−3, τ−1], is in the top tercile within cohort τ. HighRival equals one if the two firms are TNIC-2 peers in τ−1 with above-median product similarity among TNIC-2 pairs. The table estimates Equation (7) in Columns (1)–(3) and adds *FocalSigGrant* * AR^{+} and *FocalSigGrant* * AR^{-} in Columns (4)–(6). Table IA1-2 reports the four combinations of *HighTech* and *HighRival.* Controls and fixed effects are as in Table 2. Standard errors are double clustered at the peer firm and year level; t-statistics are in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Ex Ante Classification, with and without Market Reactions

Variable	*R&D*	*Acquisition*	*Exploration Ratio*	*R&D*	*Acquisition*	*Exploration Ratio*
	(1)	(2)	(3)	(4)	(5)	(6)
*FocalSigGrant * HighTech*	0.003*	0.012	−0.002	0.002	0.007	−0.001
	(1.78)	(1.19)	(−0.44)	(1.02)	(0.57)	(−0.29)
*FocalSigGrant * HighRival*	−0.002	−0.007	0.005*	−0.001	−0.004	0.003
	(−0.58)	(−0.52)	(1.69)	(−0.36)	(−0.30)	(1.04)
*FocalSigGrant * AR^{+}*				0.005**	0.026*	0.003
				(2.18)	(1.91)	(0.12)
*FocalSigGrant * AR^{-}*				0.001	0.008	0.011**
				(0.24)	(0.39)	(1.98)
FocalSigGrant	0.001	0.003	0.001	0.001	0.002	0.001
	(0.42)	(0.33)	(0.27)	(0.33)	(0.22)	(0.24)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm x Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Ind x Year x Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R-squared	0.517	0.442	0.647	0.518	0.443	0.648
Observations	238,416	238,416	238,416	238,416	238,416	238,416

Table 9: Triple-Difference Heterogeneity Analyses (Firm Relationship)

This table reports the estimation results for Equation (5). The dependent variables are *R&D*, *Acquisition*, and *ExplorationRatio*. The moderators are *CitShareOverlap*, measured over [τ−5, τ−1], and *PMR*. *CitShareOverlap* is standardized to have zero mean and unit standard deviation; *PMR* is an indicator and is mean-centered. All lower-order terms are included. All firm-level controls (i.e., *Log(Asset), Tobin's Q, Cash, Leverage, Profitability,* and *Age*) are included in all columns. Firm × cohort and industry × year × cohort fixed effects are included. Standard errors are double clustered at the peer firm and year level. The reported coefficients are cohort-size-weighted averages of the cohort-specific treatment effects. The t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Variable	*R&D*	*Acquisition*	*ExplorationRatio*
	(1)	(2)	(3)
*FocalPatentGrant * CitShareOverlap * AR⁺*	0.004**	0.016	0.001
	(2.34)	(1.45)	(0.31)
*FocalPatentGrant * CitShareOverlap * AR⁻*	0.001	0.001	0.001
	(0.46)	(0.08)	(0.29)
*FocalPatentGrant * PMR * AR⁺*	−0.002	−0.006	−0.001
	(−0.83)	(−0.46)	(−0.17)
*FocalPatentGrant * PMR * AR⁻*	0.001	0.005	−0.002
	(0.39)	(0.29)	(−0.21)
Lower-order terms			
FocalPatentGrant	0.002	0.003	0.001
	(1.15)	(0.19)	(0.31)
*FocalPatentGrant * AR⁺*	0.006**	0.028**	0.003
	(2.18)	(2.09)	(0.15)
*FocalPatentGrant * AR⁻*	0.001	0.009	0.012**
	(0.88)	(0.44)	(2.01)
*FocalPatentGrant * CitShareOverlap*	0.001	0.002	0.000
	(0.64)	(0.37)	(0.22)
*FocalPatentGrant * PMR*	−0.001	−0.002	0.001
	(−0.42)	(−0.22)	(0.27)
Controls	Yes	Yes	Yes
Firm x Cohort FE	Yes	Yes	Yes
Ind x Year x Cohort FE	Yes	Yes	Yes
Adj. R-squared	0.519	0.444	0.649
Observations	238,416	238,416	238,416

Table 10: Triple-Difference Heterogeneity Analyses (Constraints and Enablers)

This table reports the estimation results for Equation (5). The dependent variables are *R&D*, *Acquisition*, and *ExplorationRatio*. The moderators are *Thicketing*, *LitigationRisk*, and *PatentConc_Ind* in Panel A and *Cash, Leverage,* and *InnovCap* in Panel B. All moderators are standardized to have zero mean and unit standard deviation. *Cash, Leverage, InnovCap, LitigationRisk*, and *PatentConc_Ind* are measured before the grant; *Thicketing* compares the focal firm's filing share in the 18 months after and before the grant (Section 3.3.5). All lower-order terms are included. All firm-level controls (i.e., *Log(Asset), Tobin's Q, Cash, Leverage, Profitability,* and *Age*) are included in all columns. Firm × cohort and industry × year × cohort fixed effects are included. Standard errors are double clustered at the peer firm and year level. The reported coefficients are cohort-size-weighted averages of the cohort-specific treatment effects. The t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A: External Constraints

Variable	*R&D*	*Acquisition*	*ExplorationRatio*
	(1)	(2)	(3)
*FocalPatentGrant * Thicketing * AR⁺*	−0.003**	−0.012	0.001
	(−2.16)	(−1.40)	(0.42)
*FocalPatentGrant * Thicketing * AR⁻*	−0.001	−0.008	0.005**
	(−0.65)	(−0.73)	(2.04)
*FocalPatentGrant * LitigationRisk * AR⁺*	−0.003*	−0.011	−0.001
	(−1.85)	(−1.36)	(−0.32)
*FocalPatentGrant * LitigationRisk * AR⁻*	−0.001	−0.001	0.006**
	(−0.47)	(−0.09)	(2.54)
*FocalPatentGrant * PatentConc_Ind * AR⁺*	−0.003**	−0.012	0.001
	(−2.27)	(−1.45)	(0.38)
*FocalPatentGrant * PatentConc_Ind * AR⁻*	0.001	0.002	−0.002*
	(0.52)	(0.43)	(−1.71)
Lower-order terms			
FocalPatentGrant	0.002	0.003	0.001
	(1.18)	(0.20)	(0.28)
*FocalPatentGrant * AR⁺*	0.006**	0.029**	0.004
	(2.19)	(2.12)	(0.17)
*FocalPatentGrant * AR⁻*	0.001	0.010	0.012**
	(0.91)	(0.47)	(2.07)
*FocalPatentGrant * Thicketing*	−0.001	−0.002	0.001
	(−0.71)	(−0.34)	(0.55)
*FocalPatentGrant * LitigationRisk*	−0.001	−0.002	0.000
	(−0.66)	(−0.26)	(0.18)
*FocalPatentGrant * PatentConc_Ind*	−0.001	−0.001	−0.001
	(−0.53)	(−0.20)	(−0.24)
Controls	Yes	Yes	Yes
Firm x Cohort FE	Yes	Yes	Yes
Ind x Year x Cohort FE	Yes	Yes	Yes
Adj. R-squared	0.523	0.447	0.652
Observations	238,416	238,416	238,416

Panel B: Internal Enablers

Variable	*R&D*	*Acquisition*	*ExplorationRatio*
	(1)	(2)	(3)
*FocalPatentGrant * Cash * AR⁺*	0.004***	0.007	0.000
	(3.02)	(0.64)	(0.03)
*FocalPatentGrant * Cash * AR⁻*	0.003***	0.014	0.003***
	(3.21)	(1.45)	(3.32)
*FocalPatentGrant * Leverage * AR⁺*	−0.003**	−0.013	−0.001
	(−2.12)	(−1.30)	(−0.53)
*FocalPatentGrant * Leverage * AR⁻*	−0.002**	−0.011	−0.003**
	(−1.99)	(−1.59)	(−2.13)
*FocalPatentGrant * InnovCap * AR⁺*	0.004**	0.017*	0.000
	(2.14)	(1.84)	(0.10)
*FocalPatentGrant * InnovCap * AR⁻*	0.002*	0.005	0.007***
	(1.92)	(0.90)	(3.98)
Lower-order terms			
FocalPatentGrant	0.002	0.003	0.001
	(1.19)	(0.21)	(0.33)
*FocalPatentGrant * AR⁺*	0.006**	0.029**	0.004
	(2.24)	(2.15)	(0.16)
*FocalPatentGrant * AR⁻*	0.001	0.009	0.012**
	(0.86)	(0.44)	(2.05)
*FocalPatentGrant * Cash*	0.001	0.002	0.000
	(0.92)	(0.29)	(0.15)
*FocalPatentGrant * Leverage*	−0.001	−0.002	−0.001
	(−0.77)	(−0.37)	(−0.36)
*FocalPatentGrant * InnovCap*	0.001	0.003	0.001
	(0.85)	(0.41)	(0.47)
Controls	Yes	Yes	Yes
Firm x Cohort FE	Yes	Yes	Yes
Ind x Year x Cohort FE	Yes	Yes	Yes
Adj. R-squared	0.524	0.448	0.650
Observations	238,416	238,416	238,416

Table 11: Innovation Output: Patent Filing and Directional Alignment

This table reports the estimation results for Equation (2) using innovation output measures as dependent variables. The dependent variables are *PatentFiling* and *CitShareOverlap*. The main explanatory variables are $FocalPatentGrant * AR^{+}$ and $FocalPatentGrant * AR^{-}$. All the firm-level controls (i.e., *Log(Asset), Tobin's Q, Cash, Leverage, Profitability,* and *Age*) are included in columns (3) and (4). Firm-cohort fixed effects as well as industry-year-cohort fixed effects are controlled for in the regression. All standard errors are double clustered at peer firm and year level. The reported coefficients are cohort-size-weighted averages of the cohort-specific treatment effects. The t-statistics in parentheses are calculated based on the clustered standard errors. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Variable	*PatentFiling*	*CitShareOverlap*	*PatentFiling*	*CitShareOverlap*
	(1)	(2)	(3)	(4)
FocalPatentGrant	0.031	0.000	0.024	0.000
	(0.42)	(0.18)	(0.37)	(0.21)
*FocalPatentGrant * AR⁺*	0.208**	0.003**	0.142*	0.003**
	(2.15)	(2.43)	(1.75)	(2.32)
*FocalPatentGrant * AR⁻*	0.048	−0.004**	0.028	−0.005**
	(0.46)	(−2.41)	(0.89)	(−2.01)
Controls	No	No	Yes	Yes
Firm x Cohort FE	Yes	Yes	Yes	Yes
Ind x Year x Cohort FE	Yes	Yes	Yes	Yes
Adj. R-squared	0.327	0.215	0.318	0.237
Observations	240,138	240,138	238,416	238,416

Table 12: Moving Into or Out of the Technology Covered by the Focal Firm's Patent

This table reports the estimation results for Equation (2) using technology-specific outcomes. *FocalGroupShare* is the share of the peer's patent applications in the event year that are classified in the CPC main group of the focal firm's significant patent. *CitesFocalField* (*CitesFocalOther*) is the share of the peer's applications that cite at least one patent of the focal firm inside (outside) that main group. Peer-years without patent applications are excluded. All firm-level controls (i.e., Log(Asset), Tobin's Q, Cash, Leverage, Profitability, and Age) are included. Firm-cohort fixed effects as well as industry-year-cohort fixed effects are controlled for in the regressions. All standard errors are double clustered at the peer firm and year level. The t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Variable	*FocalGroupShare*	*CitesFocalField*	*CitesFocalOther*
	(1)	(2)	(3)
FocalPatentGrant * AR^+	0.011**	0.009***	0.001
	(2.36)	(2.83)	(0.48)
FocalPatentGrant * AR^-	−0.010**	−0.004*	0.000
	(−2.29)	(−1.71)	(0.06)
FocalPatentGrant	0.001	0.001	0.001
	(0.33)	(0.64)	(0.41)
Mean of dep. var.	0.046	0.021	0.034
Controls	Yes	Yes	Yes
Firm x Cohort FE	Yes	Yes	Yes
Ind x Year x Cohort FE	Yes	Yes	Yes
Adj. R-squared	0.391	0.214	0.198
Observations	143,712	143,712	143,712

Table 13: Focal Firm's Patent Disclosure and Innovative Efforts

This table reports the effect of the focal firm's patent application disclosure on peers' innovative efforts. The main explanatory variables are *FocalPatentDisc, FocalPatentDisc* × AR^{+}, and *FocalPatentDisc* × AR^{-}, where AR^{+} *(*AR^{-}*)* is defined from peers' abnormal returns around the publication date. The dependent variables are R&D and Acquisition. All firm-level controls (i.e., *Log(Asset), Tobin's Q, Cash, Leverage, Profitability,* and *Age*) are included in Columns (3) and (4). Firm × cohort and industry × year × cohort fixed effects are included. Standard errors are double clustered at the peer firm and year level. The reported coefficients are cohort-size-weighted averages of the cohort-specific treatment effects. The t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Variable	*R&D*	*Acquisition*	*R&D*	*Acquisition*
	(1)	(2)	(3)	(4)
FocalPatentDisc	0.001	0.002	0.001	0.001
	(0.42)	(0.15)	(0.37)	(0.11)
FocalPatentDisc * AR^{+}	0.001	−0.001	0.002	0.001
	(0.71)	(−0.08)	(0.88)	(0.16)
FocalPatentDisc * AR^{-}	−0.001	0.002	−0.001	0.002
	(−0.46)	(0.22)	(−0.52)	(0.19)
Log(Asset)			0.021**	0.003
			(2.35)	(1.18)
Tobin's Q			0.016	−0.013
			(0.94)	(−0.88)
Cash			0.048***	0.010
			(3.41)	(0.79)
Leverage			−0.063*	−0.007
			(−1.88)	(−0.50)
Profitability			0.027	−0.003
			(1.31)	(−0.11)
Age			0.008	0.001
			(0.08)	(0.04)
Controls	No	No	Yes	Yes
Firm x Cohort FE	Yes	Yes	Yes	Yes
Ind x Year x Cohort FE	Yes	Yes	Yes	Yes
Adj. R-squared	0.528	0.421	0.519	0.460
Observations	207,382	207,382	205,889	205,889

Table 14: Focal Firm's Patent Grant Intensity and Innovative Efforts

This table reports how the number of patents granted to the focal firm affects peers′ innovative efforts. The variable #[*SignificantFocalPatent*] counts the number of the focal firm's significant patent grants that the peer experiences in a given year. The dependent variables are *R&D* and *Acquisition* (innovative effort variables). All the firm-level controls (i.e., *Log(Asset), Tobin's Q, Cash, Leverage, Profitability,* and *Age*) are included in columns (3) and (4). Firm fixed effects as well as industry × year fixed effects are controlled for in the regression. All standard errors are double clustered at peer firm and year level. The t-statistics in parentheses are calculated based on the clustered standard errors. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Variable	*R&D*	*Acquisition*	*R&D*	*Acquisition*
	(1)	(2)	(3)	(4)
#[SignificantFocalPatent]	0.002	0.016	0.003*	0.015
	(1.55)	(1.32)	(1.82)	(1.42)
Log(Asset)			0.016**	0.002
			(2.13)	(1.40)
Tobin's Q			0.021	−0.008
			(0.63)	(−0.36)
Cash			0.021***	0.020
			(2.87)	(0.59)
Leverage			−0.053	−0.001
			(−1.18)	(−0.15)
Profitability			0.067	−0.001
			(1.40)	(−0.07)
Age			0.006	0.001
			(0.75)	(0.09)
Controls	No	No	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Ind x Year FE	Yes	Yes	Yes	Yes
Adj. R-squared	0.344	0.428	0.351	0.413
Observations	58,612	58,612	58,191	58,191

Table 15: Placebo Grant Dates

This table reports a placebo test using artificial grant dates for the focal firm's patents. For each significant patent grant by the focal firm, the treatment date is shifted forward by 90 calendar days, CARs and the AR^+ and AR^- classifications are recomputed at those dates, and the baseline specifications are re-estimated. Since outcomes are annual, the event year changes only for late-year grants. The dependent variables are *R&D, Acquisition, PatentFiling,* and *ExplorationRatio*. The main explanatory variables are $PlaceboGrant * AR^+$ and $PlaceboGrant * AR^-$. All firm-level controls (i.e., *Log(Asset), Tobin's Q, Cash, Leverage, Profitability,* and *Age*) are included in all columns. Firm-cohort fixed effects as well as industry-year-cohort fixed effects are controlled for in the regression. All standard errors are double clustered at the peer firm and year level. The reported coefficients are cohort-size-weighted averages of the cohort-specific treatment effects. The t-statistics in parentheses are calculated based on clustered standard errors. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Variable	*R&D*	*Acquisition*	*PatentFiling*	*ExplorationRatio*
	(1)	(2)	(3)	(4)
*PlaceboGrant * AR⁺*	0.012	0.004	0.024	0.002
	(0.42)	(0.41)	(0.31)	(0.27)
*PlaceboGrant * AR⁻*	−0.002	0.002	−0.011	0.003
	(−0.33)	(0.17)	(−0.12)	(0.41)
Controls	Yes	Yes	Yes	Yes
Firm x Cohort FE	Yes	Yes	Yes	Yes
Ind x Year x Cohort FE	Yes	Yes	Yes	Yes
Adj. R-squared	0.514	0.439	0.302	0.651
Observations	238,416	238,416	238,416	238,416

Table 16: Alternative Peer Definitions

This table examines whether the main findings depend on the construction of peer firms. Columns (1)–(2) redefine peers using the TNIC product-market network developed by Hoberg and Phillips (2016). Columns (3)–(4) define peers as firms operating within the same Fama–French 48 industry. For each alternative peer definition, peer-focal pairs and the AR^+ and AR^- classifications are reconstructed using the abnormal returns of the redefined peers, and clean control pairs, including never-treated pairs, are constructed as in the baseline specification. The dependent variables are *R&D* and *Acquisition*. All firm-level controls (i.e., *Log(Asset), Tobin's Q, Cash, Leverage, Profitability,* and *Age*) are included in all specifications. Firm-cohort fixed effects as well as industry-year-cohort fixed effects are controlled for in the regressions. All standard errors are double clustered at the peer firm and year level. The reported coefficients are cohort-size-weighted averages of the cohort-specific treatment effects. The t-statistics in parentheses are calculated based on clustered standard errors. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Variable	*TNICPeer (R&D)*	*TNICPeer (Acquisition)*	*IndustryPeer (R&D)*	*IndustryPeer (Acquisition)*
	(1)	(2)	(3)	(4)
*FocalPatentGrant * AR⁺*	0.004*	0.020	0.003*	0.015
	(1.89)	(1.53)	(1.74)	(1.35)
*FocalPatentGrant * AR⁻*	−0.001	0.006	0.000	0.003
	(−0.28)	(0.29)	(0.04)	(0.20)
Controls	Yes	Yes	Yes	Yes
Firm x Cohort FE	Yes	Yes	Yes	Yes
Ind x Year x Cohort FE	Yes	Yes	Yes	Yes
Adj. R-squared	0.492	0.421	0.468	0.403
Observations	151,327	151,327	486,912	486,912

Table 17: Excluding Earnings Announcements Around Patent Grant Dates

This table reports the baseline estimates after excluding the focal firm's patent grants that occur within five trading days of a firm's earnings announcement. The purpose of this test is to ensure that the abnormal stock returns used to construct the AR^+ and AR^- classifications are not driven by earnings-related information releases. The dependent variables are *R&D* and *Acquisition*. All firm-level controls (i.e., *Log(Asset), Tobin's Q, Cash, Leverage, Profitability,* and *Age*) are included in all specifications. Firm-cohort fixed effects as well as industry-year-cohort fixed effects are controlled for in the regressions. All standard errors are double clustered at the peer firm and year level. The reported coefficients are cohort-size-weighted averages of the cohort-specific treatment effects. The t-statistics in parentheses are calculated based on clustered standard errors. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Variable	*R&D*	*Acquisition*
	(1)	(2)
*FocalPatentGrant * AR⁺*	0.005**	0.027**
	(2.12)	(2.00)
*FocalPatentGrant * AR⁻*	0.001	0.007
	(0.61)	(0.37)
Controls	Yes	Yes
Firm x Cohort FE	Yes	Yes
Ind x Year x Cohort FE	Yes	Yes
Adj. R-squared	0.511	0.437
Observations	221,504	221,504

Table 18: Peer Responses and Patentability Uncertainty

This table examines whether peer responses are driven primarily by the resolution of patentability uncertainty at the grant date. Patentability uncertainty is proxied by *HighUncertainty*, an indicator equal to one if the patent's application-to-grant delay is above the sample median. The dependent variables are *R&D* and *Acquisition*. The main explanatory variables are *FocalPatentGrant* $*$ AR^{+}, *FocalPatentGrant* $*$ AR^{-}, and their interactions with *HighUncertainty*. *HighUncertainty* is mean-centered, so the coefficients on *FocalPatentGrant* $\times$ AR^{+} *(*AR^{-}*)* measure the average effect across patents. All firm-level controls (i.e., *Log(Asset), Tobin's Q, Cash, Leverage, Profitability,* and *Age*) are included in all specifications. Firm-cohort fixed effects as well as industry-year-cohort fixed effects are controlled for in the regressions. All standard errors are double clustered at the peer firm and year level. The reported coefficients are cohort-size-weighted averages of the cohort-specific treatment effects. The t-statistics in parentheses are calculated based on clustered standard errors. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Variable	*R&D*	*Acquisition*
	(1)	(2)
*FocalPatentGrant * AR⁺*	0.006**	0.029**
	(2.19)	(2.07)
*FocalPatentGrant * AR⁻*	0.001	0.010
	(0.90)	(0.45)
*FocalPatentGrant * AR⁺ * HighUncertainty*	0.001	0.004
	(0.19)	(0.13)
*FocalPatentGrant * AR⁻ * HighUncertainty*	−0.001	0.002
	(−0.46)	(0.04)
Lower-order terms		
FocalPatentGrant	0.002	0.003
	(1.20)	(0.20)
*FocalPatentGrant * HighUncertainty*	0.000	−0.001
	(0.12)	(−0.08)
Controls	Yes	Yes
Firm x Cohort FE	Yes	Yes
Ind x Year x Cohort FE	Yes	Yes
Adj. R-squared	0.521	0.441
Observations	238,416	238,416

Table 19: Robustness to Overlapping Network Exposure

This table examines repeated peer-year outcomes and overlapping exposure. Column (1) reports the baseline. Column (2) excludes peer-years exposed to other connected grants during the event window. Column (3) applies the one-event restriction described in Section 5.7. Column (4) uses inverse duplication weights so that repeated peer-year observations have total weight one. Controls and fixed effects match the baseline. Standard errors are double clustered at the peer firm and year level; t-statistics are in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Panel A: Innovation Responses (R&D)

Variable	*Baseline*	*CleanWindow*	*One-event*	*Inverse Weights*
	(1)	(2)	(3)	(4)
*FocalPatentGrant * AR⁺*	0.006**	0.005*	0.005*	0.005**
	(2.21)	(1.79)	(1.72)	(2.32)
*FocalPatentGrant * AR⁻*	0.001	0.000	0.000	0.001
	(0.32)	(0.71)	(0.18)	(0.61)
Controls	Yes	Yes	Yes	Yes
Firm x Cohort FE	Yes	Yes	Yes	Yes
Ind x Year x Cohort FE	Yes	Yes	Yes	Yes
Adj. R-squared	0.517	0.509	0.531	0.511
Observations	238,416	112,842	64,278	238,416

Panel B: Innovation Responses (Acquisition)

Variable	*Baseline*	*CleanWindow*	*One-event*	*Inverse Weights*
	(1)	(2)	(3)	(4)
*FocalPatentGrant * AR⁺*	0.029**	0.026	0.025	0.028*
	(2.17)	(1.49)	(1.29)	(1.73)
*FocalPatentGrant * AR⁻*	0.010	0.011	−0.009	0.009
	(0.46)	(0.73)	(−0.21)	(0.37)
Controls	Yes	Yes	Yes	Yes
Firm x Cohort FE	Yes	Yes	Yes	Yes
Ind x Year x Cohort FE	Yes	Yes	Yes	Yes
Adj. R-squared	0.442	0.436	0.452	0.438
Observations	238,416	112,842	64,278	238,416

Table 20: Alternative Asset Pricing Models and Market-Reaction Classifications

This table examines whether the classification of the focal firm's patent grants into AR^+ and AR^- grants is sensitive to the choice of asset-pricing model used to estimate abnormal returns. Column (1) reports the baseline market-model classification. Column (2) reconstructs abnormal returns using the Fama–French five-factor model, while Column (3) uses the q-factor model. AR^+ and AR^- indicators are redefined separately under each return-generating model, and the baseline regressions are re-estimated accordingly. The dependent variables are *R&D* and *Acquisition*. All firm-level controls (i.e., *Log(Asset), Tobin's Q, Cash, Leverage, Profitability,* and *Age*) are included in all specifications. Firm-cohort fixed effects as well as industry-year-cohort fixed effects are controlled for in the regressions. All standard errors are double clustered at the peer firm and year level. The reported coefficients are cohort-size-weighted averages of the cohort-specific treatment effects. The t-statistics in parentheses are calculated based on clustered standard errors. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A: Distribution of Significant Patent Grants

	Market Model	*FF5 Model*	*q-Factor Model*
	(1)	(2)	(3)
AR^+ Share	63.3%	62.1%	61.7%
AR^- Share	36.7%	37.9%	38.3%
Significant Patents	6,341	6,218	6,104

Panel B: Innovation Responses (R&D)

Variable	*Market Model*	*FF5 Model*	*q-Factor Model*
	(1)	(2)	(3)
*FocalPatentGrant * AR⁺*	0.006**	0.006**	0.005**
	(2.21)	(2.17)	(2.08)
*FocalPatentGrant * AR⁻*	0.001	0.001	0.000
	(0.32)	(0.74)	(0.18)
Controls	Yes	Yes	Yes
Firm x Cohort FE	Yes	Yes	Yes
Ind x Year x Cohort FE	Yes	Yes	Yes
Adj. R-squared	0.517	0.514	0.511
Observations	238,416	236,822	234,971

Panel C: Innovation Responses (Acquisition)

Variable	*Market Model*	*FF5 Model*	*q-Factor Model*
	(1)	(2)	(3)
*FocalPatentGrant * AR⁺*	0.029**	0.028**	0.027*
	(2.17)	(2.04)	(1.70)
*FocalPatentGrant * AR⁻*	0.010	0.008	0.007
	(0.46)	(0.36)	(0.30)
Controls	Yes	Yes	Yes
Firm x Cohort FE	Yes	Yes	Yes
Ind x Year x Cohort FE	Yes	Yes	Yes
Adj. R-squared	0.442	0.439	0.436
Observations	238,416	236,822	234,971

Internet Appendix

Build On or Branch Out?
Patent Grants and Peers' Innovation Policies

Seiwoong Hong

October 31, 2026

Section IA.1 reports alternative return measures and event-attribution checks for the AR^{+} and AR^{-} classification (Table IA1-1) and the responses for the four combinations of ex ante technological proximity and product-market rivalry (Table IA1-2), as discussed in Sections 3.3.1 and 4.8 of the paper.

Section IA.2 reports the dynamic responses of peers' cash holdings, leverage, net equity issuance, and capital expenditures (Figure IA2-1), as discussed in Section 4.7 of the paper.

Section IA.3 describes the construction of the patent-text similarity measure and reports the corresponding alignment estimates (Table IA3-1), as discussed in Section 4.10 of the paper.

Section IA.4 reports responses by the size of the peer's market reaction and the economic magnitude of the R&D response (Table IA4-1), as discussed in Sections 2.2 and 4.3 of the paper.

Section IA.5 plots the implied R&D response to AR^{+} patent grants across percentiles of cash holdings and innovative capability (Figure IA5-1), as discussed in Section 4.9 of the paper.

Section IA.6 reports the model that predicts application significance before the examiner's decision, together with its sample construction, predictors, and data sources (Tables IA6-1 and IA6-2), as discussed in Section 3.4.2 of the paper.

IA.1 Return Measurement and Pre-Grant Classification

Table IA1-1 checks the return measurement and attribution of the event. Adding the peer's standardized grant-day CAR as a control leaves AR^{+} coefficients of 0.004 for R&D and 0.021 for acquisitions, both significant at the 5% level. Restricting grant days to at most two patents granted to the focal firm yields 0.007 and 0.030. Using [0, +1] CARs or industry-adjusted CARs produces R&D estimates of 0.005–0.006 and acquisition estimates of 0.024–0.025, close to the baseline estimates of 0.006 and 0.029. These exercises preserve the central pattern while addressing distinct concerns about return magnitude, multiple grants, and contemporaneous industry news.

Table IA1-2 reports the four combinations of HighTech and HighRival. R&D and acquisition responses are concentrated in the high-proximity, low-rivalry group, which accounts for 29.4% of significant-grant pair-years. Exploration responses are concentrated in the low-proximity, high-rivalry group. These estimates are significant only at the 10% level. Estimates for pairs high on both dimensions are small and statistically insignificant, consistent with, but not sufficient to establish, offsetting forces.

Table IA1-1: Return Measurement and Event Attribution

Panel A uses R&D and Panel B uses Acquisition. Column (1) adds the peer's standardized grant-day CAR; Column (2) limits grant days to at most two patents granted to the focal firm; Column (3) uses CARs over [0, +1]; Column (4) uses industry-adjusted CARs. Controls, fixed effects, and clustering are as in Table 2. T-statistics are in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Panel A: Innovation Responses (R&D)

Variable	*Peer CAR control*	*≤ 2 patents on grant day*	*CAR[0,+1]*	*Industry-adjusted CAR*
	(1)	(2)	(3)	(4)
*FocalPatentGrant * AR⁺*	0.004**	0.007**	0.006**	0.005**
	(2.01)	(2.11)	(2.09)	(2.01)
*FocalPatentGrant * AR⁻*	0.001	0.001	0.000	0.001
	(0.27)	(0.22)	(0.08)	(0.31)
FocalPatentGrant	0.002	0.002	0.002	0.002
	(0.58)	(0.49)	(0.63)	(0.60)
Peer CAR (standardized)	0.001*			
	(1.71)			
Controls	Yes	Yes	Yes	Yes
Firm x Cohort FE	Yes	Yes	Yes	Yes
Ind x Year x Cohort FE	Yes	Yes	Yes	Yes
Adj. R-squared	0.518	0.522	0.516	0.517
Observations	238,416	151,208	236,950	237,128

Panel B: Innovation Responses (Acquisition)

Variable	*Peer CAR control*	*≤ 2 patents on grant day*	*CAR[0,+1]*	*Industry-adjusted CAR*
	(1)	(2)	(3)	(4)
*FocalPatentGrant * AR⁺*	0.021**	0.030**	0.025**	0.024**
	(2.36)	(2.37)	(2.49)	(2.56)
*FocalPatentGrant * AR⁻*	0.008	0.007	0.007	0.008
	(0.58)	(0.41)	(0.47)	(0.55)
FocalPatentGrant	0.003	0.002	0.003	0.003
	(0.27)	(0.21)	(0.28)	(0.27)
Peer CAR (standardized)	0.002**			
	(2.03)			
Controls	Yes	Yes	Yes	Yes
Firm x Cohort FE	Yes	Yes	Yes	Yes
Ind x Year x Cohort FE	Yes	Yes	Yes	Yes
Adj. R-squared	0.443	0.448	0.441	0.442
Observations	238,416	151,208	236,950	237,128

Table IA1-2: Technological Proximity and Product-Market Rivalry

This table reports responses to significant patent grants to the focal firm for the four combinations of *HighTech* and *HighRival* defined in Section 3.3.7. It uses the specification underlying Table 8 with indicators for each combination. Shares of significant-grant pair-years are in brackets. Controls, fixed effects, and clustering are as in Table 2. T-statistics are in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Four Combinations of Technological Proximity and Product-Market Rivalry

Variable	*R&D*	*Acquisition*	*ExplorationRatio*
	(1)	(2)	(3)
*FocalSigGrant * HighTech * LowRival [29.4%]*	0.004*	0.015*	−0.001
	(1.81)	(1.87)	(−0.21)
*FocalSigGrant * HighTech * HighRival [4.8%]*	0.001	0.008	0.003
	(0.33)	(0.36)	(0.68)
*FocalSigGrant * LowTech * LowRival [61.5%]*	0.001	0.003	0.001
	(0.42)	(0.47)	(0.27)
*FocalSigGrant * LowTech * HighRival [4.3%]*	−0.001	−0.004	0.007*
	(−0.31)	(−0.29)	(1.66)
Controls	Yes	Yes	Yes
Firm x Cohort FE	Yes	Yes	Yes
Ind x Year x Cohort FE	Yes	Yes	Yes
Adj. R-squared	0.517	0.442	0.647
Observations	238,416	238,416	238,416

IA.2 Financing and Investment Policies

In addition to examining peers' innovative efforts, I explore whether the focal firm's patent grants generate broader adjustments in peers' financial and operating policies. A focal firm's innovation may influence a peer's strategic posture in ways not fully captured by R&D spending or acquisition activity. For example, peers may revise their financial structure, accumulate precautionary liquidity, or shift real investment in response to perceived changes in technological competition. This section documents how a range of peer outcomes evolve following the focal firm's patent grants.

Equity issuance is a natural margin to examine. Following AR^+ patent grants, peers may need external funds to finance the expansion documented above, and equity is well suited to financing innovation because innovation relies on intangible assets that are difficult to pledge as collateral (Acharya and Xu, 2017; Brown et al., 2009; Brown and Petersen, 2011; Hall, 2002). Higher valuations after AR^+ grants may also make issuance attractive regardless of the patent's content (Baker et al., 2003). Section 4.7 separates the two explanations.

I focus on four firm-level outcomes: cash holdings, leverage, net equity issuance, and capital expenditures (CAPEX). Cash holdings capture changes in precautionary liquidity, which may increase when firms perceive greater technological uncertainty or competitive pressure. Leverage and equity issuance reflect adjustments in external financing. CAPEX provides a measure of real investment outside the R&D channel and can indicate whether firms reallocate resources toward physical assets or production capabilities instead of—or in addition to—innovative effort.

I intentionally do not examine outcomes such as sales growth or operating performance (e.g., ROA). While economically important, these measures typically adjust only with a delay, reflecting the time needed for new products to reach the market, competitive dynamics to unfold, and financial results to appear in accounting data. Such product-market outcomes are unlikely to respond contemporaneously. Focusing instead on financial and investment policies provides a clearer view of the peer's near-term strategic adjustments following the focal firm's innovation.

Figure IA2-1 summarizes the dynamic responses of these four firm-level outcomes—cash holdings, leverage, equity issuance, and CAPEX—around the focal firm's patent grants, separately for positive peer reactions (AR^{+}) and negative peer reactions (AR^{-}), both conditional on a positive focal-firm reaction. The AR^{+} vs. AR^{-} comparison captures how peers' financial and operating policies respond to the focal firm's innovations that have either favorable or unfavorable competitive implications, both of which matter economically for the peers. The dynamic event-time specifications used here follow the same empirical framework applied in Section 4.6.

The top row plots the responses for AR^{+} grants, where the market interprets the focal firm's patent as having favorable implications for the peer firm. Across Panels A, B, and D, the estimates fluctuate irregularly around zero, and the confidence intervals consistently overlap zero. These patterns indicate that peers do not make statistically meaningful adjustments in cash holdings, leverage, or CAPEX in the immediate aftermath of a focal firm's positively interpreted innovation. The only clear response appears in Panel C: equity issuance rises sharply beginning in the year after the focal firm's patent is granted, with coefficients that are statistically significant in years +1 through +3. Because a similar increase follows equally large price movements without a patent grant (Section 4.7), this response is better interpreted as market timing than as patent-specific financing.

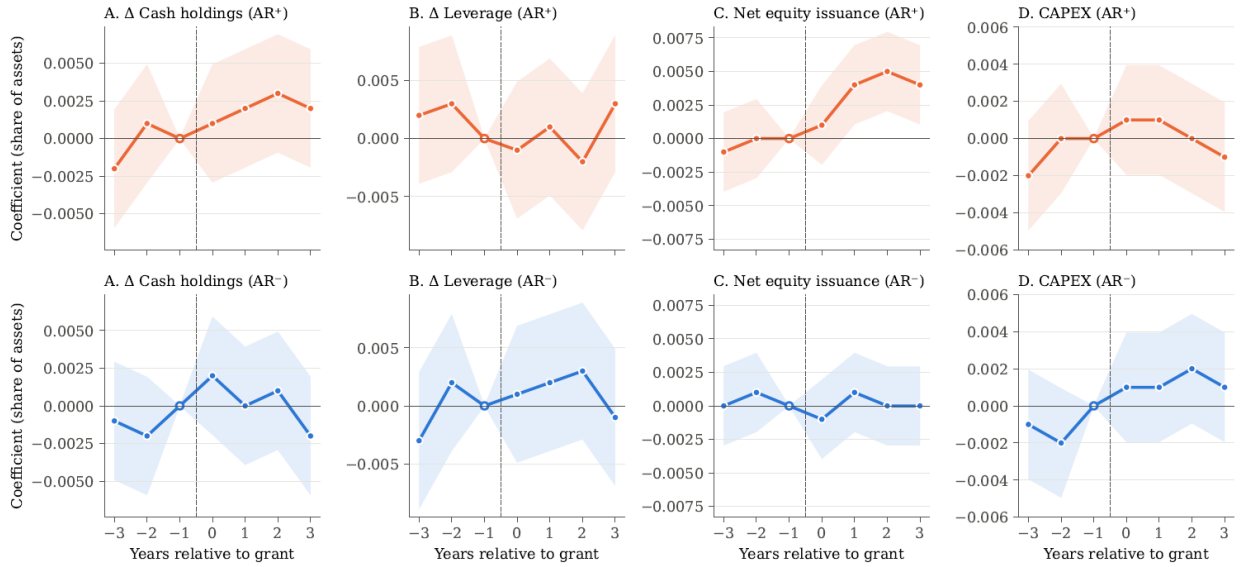
The magnitude of the equity response is economically meaningful: net equity issuance rises by about 0.4–0.5 percentage points of total assets per year in the three years following AR^{+} patent grants, roughly half of the sample mean (0.8%). Despite this sizable financing inflow, cash holdings themselves exhibit little systematic movement. This contrast is not inconsistent. Equity issuance is a flow variable that responds immediately when firms seek to expand financial flexibility, whereas cash holdings reflect the net effect of multiple concurrent uses and sources of funds. Newly raised equity may be allocated toward a range of purposes—including innovation-related spending, organizational capacity building, or general corporate needs—that do not necessarily result in a visible buildup of cash. Thus, the absence of a pronounced increase in liquidity does not contradict the strong equity response; rather, it is consistent with peers using higher valuations to raise equity.

The bottom row shows the corresponding responses for AR^- grants, where the focal firm's patent grants convey adverse implications for the peers' competitive position. Here, the movements in Panels A through D remain small, irregular, and consistently within wide confidence bands that overlap zero. Peers neither accumulate nor deplete liquidity in a systematic way (Panel A), nor do they adjust leverage, equity issuance, or CAPEX in a directionally consistent manner (Panels B through D). Overall, the AR^- patterns are noisy and centered near zero, suggesting the absence of any meaningful financial or real-investment responses to the focal firm's innovations that have unfavorable implications.

Taken together, the figure indicates that peers react through financial policy adjustments only when the focal firm's innovations are perceived to have positive implications. Patent grant shocks that have positive implications generate a pronounced increase in equity issuance while other balance-sheet and investment policies remain unchanged. In contrast, the focal firm's innovations that have negative implications do not trigger systematic changes in liquidity, financing, or physical investment. Because the equity response also follows large price movements without a grant, it appears to reflect higher valuations rather than the information in the patent. These financing patterns are also consistent with the broader literature on corporate investment opportunities. In particular, Hoberg and Phillips (2025) document that firms expanding the scope of their activities increase both R&D investment and equity issuance while relying relatively little on debt financing. The results presented here exhibit a similar pattern.

Figure IA2-1: Dynamic Response of Peers' Financial and Investment Policies after Focal Firm's Patent Grant

This figure plots the coefficients from the event-time specification in Section 4.6 for changes in peers' cash holdings (Panel A), changes in leverage (Panel B), net equity issuance (Panel C), and CAPEX (Panel D), all scaled by total assets. The top row reports responses to AR^{+} patent grants and the bottom row responses to AR^{-} patent grants. Year −1 is the omitted reference year. Shaded areas are 95% confidence intervals based on standard errors double clustered at the peer firm and year level.



IA.3 Textual Measures of Technological Alignment

Because citation overlap captures similarity in knowledge sources rather than in technological content, I also measure the textual similarity between each significant patent granted to the focal firm and the peer's subsequent inventions, using a term frequency–inverse document frequency (TF-IDF) representation of patent text, following Kelly et al. (2021). Each patent is represented by the text of its abstract and claims. I convert the text to lowercase, remove punctuation, numbers, and standard English stop words, apply Porter stemming, and drop terms that appear in fewer than ten patents. Each term *w* in patent *p* receives the weight $TF_{w,p} * IDF_w$, where $TF_{w,p}$ is the share of term *w* among all terms in patent *p*, and $IDF_w = \log\left(\frac{N_{\{<g\}}}{(1+n_{\{w,<g\}})}\right)$ is computed only from the $N_{\{<g\}}$ patents granted before the focal firm's patent grant date *g*, $n_{\{w,<g\}}$ of which contain term *w*. Computing the weights from earlier patents only, as in the backward IDF of Kelly et al. (2021), ensures that the measure does not use information that became available after the grant. The similarity between the focal firm's patent f and a peer patent q is the cosine of the angle between their unit-normalized TF-IDF vectors, which equals zero when the two patents share no terms and one when their weighted term proportions are identical.

I then define $TextSimilarity_{ij,t}$ as the average cosine similarity between the focal firm j's significant patent granted in year τ and all patents filed by peer i in event year t ∈ [τ − 3, τ + 3]. When a grant event involves several patents granted to the focal firm, I average over all pairs of patents belonging to the focal firm and the peer. Peer-years without any patent filing are missing, so the number of observations is smaller than in the baseline sample. Because the outcome is observed both before and after the grant within each cohort, the stacked difference-in-differences specification compares how textually close the peer's new patents are to the focal firm's patent after the grant with how close they were before the grant, relative to control peers.

The distribution of *TextSimilarity* is in line with existing text-based measures of patent similarity. In my sample, the mean is 0.120 and the median is 0.108 (Table 1), comparable to the median of 7.8% and mean of 10.2% that Kelly et al. (2021) report for pairwise patent similarity and modestly higher, as expected,

because technological peers share CPC subclasses and citation links with the focal firm. As a validation, peer patents that cite the focal firm's patent have an average similarity of 0.213, compared with 0.114 for peer patents that do not.

With controls, TextSimilarity increases by 0.019 after AR^{+} grants and decreases by 0.015 after AR^{-} grants, both significant at the 5% level (Table IA3-1). Relative to a mean of 0.120, these coefficients represent changes of about 15.8% and −12.5%. The estimates reinforce the directional pattern in citation overlap. Text similarity is observed only when a peer files a patent. These estimates therefore describe alignment among filing peer-years; they do not measure changes in the probability of filing or the quality of resulting inventions.

Table IA3-1: Patent Text Similarity and Technological Alignment

This table reports the estimation results for changes in the textual similarity between the focal firm's patents and peers' subsequent patents. The dependent variable is *TextSimilarity*, the average TF-IDF cosine similarity between the focal firm's significant patent and the patents filed by the peer in each event year t $\in [\tau - 3, \tau + 3]$. Patent text consists of abstracts and claims, and IDF weights are computed from patents granted before the focal firm's patent grant date (Kelly et al., 2021). Peer-years without patent filings are excluded. The main explanatory variables are $FocalPatentGrant * AR^{+}$ and $FocalPatentGrant * AR^{-}$. All firm-level controls (i.e., *Log(Asset), Tobin's Q, Cash, Leverage, Profitability,* and *Age*) are included in Column (2). Firm-cohort fixed effects as well as industry-year-cohort fixed effects are controlled for in the regression. All standard errors are double clustered at the peer firm and year level. The reported coefficients are cohort-size-weighted averages of the cohort-specific treatment effects. The t-statistics in parentheses are calculated based on clustered standard errors. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Variable	*TextSimilarity*	*TextSimilarity*
	(1)	(2)
*FocalPatentGrant * AR⁺*	0.024***	0.019**
	(3.14)	(2.42)
*FocalPatentGrant * AR⁻*	−0.018**	−0.015**
	(−2.27)	(−2.08)
Controls	No	Yes
Firm x Cohort FE	Yes	Yes
Ind x Year x Cohort FE	Yes	Yes
Adj. R-squared	0.284	0.312
Observations	144,738	143,712

IA.4 Response Size and Economic Magnitude

If the market reaction measures how strongly a patent granted to the focal firm affects a peer, as in the model in Appendix B, responses should grow with the size of that reaction rather than switch on at the two-standard-deviation threshold. To examine this, I split grants with a standardized CAR above 2 for the focal firm by the size of the peer's standardized CAR and interact *FocalPatentGrant* with an indicator for each category in the baseline specification. The categories between 1 and 2 standard deviations, which are not classified as AR^+ or AR^- in the baseline, are included as treated events. Section 5.2 provides a complementary analysis of exposure intensity using the number of significant patent grants within a year.

Table IA4-1, Panel A, reports larger responses for larger peer reactions. AR^+ R&D coefficients rise from 0.002 for standardized CARs between 1 and 2 to 0.005 for CARs between 2 and 3 and 0.009 for CARs above 3; the latter two are significant at the 5% level. Acquisition responses rise from 0.8 to 2.4 to 4.1 percentage points, with $p = 0.050$ for the difference between extreme categories. On the negative side, exploration coefficients rise from 0.004 to 0.010 to 0.017 as returns become more negative. The pattern supports a graded response, but the estimates do not establish a discontinuity or a causal effect of return magnitude.

Panel B scales the R&D coefficient of 0.006 by the affected peers' average assets and four post-grant years. This implies about $21.6 million per AR^+ peer-focal pair and $67.6 billion across 3,129 pairs, in 2021 dollars. The aggregate calculation can count the same peer's response more than once when exposures overlap. It is an illustrative extrapolation, not an estimate of unique aggregate R&D created by patent grants.

Table IA4-1: Dose-Response and Economic Magnitude

Panel A re-estimates the baseline specification after splitting grants with a standardized CAR above 2 for the focal firm by the peer's standardized CAR z. The categories $1 < z \leq 2$ and $-2 \leq z < -1$ are weaker reactions that are not classified as AR^+ or AR^- in the baseline; including them as treated events increases the stacked sample. Each peer-focal-year is assigned to the category of the grant with the largest absolute peer reaction, so the categories are mutually exclusive; together with the 5,052 peer-focal-years with $|z| \leq 1$, they sum to the 18,912 peer-focal-years in Table 1, Panel B. Panel B combines the estimates in Table 2 with sample averages; dollar values are in 2021 dollars. All firm-level controls are included. All standard errors are double clustered at the peer firm and year level. The t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A: Responses by the Size of the Peer's Market Reaction

Variable	*R&D*	*Acquisition*	*ExplorationRatio*
	(1)	(2)	(3)
*FocalPatentGrant * (1 < z ≤ 2)*	0.002	0.008	0.000
	(1.21)	(1.30)	(0.04)
*FocalPatentGrant * (2 < z ≤ 3)*	0.005**	0.024**	0.003
	(2.01)	(2.33)	(0.12)
*FocalPatentGrant * (z > 3)*	0.009**	0.041***	0.006
	(2.12)	(2.64)	(0.19)
*FocalPatentGrant * (−2 ≤ z < −1)*	0.000	0.002	0.004
	(0.07)	(0.21)	(1.18)
*FocalPatentGrant * (−3 ≤ z < −2)*	0.001	0.008	0.010*
	(0.24)	(0.47)	(1.78)
*FocalPatentGrant * (z < −3)*	0.001	0.012	0.017**
	(0.19)	(0.52)	(2.08)
FocalPatentGrant	0.002	0.003	0.001
	(0.57)	(0.26)	(0.29)
p-value: (z > 3) = (1 < z ≤ 2)	0.13	0.050	
p-value: (z < −3) = (−2 ≤ z < −1)			0.14
Controls	Yes	Yes	Yes
Firm x Cohort FE	Yes	Yes	Yes
Ind x Year x Cohort FE	Yes	Yes	Yes
Adj. R-squared	0.519	0.444	0.649
Observations	312,640	312,640	312,640

Panel B: Economic Magnitude of the R&D Response to AR^+ Patent Grants

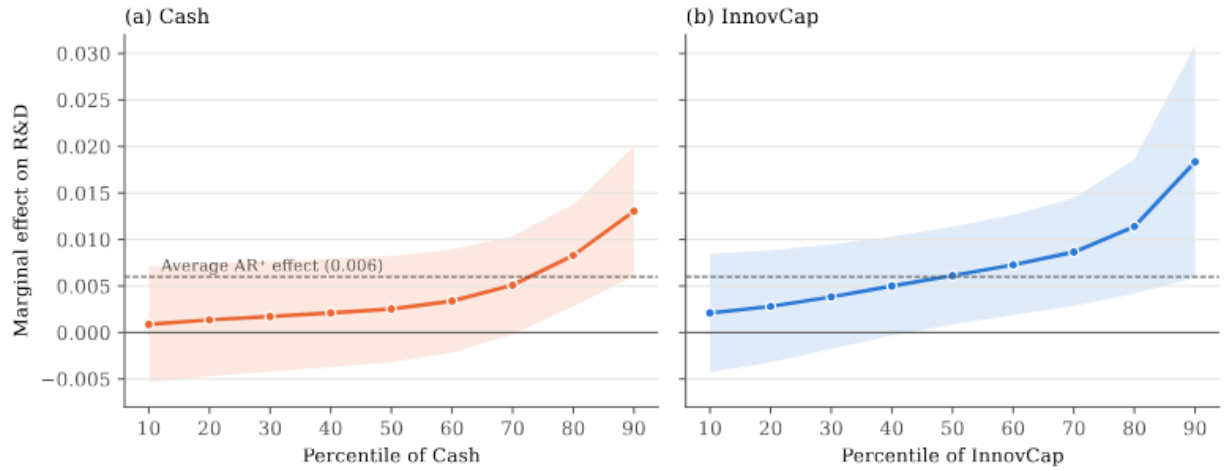
Item	*Value*	*Calculation*
AR^+ peer-focal-years	3,129	Table 1
Mean total assets of affected peers	\$0.90 billion	Compustat
Annual R&D response (share of assets)	0.006	Table 2, Column (3)
Post-grant years	4	Figure 5
Induced R&D per AR^+ pair	\$21.6 million	0.006 × 0.90 × 4
Induced R&D, all AR^+ pairs (2001–2021)	\$67.6 billion	3,129 × 21.6

IA.5 Marginal Responses by Financial and Innovative Capacity

Figure IA5-1 plots the implied AR^+ R&D response across percentiles of *Cash* and *InnovCap* using Table 10, Panel B. The estimates rise from about 0.001 to 0.013 between the 10th and 90th cash percentiles, and from about 0.003 to 0.015 across the same capability percentiles. These plots summarize the fitted interactions; they are not additional independent tests.

Figure IA5-1: Marginal Effect of Focal Firm's Patent Grant across Percentiles of Cash and Innovative Capability

This figure plots the marginal effect of AR^+ patent grants on peers' R&D across deciles of *Cash* (Panel a) and *InnovCap* (Panel b), implied by the estimates in Panel B of Table 10. The marginal effect at each decile equals the coefficient on *FocalPatentGrant* × AR^+ plus the coefficient on the corresponding triple interaction multiplied by the standardized value of the moderator at that decile. Shaded areas are 95% confidence intervals based on standard errors double clustered at the peer firm and year level. The dashed line marks the average effect of AR^+ patent grants (0.006).



IA.6 Predicting Application Significance

Tables IA6-1 and IA6-2 document the pre-decision significance model, sample construction, predictors, and data sources used in Section 3.4.2.

Table IA6-1: Predicting Significant Grants from Pre-Decision Information

This table reports the model used to predict, before the examiner's decision, whether an application would be a significant grant. A granted application is significant if the focal firm's standardized five-day CAR around the grant date exceeds 2 (Section 3.3.1); applications granted to the same firm on the same day share the same label. Panel A reports a logit model estimated on granted applications in the final sample of Panel B, with CPC subclass-by-filing-year fixed effects. All predictors are measured with information available before the examiner's decision and are standardized; Table IA6-2 defines each predictor, lists its data source, and explains why it is expected to predict significance. Predicted significance for granted and abandoned applications is computed with coefficients estimated on other filing-year cohorts (leave-cohort-out). Panel B describes the sample. Because abandoned applications are publicly observable only if the application was published, the sample is restricted to applications filed on or after November 29, 2000 (the effective date of the American Inventors Protection Act) that did not request non-publication. Applications are classified as granted or abandoned using their final disposition; applications still pending at the end of 2018 are excluded so that outcomes can be observed for three years after the decision. Standard errors are clustered at the focal firm level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A: Logit Model of Significant Grants

Predictor (standardized)	*Significant grant*
	(1)
Log(1 + independent claims in published application)	0.12**
	(2.44)
Foreign priority claim	0.05
	(1.41)
Novelty of published application text	0.18***
	(3.27)
Citations to published application before decision	0.09**
	(2.04)
Focal firm's significant-grant rate, prior 3 years	0.24***
	(4.63)
Log(market capitalization), prior year end	−0.22***
	(−3.87)
CPC subclass x Filing-year FE	Yes
Pseudo R-squared	0.039
Out-of-sample AUC	0.61
Share of significant grants, all granted applications	0.046
Share of significant grants, top quintile of predicted significance	0.074
Share of all significant grants in top quintile	0.33
Observations (granted applications)	798,650

Table IA6-1 (continued)

Panel B: Sample Construction

Step	*Applications*
Utility applications assigned to sample firms, filed Nov 29, 2000 – Dec 31, 2016	1,598,240
Less: applications with a non-publication request (not published)	1,362,710
Less: no final disposition (granted or abandoned) by Dec 31, 2018	1,281,560
Less: examiner or art unit missing, or fewer than 10 other applications for the examiner in the art unit-year	1,207,930
Less: published claims or text unavailable	1,193,480
Less: focal firm without stock returns over the event and estimation windows	1,142,870
of which granted	798,650
of which abandoned	344,220
Top quintile of predicted significance	228,574
Highest-predicted application per focal firm-year (IV sample, Table 4)	8,126

Table IA6-2: Predictors of Significant Grants: Definitions, Data Sources, and Rationale

This table defines the predictors in Table IA6-1, lists their data sources, and summarizes why each is expected to predict whether a grant moves the patentee's stock price. All predictors are measured before the examiner's decision. The bottom rows list the data sources used to construct the sample in Panel B of Table IA6-1.

Predictor and definition	*Data source*	*Why it predicts a significant grant (expected sign)*
Independent claims: log(1 + number of independent claims in the pre-grant publication)	PatentsView pre-grant publication claims (independent claims identified by the dependency field); USPTO Patent Claims Research Dataset (Marco et al., 2019)	(+) Independent claims define the scope of the right to exclude. Broader claimed scope is associated with higher patent value, renewal, and litigation (Lanjouw and Schankerman, 2001, 2004), so grants of such applications are more likely to change expected profits.
Foreign priority: indicator that the application claims priority to a foreign application	PatEx foreign priority data (Graham et al., 2018)	(+) Seeking protection in several jurisdictions is costly and is undertaken for more valuable inventions; patent family size predicts patent value (Putnam, 1996; Harhoff et al., 2003).
Novelty: text-based novelty of the pre-grant publication relative to earlier patents	Abstract and claims of the pre-grant publication (PatentsView); backward-IDF similarity measure of Kelly et al. (2021)	(+) Kelly et al. (2021) show that text-based novelty identifies breakthrough patents that are associated with higher market value; the measure is computed only from documents published before the application.
Pre-decision citations: number of granted patents citing the pre-grant publication before the examiner's decision	PatentsView citations to pre-grant publications (application citations by granted patents), with citing grant dates	(+) Forward citations measure technological importance and are correlated with market value (Hall et al., 2005); counting only citations before the decision keeps the measure ex ante.
Firm significance rate: share of the focal firm's grant days with standardized CAR above 2 over the prior three years	CRSP daily stock returns; DISCERN firm–patent links (Arora et al., 2021); PatentsView grant dates	(+) Whether a grant moves the stock price depends on how material the firm's patents are to its value and on investor attention to its patents, both of which are persistent firm characteristics.
Firm size: log market capitalization at the end of the prior year	CRSP	(−) A patent of given value is a smaller share of a larger firm's market value, so its grant is less likely to produce a return two standard deviations above the firm's normal variation (Kogan et al., 2017).
Sample construction (Table IA6-1, Panel B)	PatEx (application filing dates, publication status, final disposition, examiner and art unit); DISCERN (assignee–firm match); PatentsView (published claims and text); CRSP (stock returns)	